

Multivariate Real Analysis

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In here, we extend the results of univariate real analysis to multivariate and/or vector-valued functions. In practice, multivariate calculus is used, and there are many new results that arise in the multivariate case. The case for computing limits and continuity on multivariate functions is very straightforward, since these are topological properties. Generally, to prove that a limit is something, we just use ϵ - δ , and to actually *compute* what the limit might be, we can just compute $f(a)$ assuming that it is continuous, or take some sort of path p to take the univariate limit. We can also show that the multivariate limit doesn't exist by taking two sequences or two path functions where the limits do not equal each other.

However, the definition of the derivative and the integral will need to be generalized. Note that both the domain and codomain of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ are vector spaces. However, Euclidean spaces have a lot of structure on them, and it is nice to identify the essential properties we need from these sets. It is essential that we work in vector spaces because when we define a derivative, we usually see some form that looks like

$$\frac{f(x+h) - f(x)}{h} \quad (1)$$

Note the operations used here. First, we want a notion of addition in the domain ($x+h$) and the codomain $f(x+h) - f(x)$, along with some scalar multiplication when we multiply by $1/h$. A vector space precisely supports these operations and therefore is a natural choice. It is immediate that to define convergence, we definitely need a topology. We will see later that we want to define multivariate derivatives by adding a norm to this term, requiring the use of a normed vector space. Completion is clearly essential as we have seen in single-variable analysis.

Definition 0.1 (Banach Space)
A Banach space is a normed complete vector space.

Note that by extending the dimension, we have essentially lost the ordering $\leq$ on these spaces, along with the field properties. Therefore, we will need to adapt our definitions accordingly.

Differentiation

1.1 Partial, Directional, and Path Derivatives

A simple way to extract derivatives is to fix all of the input components except for one, and treat f as a single-variable function. This results in—for now—a completely separate notion of a derivative.

Definition 1.1 (Partial Derivative)

Given function $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$, the i th **partial derivative** of f at $a \in E$ is defined

$$\left. \frac{\partial f}{\partial x_i} \right|_a := \lim_{h \rightarrow 0} \frac{f(a + he_i) - f(a)}{h} \quad (2)$$

where e_i is the i th basis vector of $\mathbb{R}^n$.

Since we are fixing every other parameter except for the i th one, we can find the partial derivative by differentiating the function w.r.t. x_i and fixing all other variables.

Example 1.1 (Partial Derivatives of Saddle Points)

To find all partial derivatives of the function

$$f(x, y) = x^2 - y^2 \quad (3)$$

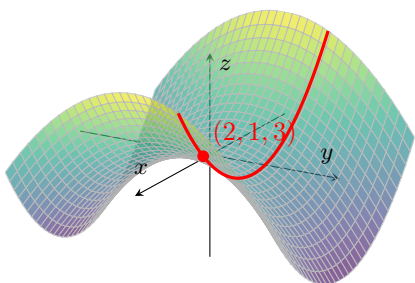
We compute

$$\frac{\partial f}{\partial x} = 2x \quad (4)$$

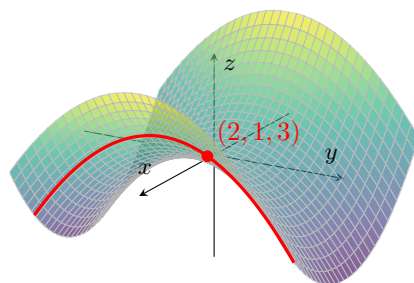
$$\frac{\partial f}{\partial y} = -2y. \quad (5)$$

At the point $(2, 1)$, these become

$$\frac{\partial f}{\partial x}(2, 1) = 4 \quad \text{and} \quad \frac{\partial f}{\partial y}(2, 1) = -2. \quad (6)$$



(a) Slicing the saddle surface along the plane $y = 1$, the resulting curve has slope 4 at $(2, 1, 3)$.



(b) Slicing along the plane $x = 2$ gives a curve with slope -2 at the same point.

Figure 1: Two cross-sections of the saddle surface $z = x^2 - y^2$ through the point $(2, 1, 3)$. Each slice isolates one partial derivative by holding the other variable fixed.

Example 1.2 (Partial Derivatives Using Differentiation Rules)

To find all partial derivatives of the function

$$f(x, y, z) = x \ln(y^2 + z^2), \quad (7)$$

We compute

$$\frac{\partial f}{\partial x} = \ln(y^2 + z^2) \quad (8)$$

$$\frac{\partial f}{\partial y} = x \cdot \frac{1}{y^2 + z^2} \cdot (2y) = \frac{2xy}{y^2 + z^2} \quad (9)$$

$$\frac{\partial f}{\partial z} = x \cdot \frac{1}{y^2 + z^2} \cdot (2z) = \frac{2xz}{y^2 + z^2} \quad (10)$$

Example 1.3 (Partial Derivatives Using Limit Definition)

Let

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2} & (x, y) \neq (0, 0), \\ 0 & (x, y) = (0, 0). \end{cases} \quad (11)$$

We compute the partial derivatives $f_x(0, 0)$ and $f_y(0, 0)$ using the limit definition.

First, for the partial derivative with respect to x , we have

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} \quad (12)$$

$$= \lim_{h \rightarrow 0} \frac{\frac{h(0)}{h^2 + 0^2} - 0}{h} \quad (13)$$

$$= \lim_{h \rightarrow 0} \frac{0}{h} \quad (14)$$

$$= 0. \quad (15)$$

Similarly, for the partial derivative with respect to y , we have

$$f_y(0, 0) = \lim_{h \rightarrow 0} \frac{f(0, h) - f(0, 0)}{h} \quad (16)$$

$$= \lim_{h \rightarrow 0} \frac{\frac{0(h)}{0^2 + h^2} - 0}{h} \quad (17)$$

$$= \lim_{h \rightarrow 0} \frac{0}{h} \quad (18)$$

$$= 0. \quad (19)$$

Therefore,

$$f_x(0, 0) = 0 \quad \text{and} \quad f_y(0, 0) = 0. \quad (20)$$

Note that even though the limit of $f(x, y)$ as $(x, y) \rightarrow (0, 0)$ does not exist, the partial derivatives at $(0, 0)$ still exist.

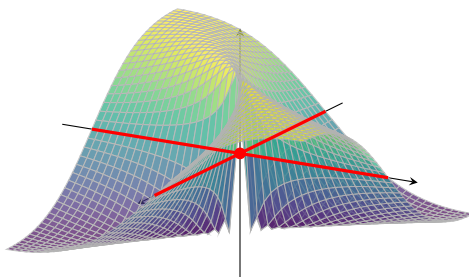


Figure 2: Both coordinate cross-sections of $f(x, y) = \frac{xy}{x^2 + y^2}$ through the origin lie on the line $z = 0$, which shows why $f_x(0, 0) = 0$ and $f_y(0, 0) = 0$ even though the full two-variable limit does not exist.

The partial derivative looks at the function as it is approaching a along an axis, but we can consider any direction in the domain.

Definition 1.2 (Directional Derivative)

Given function $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$, the **directional derivative** of f at $a \in E$ in direction $u \in \mathbb{R}^n$ is defined

$$D_v f_a := \lim_{h \rightarrow 0} \frac{f(a + hu) - f(a)}{h} \quad (21)$$

It should be obvious that if all directional derivatives exist, then the partials exist (since we can just set the directional vectors to be the unit vectors).

When computing directional derivatives, it is convenient to normalize the directional vector v to be unit length so that it coincides with the partial derivatives. We don't technically need to set $\|v\| = 1$, but if we have two vectors v and a scaled cv , then the directional derivatives will also be scaled ($\nabla_{cv} f(a) = c \nabla_v f(a)$), so we will only work with unit directional vectors. Some say that this restriction is undesirable, since it loses the linearity of the function $v \mapsto \partial_v f(a)$.

Example 1.4

Let

$$f(x, y) = x^2 y + 4y^2. \quad (22)$$

We compute the directional derivative of f at the point $(2, 1)$ in the direction of the vector

$$v = (1, 1).$$

First, we normalize v . Since

$$\|v\| = \sqrt{1^2 + 1^2} = \sqrt{2}, \quad (23)$$

the unit direction vector is

$$u = \frac{v}{\|v\|} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right). \quad (24)$$

By the limit definition of the directional derivative,

$$D_u f(2, 1) = \lim_{h \rightarrow 0} \frac{f\left(\left(2, 1\right) + hu\right) - f(2, 1)}{h} \quad (25)$$

$$= \lim_{h \rightarrow 0} \frac{f\left(2 + \frac{h}{\sqrt{2}}, 1 + \frac{h}{\sqrt{2}}\right) - f(2, 1)}{h}. \quad (26)$$

Now,

$$f(2, 1) = 2^2(1) + 4(1)^2 = 8. \quad (27)$$

Also,

$$f\left(2 + \frac{h}{\sqrt{2}}, 1 + \frac{h}{\sqrt{2}}\right) = \left(2 + \frac{h}{\sqrt{2}}\right)^2 \left(1 + \frac{h}{\sqrt{2}}\right) + 4\left(1 + \frac{h}{\sqrt{2}}\right)^2 \quad (28)$$

$$= 8 + \frac{16h}{\sqrt{2}} + \frac{9h^2}{2} + \frac{h^3}{2\sqrt{2}}. \quad (29)$$

Therefore,

$$D_u f(2, 1) = \lim_{h \rightarrow 0} \frac{8 + \frac{16h}{\sqrt{2}} + \frac{9h^2}{2} + \frac{h^3}{2\sqrt{2}} - 8}{h} \quad (30)$$

$$= \lim_{h \rightarrow 0} \left(\frac{16}{\sqrt{2}} + \frac{9h}{2} + \frac{h^2}{2\sqrt{2}} \right) \quad (31)$$

$$= \frac{16}{\sqrt{2}} \quad (32)$$

$$= 8\sqrt{2}. \quad (33)$$

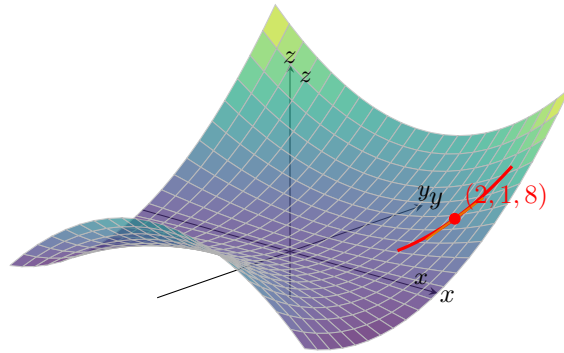


Figure 3: The surface $z = x^2y + 4y^2$ together with the red curve obtained by moving from $(2, 1)$ in the direction $u = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$. The orange line is the tangent line, whose slope is the directional derivative $D_u f(2, 1) = 8\sqrt{2}$.

Example 1.5

Let

$$f(x, y, z, w) = x^2y + z \cos(w). \quad (34)$$

We compute the directional derivative of f at the point $(1, 2, 0, 0)$ in the direction of the vector

$$v = (1, 1, 1, 1).$$

First, we normalize v . Since

$$\|v\| = \sqrt{1^2 + 1^2 + 1^2 + 1^2} = 2, \quad (35)$$

the unit direction vector is

$$u = \frac{v}{\|v\|} = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right). \quad (36)$$

By the limit definition of the directional derivative,

$$D_u f(1, 2, 0, 0) = \lim_{h \rightarrow 0} \frac{f((1, 2, 0, 0) + hu) - f(1, 2, 0, 0)}{h} \quad (37)$$

$$= \lim_{h \rightarrow 0} \frac{f\left(1 + \frac{h}{2}, 2 + \frac{h}{2}, \frac{h}{2}, \frac{h}{2}\right) - f(1, 2, 0, 0)}{h}. \quad (38)$$

Now,

$$f(1, 2, 0, 0) = 1^2(2) + 0 \cos(0) = 2. \quad (39)$$

Also,

$$f\left(1 + \frac{h}{2}, 2 + \frac{h}{2}, \frac{h}{2}, \frac{h}{2}\right) = \left(1 + \frac{h}{2}\right)^2 \left(2 + \frac{h}{2}\right) + \frac{h}{2} \cos\left(\frac{h}{2}\right) \quad (40)$$

$$= \left(1 + h + \frac{h^2}{4}\right) \left(2 + \frac{h}{2}\right) + \frac{h}{2} \cos\left(\frac{h}{2}\right) \quad (41)$$

$$= 2 + \frac{5h}{2} + h^2 + \frac{h^3}{8} + \frac{h}{2} \cos\left(\frac{h}{2}\right). \quad (42)$$

Therefore,

$$D_u f(1, 2, 0, 0) = \lim_{h \rightarrow 0} \frac{2 + \frac{5h}{2} + h^2 + \frac{h^3}{8} + \frac{h}{2} \cos\left(\frac{h}{2}\right) - 2}{h} \quad (43)$$

$$= \lim_{h \rightarrow 0} \left(\frac{5}{2} + h + \frac{h^2}{8} + \frac{1}{2} \cos\left(\frac{h}{2}\right) \right) \quad (44)$$

$$= \frac{5}{2} + \frac{1}{2} \quad (45)$$

$$= 3. \quad (46)$$

But be careful! The existence of partials does not imply the existence of all directional derivatives!

Example 1.6 (Existence of Partial $\not\Rightarrow$ Existence of Directional Derivatives)

Consider the function

$$f(x_1, x_2) = \begin{cases} \frac{x_1 x_2}{x_1^2 + x_2^2} & \text{if } (x_1, x_2) \neq (0, 0) \\ 0 & \text{if } (x_1, x_2) = (0, 0) \end{cases}$$

The partial derivatives exist everywhere. Away from the origin we can simply compute

$$\begin{aligned} \frac{\partial f}{\partial x_1} &= \frac{x_2(x_1^2 + x_2^2) - x_1 x_2 \cdot 2x_1}{(x_1^2 + x_2^2)^2} = \frac{-x_1^2 x_2 + x_2^3}{(x_1^2 + x_2^2)^2} \\ \frac{\partial f}{\partial x_2} &= \frac{x_1(x_1^2 + x_2^2) - x_1 x_2 \cdot 2x_2}{(x_1^2 + x_2^2)^2} = \frac{x_1^3 - x_1 x_2^2}{(x_1^2 + x_2^2)^2} \end{aligned}$$

As for the partials at the origin, we must compute using the limit rule.

$$\begin{aligned} \partial_{x_1} f(0) &= \lim_{h \rightarrow 0} \frac{f(0 + he_1) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{f(h, 0)}{h} = 0 \\ \partial_{x_2} f(0) &= \lim_{h \rightarrow 0} \frac{f(0 + he_2) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{f(0, h)}{h} = 0 \end{aligned}$$

However, the directional derivative taken in direction $v = (1, 1)$ gives

$$\begin{aligned}\nabla_{(1,1)}f(0,0) &= \lim_{h \rightarrow 0} \frac{f(0+h(1,1)) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(h,h)}{h} \\ &= \lim_{h \rightarrow 0} \frac{1}{2h}\end{aligned}$$

which does not have a limit as $h \rightarrow 0$. To visualize this, let's look at the values of f along various lines in $\mathbb{R}^2$.

1. $f = 0$ at the line $x_1 = 0$ and $x_2 = 0$, which is why the partials are 0.
2. $f = \frac{1}{2}$ at the line where $x_1 = x_2$, except for the point $(0,0)$, where $f = 0$, which is why the limit in the direction doesn't exist.

Definition 1.3 (Path Derivative)

Let $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$, $a \in E$, and $p : \mathbb{R} \rightarrow \mathbb{R}^n$ be a path function where $p(t_0) = a$. Then, the **path derivative** of f at a with respect to path p is defined

$$D_p f(a) := \lim_{t \rightarrow t_0} \frac{f(p(t)) - f(p(t_0))}{h} \quad (47)$$

Example 1.7

Let

$$f(x, y) = x^2 + y, \quad (48)$$

and let

$$p(t) = (1 + t, 1 + t^2). \quad (49)$$

Since

$$p(0) = (1, 1), \quad (50)$$

we compute the path derivative of f at $a = (1, 1)$ with respect to the path p .

By the definition of the path derivative,

$$D_p f(1, 1) = \lim_{t \rightarrow 0} \frac{f(p(t)) - f(p(0))}{t - 0} \quad (51)$$

$$= \lim_{t \rightarrow 0} \frac{f(1 + t, 1 + t^2) - f(1, 1)}{t}. \quad (52)$$

Now,

$$f(1, 1) = 1^2 + 1 = 2, \quad (53)$$

and

$$f(1 + t, 1 + t^2) = (1 + t)^2 + (1 + t^2) \quad (54)$$

$$= 1 + 2t + t^2 + 1 + t^2 \quad (55)$$

$$= 2 + 2t + 2t^2. \quad (56)$$

Therefore,

$$D_p f(1, 1) = \lim_{t \rightarrow 0} \frac{2 + 2t + 2t^2 - 2}{t} \quad (57)$$

$$= \lim_{t \rightarrow 0} \frac{2t + 2t^2}{t} \quad (58)$$

$$= \lim_{t \rightarrow 0} (2 + 2t) \quad (59)$$

$$= 2. \quad (60)$$

Example 1.8

Let

$$f(x, y) = (xy, x^2 - y), \quad (61)$$

and let

$$p(t) = (\cos t, \sin t). \quad (62)$$

Since

$$p(0) = (1, 0), \quad (63)$$

we compute the path derivative of f at $a = (1, 0)$ with respect to the path p .

By the definition of the path derivative,

$$D_p f(1, 0) = \lim_{t \rightarrow 0} \frac{f(p(t)) - f(p(0))}{t - 0} \quad (64)$$

$$= \lim_{t \rightarrow 0} \frac{f(\cos t, \sin t) - f(1, 0)}{t}. \quad (65)$$

Now,

$$f(1, 0) = (0, 1), \quad (66)$$

and

$$f(\cos t, \sin t) = (\cos t \sin t, \cos^2 t - \sin t). \quad (67)$$

Therefore,

$$D_p f(1, 0) = \lim_{t \rightarrow 0} \frac{(\cos t \sin t, \cos^2 t - \sin t) - (0, 1)}{t} \quad (68)$$

$$= \lim_{t \rightarrow 0} \left(\frac{\cos t \sin t}{t}, \frac{\cos^2 t - \sin t - 1}{t} \right). \quad (69)$$

We compute each component separately. For the first component,

$$\lim_{t \rightarrow 0} \frac{\cos t \sin t}{t} = \lim_{t \rightarrow 0} \cos t \cdot \frac{\sin t}{t} \quad (70)$$

$$= 1. \quad (71)$$

For the second component,

$$\lim_{t \rightarrow 0} \frac{\cos^2 t - \sin t - 1}{t} = \lim_{t \rightarrow 0} \left(\frac{\cos^2 t - 1}{t} - \frac{\sin t}{t} \right) \quad (72)$$

$$= \lim_{t \rightarrow 0} \left(\frac{-\sin^2 t}{t} - \frac{\sin t}{t} \right) \quad (73)$$

$$= 0 - 1 \quad (74)$$

$$= -1. \quad (75)$$

Hence,

$$D_p f(1, 0) = (1, -1). \quad (76)$$

1.2 Total Derivatives

Definition 1.4 (Total/Frechet Derivative)

Let $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $a \in E$. f is **differentiable** at a if it satisfies either of the equivalent conditions.

1. *Linear Approximation.* There exists a linear map $Df_a : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that

$$\lim_{h \rightarrow 0} \frac{\|f(a+h) - f(a) - Df_a h\|}{\|h\|} = 0 \quad (77)$$

2. *Fundamental Increment Lemma.* There exists a linear map $Df_a : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and an infinitesimal^b function $\Phi : \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined for sufficiently small nonzero $h \in \mathbb{R}^n$ such that

$$f(a+h) = f(a) + Df_a h + \Phi(h)\|h\|, \quad \lim_{h \rightarrow 0} \Phi(h) = 0 \quad (78)$$

If such a map Df_a exists, then it is unique, and we call this the **total derivative**—or **Frechet derivative**—of f at a .

^aNote that when $m = 1$, then Df_a is a linear functional, i.e. a dual vector.

^bAs in $\Phi(h) \rightarrow 0$ as $h \rightarrow 0$.

Proof. We'll first prove that the two are equivalent.

1. ($\rightarrow$). Given that linear approximation holds, we can construct the map

$$\Phi(h) := \frac{f(a+h) - f(a) - Df_a h}{\|h\|} \quad (79)$$

Therefore, by our assumption and absolute homogeneity of the norm, we have

$$0 = \lim_{h \rightarrow 0} \frac{\|f(a+h) - f(a) - Df_a h\|}{\|h\|} \quad (80)$$

$$= \lim_{h \rightarrow 0} \left\| \frac{f(a+h) - f(a) - Df_a h}{\|h\|} \right\| \quad (81)$$

$$= \lim_{h \rightarrow 0} \|\Phi(h)\| \quad (82)$$

Since the norm $v \mapsto \|v\|$ is continuous, this implies that

$$\lim_{h \rightarrow 0} \Phi(h) = 0 \quad (83)$$

2. ($\leftarrow$). Given that the fundamental increment lemma holds, we can see that for nonzero h ,

$$\Phi(h) = \frac{f(a+h) - f(a) - Df_a h}{\|h\|} \quad (84)$$

Therefore, by taking the limit of both sides, we must have

$$0 = \lim_{h \rightarrow 0} \Phi(h) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a) - Df_a h}{\|h\|} \quad (85)$$

Therefore, by continuity of the norm and absolute homogeneity,

$$0 = \|0\| = \left\| \lim_{h \rightarrow 0} \frac{f(a+h) - f(a) - Df_a h}{\|h\|} \right\| \quad (86)$$

$$= \lim_{h \rightarrow 0} \left\| \frac{f(a+h) - f(a) - Df_a h}{\|h\|} \right\| \quad (87)$$

$$= \lim_{h \rightarrow 0} \frac{\|f(a+h) - f(a) - Df_a h\|}{\|h\|} \quad (88)$$

and we are done.

Finally, to prove uniqueness of the derivative, it seems more convenient to use the fundamental increment lemma. Say that there are two derivatives, A_1 and A_2 , each with their own infinitesimal functions Φ_1, Φ_2 defined in the intersection of the open neighborhoods of a , which is also an open neighborhood. Then,

$$f(a+h) = f(a) + A_1 h + \Phi_1(h)\|h\| \quad (89)$$

$$f(a+h) = f(a) + A_2 h + \Phi_2(h)\|h\| \quad (90)$$

So, by subtracting them and rearranging, we have

$$0 = (A_1 - A_2)h + (\Phi_1 - \Phi_2)(h) \quad (91)$$

So for $h \neq 0$, we have

$$(A_2 - A_1) \frac{h}{\|h\|} = (\Phi_1 - \Phi_2)(h) \quad (92)$$

The left hand side is a linear map of h and is in fact a constant c since $\frac{h}{\|h\|}$ is a unit vector. So by taking the limits of both sides, we get

$$c = \lim_{h \rightarrow 0} (A_2 - A_1) \frac{h}{\|h\|} = \lim_{h \rightarrow 0} (\Phi_1 - \Phi_2)(h) = 0 \quad (93)$$

Therefore, $c = 0 \implies A_2 - A_1 = 0$. So $A_1 = A_2$.

The reason we want E to be open is that we require that $x+h$ to also be in D for sufficiently small h , and we can guarantee this since an ϵ -ball around x is guaranteed to be in E . As in univariate analysis, we first establish the relation to continuity.

Theorem 1.1 (Differentiability Implies Continuity)

If f is differentiable at x , then it is continuous at x .

Proof. By assumption, we have

$$\lim_{h \rightarrow 0} \frac{\|f(x+h) - f(x) - Df_x h\|}{\|h\|} = 0 \implies \lim_{h \rightarrow 0} \|f(x+h) - f(x) - Df_x h\| = 0 \quad (94)$$

since the norm is a continuous mapping. But $\lim_{h \rightarrow 0} Df_x h = 0$ since Df_x is linear, and so

$$\lim_{h \rightarrow 0} \|f(x+h) - f(x)\| = 0 \quad (95)$$

Therefore, we have 4 separate notions of derivatives. A natural question to ask is whether the existence of one derivative implies the existence of another.

Theorem 1.2 (Differentiability $\implies$ Existence of All Directionals)

If $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable at $a \in E$, then all of its directional derivatives exist. Furthermore, given direction vector $u \in \mathbb{R}^n$,

$$D_u f_a = Df_a(u) \quad (96)$$

That is, we can compute the directional derivative by plugging in the direction vector into the linear map Df_a .

Proof. Since f is differentiable at a , the limit

$$\lim_{h \rightarrow 0} \frac{\|f(a+h) - f(a) - Df_a(h)\|}{\|h\|} = 0 \quad (97)$$

for all paths through 0. Therefore, let us choose the path

$$\lim_{h \rightarrow 0} \frac{\|f(a+hu) - f(a) - Df_a(hu)\|}{\|hu\|} = \lim_{h \rightarrow 0} \left\| \frac{f(a+hu) - f(a) - Df_a(hu)}{h} \right\| = 0 \in \mathbb{R} \quad (98)$$

Since the norm is a continuous mapping, this means that

$$0 = \lim_{h \rightarrow 0} \frac{f(a+hu) - f(a) - Df_a(hu)}{h} \quad (99)$$

$$= \lim_{h \rightarrow 0} \frac{f(a+hu) - f(a) - hDf_a(u)}{h} \quad (100)$$

$$= \lim_{h \rightarrow 0} \frac{f(a+hu) - f(a)}{h} - Df_a(u) \quad (101)$$

Therefore,

$$Df_a(u) = \lim_{h \rightarrow 0} \frac{f(a+hu) - f(a)}{h} = D_u f_a \quad (102)$$

Therefore, we know that differentiability (i.e. existence of the total derivative) is the strongest. If this is the case, then the partial and directional derivatives exist, and we also know how the derivative acts on each basis vector. Therefore, we can immediately know that

Lemma 1.3 (Action of Derivative on Basis Vectors)

Let $(e_i)_i$ and (e'_j) be the standard basis vectors of $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. If $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable at a , then the partial derivatives $\frac{\partial f_j}{\partial x_i}$ exist, and

$$Df_a(e_i) = \sum_{j=1}^m \left(\frac{\partial f_j}{\partial x_i}(a) \right) e'_j \quad (103)$$

Proof. The partial derivatives existing is an immediate consequence of the previous theorem. As for the identity, we can see that

$$Df_a(e_i) = D_{e_i} f_a = D_{e_i} \left(\sum_{j=1}^m f_j e'_j \right)_a \quad (104)$$

$$= \sum_{j=1}^m D_{e_i} (f_j e'_j)_a \quad (105)$$

$$= \sum_{j=1}^m D_{e_i} (f_j)_a e'_j \quad (106)$$

$$= \sum_{j=1}^m \frac{\partial f_j}{\partial x_i}(a) e'_j \quad (107)$$

Since our vector spaces come with a basis, with this theorem, we can realize Df_x in a matrix form.

Definition 1.5 (Jacobian)

Given $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ differentiable at x , the **Jacobian** of f at x is the matrix realization of the total derivative, with entries being precisely the partial derivatives.

$$Jf_a = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{pmatrix} \quad (108)$$

Therefore, we can see that the derivative maps the i th basis vector $e_i \in \mathbb{R}^n$ to the i th column of our Jacobian matrix.

Analogous to the univariate case, a nice way to visualize the derivative is by looking at the tangent *plane*. In general, we have an *affine hyperplane*. Define this here. Unfortunately, the only types of functions that we can meaningfully visualize are those mapping $f : \mathbb{R}^2 \rightarrow \mathbb{R}$.

Definition 1.6 (Tangent Hyperplane)

If $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable at $a \in \mathbb{R}^n$, then the affine hyperplane tangent to the graph of f at $(a, f(a))$ is the set

$$P = \{(x, f(x)) : z = f(a) + Df_a(x - a)\}. \quad (109)$$

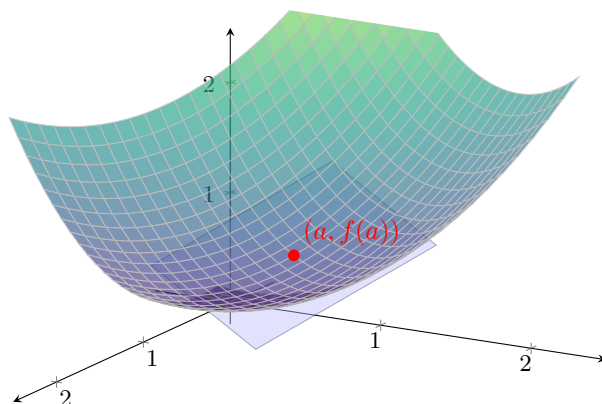


Figure 4: The tangent plane to the surface $z = f(x, y)$ at the point $(a, f(a))$. Here $f(x, y) = \frac{x^2 + y^2}{2}$ and $a = (1, 1)$, so the tangent plane is $z = x + y - 1$.

So, to prove that a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable at a , and if so, what its total derivative is, there are essentially two steps.

1. We find a candidate for M by evaluating the partials.

$$M = (\partial_{x_1} f(a) \quad \partial_{x_2} f(a) \quad \cdots \quad \partial_{x_n} f(a)) \quad (110)$$

2. We check to see if the limit is true.

$$\lim_{h \rightarrow 0} \frac{f(a + h) - f(a) - Mh}{\|h\|} = 0 \quad (111)$$

If it is, then $Df_a = M$, and the "tangent plane" of f at a is defined by the equation $y = f(a) + Mh$.

Example 1.9 (Computing Total Derivative)

The function $f(x_1, x_2) = x_1^2 + x_2^2$ is differentiable at $(1, 1)$. We let $M = (\partial_{x_1} f(1, 1), \partial_{x_2} f(1, 1)) = (2, 2)$ and see that

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a) - Mh}{\|h\|} = \lim_{h \rightarrow 0} \frac{f(1+h_1, 1+h_2) - f(1, 1) - 2h_1 - 2h_2}{\sqrt{h_1^2 + h_2^2}} \quad (112)$$

$$= \lim_{h \rightarrow 0} \frac{(1+h_1)^2 + (1+h_2)^2 - 2 - 2h_1 - 2h_2}{\sqrt{h_1^2 + h_2^2}} \quad (113)$$

$$= \lim_{h \rightarrow 0} \sqrt{h_1^2 + h_2^2} = 0 \quad (114)$$

So, $Df(1, 1) = (2, 2)$.

Example 1.10 (Computing Tangent Plane)

Let us find the equation of the tangent plane to $f(x, y) = \ln(2x + y)$ at $(-1, 3)$. Our total derivative, if it exists, is the covector of partials

$$Df = \left(\frac{\partial f}{\partial x} \quad \frac{\partial f}{\partial y} \right) = \left(\frac{2}{2x+y} \quad \frac{1}{2x+y} \right) \quad (115)$$

which is indeed continuous at a neighborhood of $(-1, 3)$ (in fact, in every neighborhood not containing 0). By continuity of partials, f is differentiable at $(-1, 3)$, with $Df_{(-1,3)} = (2, 1)$. The equation of the plane is then

$$z = f(-1, 3) + Df_{(-1,3)} \begin{pmatrix} x+1 \\ y-3 \end{pmatrix} = 2(x+1) + 1(y-3) \implies z = 2x + y - 1 \quad (116)$$

However, the converse is not true for either statements. You cannot just evaluate all the partial derivatives and assume that the total derivative exists!

Example 1.11 (Partial Derivatives Exist, but Not Differentiable)

Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by:

$$f(x, y) = \begin{cases} \frac{xy}{x^2+y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases} \quad (117)$$

At the origin $(0, 0)$, the partial derivatives are:

$$\frac{\partial f}{\partial x}(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0. \quad (118)$$

Similarly, $\frac{\partial f}{\partial y}(0, 0) = 0$. Both partial derivatives exist. However, the total derivative **cannot** exist because the function is not continuous at $(0, 0)$. As we saw in the limits section, approaching along $y = x$ gives a limit of $1/2$, while the function value is 0. A fundamental property of differentiability is that it implies continuity; therefore, f is not differentiable at the origin.

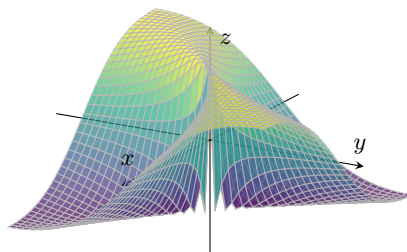


Figure 5: A function where partial derivatives exist at the origin, but the surface is “torn” there.

Example 1.12 (All Directional Derivatives Exist, but Not Differentiable)

Consider $f(x, y) = \begin{cases} \frac{x^2 y}{x^4 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$. For any direction $u = (u_1, u_2)$, the directional derivative

at the origin exists and is given by $D_u f(0, 0) = \frac{u_1^2}{u_2}$ (if $u_2 \neq 0$) or 0 (if $u_2 = 0$).

Despite having directional derivatives in every single direction, the total derivative does not exist. First, f is not continuous at the origin (approaching along $y = x^2$ gives $1/2 \neq 0$). Second, even if it were continuous, the total derivative requires that the linear map $Df(a)$ approximates the function well in *every* manner of approach, not just along straight lines. Here, the “peak” of the function follows the parabola $y = x^2$, which escapes the linear approximation provided by the directional derivatives.

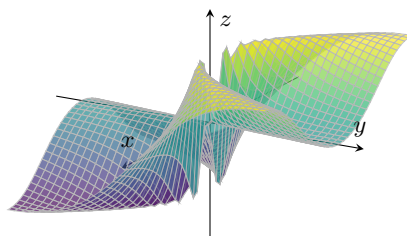


Figure 6: A function where all directional derivatives exist at the origin, yet the surface is non-differentiable.

1.3 Rules of Differentiation

Just like single variable calculus, the total derivative behaves in predictable ways: it is linear, product/quotient rules, and the chain rule.

Lemma 1.4 (Linearity of Total Derivatives)

Let $f, g : E \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ be differentiable at $a \in E$. Then, the total derivative at a is linear w.r.t. the input functions.

1. *Addition.* $D(f + g)_a = Df_a + Dg_a$
2. *Scalar Multiplication.* $D(cf)_a = cDf_a$

Proof. Let f, g be differentiable at a and $c \in \mathbb{R}$.

1. *Addition.* There exists linear functions Df_a, Dg_a and infinitesimal functions Φ_f, Φ_g such that

$$f(a + h) = f(a) + Df_a(h) + \Phi_f(h)\|h\| \quad (119)$$

$$g(a + h) = g(a) + Dg_a(h) + \Phi_g(h)\|h\| \quad (120)$$

By adding the two, we get

$$(f + g)(a + h) = (f + g)(a) + (Df_a + Dg_a)(h) + (\Phi_f + \Phi_g)(h)\|h\| \quad (121)$$

where $Df_a + Dg_a$ is also linear and $\Phi_f + \Phi_g$ is also infinitesimal. So $f + g$ is differentiable with derivative $D(f + g)_a = Df_a + Dg_a$.

2. *Scalar Multiplication.* There exists a linear function Df_a and infinitesimal Φ such that

$$f(a + h) = f(a) + Df_a(h) + \Phi(h)\|h\| \quad (122)$$

By multiplying both sides by c , we get

$$(cf)(a + h) = (cf)(a) + (cDf_a)(h) + (c\Phi)(h)\|h\| \quad (123)$$

where cDf_a is also linear and Φ also infinitesimal. So, cf is differentiable with derivative $D(cf)_a = cDf_a$.

Note that for the product and quotient rules, our scope is only for scalar valued functions since products and quotients aren't defined over codomains that are vector spaces.

Lemma 1.5 (Product Rule)

Let $f, g : E \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable at a . Then,

$$D(fg)_a = Df_a g(a) + f(a) Dg_a \quad (124)$$

Proof. By differentiability, we have

$$f(a + h) = f(a) + Df_a(h) + \Phi_f(h)\|h\| \quad (125)$$

$$g(a + h) = g(a) + Dg_a(h) + \Phi_g(h)\|h\| \quad (126)$$

By multiplying and expanding,^a we get

$$(fg)(a + h) = f(a)g(a) + f(a)Dg_a(h) + g(a)Df_a(h) \quad (127)$$

$$+ Df_a(h)Dg_a(h) + f(a)\Phi_g(h)\|h\| + g(a)\Phi_f(h)\|h\| + Df_a(h)\Phi_g(h)\|h\| \quad (128)$$

$$+ Dg_a(h)\Phi_f(h)\|h\| + \Phi_f(h)\|h\|\Phi_g(h)\|h\| \quad (129)$$

The map $fDg_a + Df_a f$ is linear, so it suffices to prove that the rest of the terms can be turned into a form of $\Phi(h)\|h\|$ where Φ is infinitesimal. We can see that by linearity,

$$Df_a(h)Dg_a(h) = \frac{Df_a(h)Dg_a(h)\|h\|}{\|h\|} = Df_a(h)Dg_a\left(\frac{h}{\|h\|}\right)\|h\| \quad (130)$$

where $\lim_{h \rightarrow 0} Df_a(h)Dg_a\left(\frac{h}{\|h\|}\right) = \left(\lim_{h \rightarrow 0} Df_a(h)\right)\left(\lim_{h \rightarrow 0} Dg_a\left(\frac{h}{\|h\|}\right)\right) = 0 \cdot \left(\lim_{h \rightarrow 0} Dg_a\left(\frac{h}{\|h\|}\right)\right) = 0$. By the limit rules, the rest of the terms are obvious.

^aI wish there was a cleaner solution such as the proof for the univariate case, but I wasn't able to derive one.

Lemma 1.6 (Quotient Rule)

Let $f, g : E \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable at a with $g(a) \neq 0$. Then,

$$D\left(\frac{f}{g}\right)_a = \frac{Df_a g(a) - f(a) Dg_a}{g(a)^2} \quad (131)$$

Proof. Just like the univariate case, we may make an educated guess that $D(1/g)_a = -\frac{Dg_a}{g^2}$. Therefore,

$$\frac{(1/g)(a+h) - (1/g)(a) - \left(-\frac{Dg_a}{g^2}\right)}{\|h\|} = \frac{\frac{g(a)^2 - g(a)g(a+h) + Dg_a(h)g(a+h)}{g(a+h)g(a)^2}}{\|h\|} \quad (132)$$

$$= \frac{1}{g(a)^2} \frac{g(a)^2 - g(a)g(a+h) + Dg_a(h)g(a+h)}{\|h\|g(a+h)} \quad (133)$$

We would like to turn this into a form that represents the limit definition of the derivative of g .

$$= \frac{1}{g(a)^2} \frac{g(a)^2 - g(a+h)^2 + g(a+h)^2 - g(a)g(a+h) + Dg_a(h)g(a+h)}{\|h\|g(a+h)} \quad (134)$$

$$= \frac{1}{g(a)^2} \left(\underbrace{\frac{g(a)^2 - g(a+h)^2}{\|h\|g(a+h)}}_{\rightarrow 0} + \frac{g(a+h)[g(a+h) - g(a) - Dg_a(h)]}{\|h\|g(a+h)} \right) \quad (135)$$

The left term goes to 0 since g is differentiable and hence continuous. The right term can be simplified by canceling out the $g(a+h)$, which can be done since g is not 0 in a neighborhood of a . So, adding the limit back in, the whole term simplifies to

$$\lim_{h \rightarrow 0} \frac{(1/g)(a+h) - (1/g)(a) - \left(-\frac{Dg_a}{g^2}\right)}{\|h\|} = \lim_{h \rightarrow 0} \frac{1}{g(a)^2} \frac{g(a+h) - g(a) - Dg_a(h)}{\|h\|} \quad (136)$$

$$= \frac{1}{g(a)^2} \lim_{h \rightarrow 0} \frac{\|g(a+h) - g(a) - Dg_a(h)\|}{\|h\|} = 0 \quad (137)$$

by the definition of g being differentiable at a .

Theorem 1.7 (Chain Rule)

Let $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ be differentiable at $a \in E$ and $g : E' \subset \mathbb{R}^m \rightarrow \mathbb{R}^p$ be differentiable at $g(a) \in E'$. Then, $g \circ f$ is differentiable at a , and

$$D(g \circ f)_a = Dg_{f(a)} \circ Df_a \quad (138)$$

Proof. Since f is differentiable at a and g at $f(a)$, we can write

$$f(a+h) = f(a) + Df_a(h) + \Phi_f(h)\|h\| \quad (139)$$

$$g(f(a)+t) = g(f(a)) + Dg_{f(a)}(t) + \Phi_g(t)\|t\| \quad (140)$$

Therefore, we wish to show that

$$\lim_{h \rightarrow 0} \frac{g(f(a+h)) - g(f(a)) - (Dg_{f(a)} \circ Df_a)(h)}{\|h\|} = 0 \quad (141)$$

Let's remove the limit and focus on the term. First substitute $f(a+h)$ and expand the term for

$$g(f(a) + t).$$

$$= \frac{g(f(a) + \overbrace{Df_a(h) + \Phi_f(h)\|h\|}^t) - g(f(a)) - (Dg_{f(a)} \circ Df_a)(h)}{\|h\|} \quad (142)$$

$$= \frac{g(f(a)) + Dg_{f(a)}(Df_a(h) + \Phi_f(h)\|h\|) + \Phi_g(Df_a(h) + \Phi_f(h)\|h\|)\|h\| - g(f(a)) - Dg_{f(a)}(Df_a(h))}{\|h\|} \quad (143)$$

$$= \frac{(Dg_{f(a)} \circ Df_a)(h) + Dg_{f(a)}(\Phi_f(h)\|h\|) + \Phi_g(Df_a(h) + \Phi_f(h)\|h\|)\|h\| - (Dg_{f(a)} \circ Df_a)(h)}{\|h\|} \quad (144)$$

$$= Dg_{f(a)}(\Phi_f(h)) + \Phi_g(Df_a(h) + \Phi_f(h)\|h\|) \rightarrow 0 \quad (145)$$

where the final step follows from linearity of $Dg_{f(a)}$. As $h \rightarrow 0$, the limit of the left term is 0 since linear maps are continuous, and the right term goes to 0 since Φ_g is infinitesimal. We are done.

Therefore, given the composition of function $f \circ g$, we have two methods of finding the derivative matrix of $f \circ g$ at point x_0 . First is to explicitly compute $f \circ g$ and find its $m \times p$ derivative matrix $D(f \circ g)$, and plug in a to get $D(f \circ g)_a$. The second way is to use the chain rule to find the individual total derivatives $Df_{g(a)}$ and Dg_a , and multiply them together.

1.4 Continuously Differentiable Functions

An even stronger condition beyond differentiability is continuous partials, and we often prove continuity of partials to prove differentiability.

Definition 1.7 (Continuously Differentiable)

A function $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **continuously differentiable** at $a \in E$ if it satisfies either of the equivalent definitions:

1. f is differentiable at a and the map $a \in \mathbb{R}^n \rightarrow Df_a \in \mathbb{R}^{m \times n}$ is continuous.
2. f is differentiable at a and all the partial derivatives $a \in \mathbb{R}^n \rightarrow \frac{\partial f_j}{\partial x_i}(a) \in \mathbb{R}$ is continuous.

Proof. We prove bidirectionally.

1. (1 $\rightarrow$ 2). Let $\phi_{ij} : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ be the projection of the matrix onto the ij th element. This is continuous and we can see that

$$\frac{\partial f_j}{\partial x_i}(a) = \phi_{ij}(Df_a) \quad (146)$$

so all partial derivatives are continuous.

2. (2 $\rightarrow$ 1). Let $\epsilon > 0$. If all partial derivatives are continuous, then there exists a $\delta > 0$ such that

$$\|x - a\| < \delta \implies \left\| \frac{\partial f_j}{\partial x_i}(x) - \frac{\partial f_j}{\partial x_i}(a) \right\| < \frac{\epsilon}{mn} \implies \|Df_a\|_1 = \sum_{i,j} (Df_a)_{ij} < \epsilon \quad (147)$$

So $a \rightarrow Df_a$ is continuous with respect to the L_1 norm on $\mathbb{R}^{m \times n}$. But since all L_p norms are equivalent in finite-dimensional Euclidean space, it is continuous with respect to all norms.

Theorem 1.8 (Continuous Partial $\implies$ Differentiability)

Given a function $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}$ and a point $a \in E$, if all the partials $\partial_{x_i} f$ exist and are continuous at a , then f is differentiable at a .

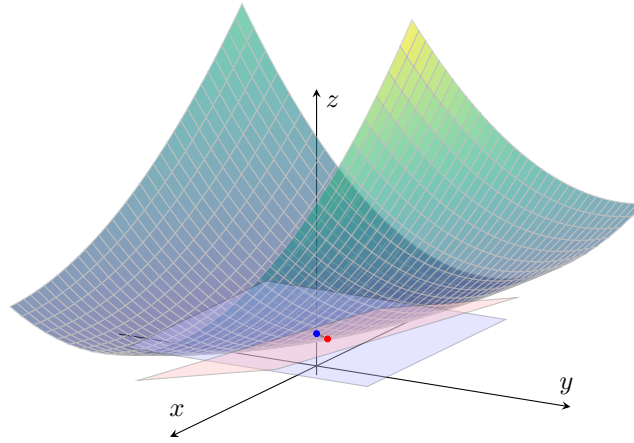


Figure 7: We can also visualize this theorem. Since the partials are continuous, then the tangent subspace, which is determined by the span of the tangent vectors determined by the partials, also changes continuously, and therefore the total derivative within a neighborhood of a exists.

Proof. The idea is to bound the perturbations in each component using the mean value theorem. Let us consider real valued $f : \mathbb{R}^n \rightarrow \mathbb{R}$ for now. Given $a, a + h \in \mathbb{R}^n$, let us construct a stepwise path from a to $a + h$. Let us denote $\phi_k(h) = (h_1, h_2, \dots, h_k, 0, \dots, 0)$. Then, we can construct the sequence

$$a = a + \phi_0(h), a + \phi_1(h), a + \phi_2(h), \dots, a + \phi_k(h) = a + h \quad (148)$$

We take the matrix of partials

$$A = \left(\frac{\partial f}{\partial x_1}(a) \quad \dots \quad \frac{\partial f}{\partial x_n}(a) \right) \quad (149)$$

and wish to show that this is indeed the total derivative. We see that

$$f(a + h) - f(a) - Ah = f(a + h) - f(a) - \sum_{i=1}^n \frac{\partial f}{\partial x_i}(a) h_i \quad (150)$$

$$= \sum_{i=1}^n \frac{\partial f}{\partial x_i}(a) h_i + (f(a + \phi_i(h)) - f(a + \phi_{i-1}(h))) \quad (151)$$

Since $h_i \mapsto f(a + \phi_i(h)) - f(a + \phi_{i-1}(h))$ is a univariate differentiable function, by MVT, there exists a $c_i \in (a_i, a_i + h_i)$ s.t. $\frac{\partial f}{\partial x_i}(c_i) = \frac{f(a + \phi_i(h)) - f(a + \phi_{i-1}(h))}{h_i}$. Therefore, the above equals

$$= \sum_{i=1}^n \left(\frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i) \right) h_i \quad (152)$$

So,

$$\|f(a + h) - f(a) - Ah\| = \left\| \sum_{i=1}^n \left(\frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i) \right) h_i \right\| \quad (153)$$

Since $|h_i| \leq \|h\|$,

$$\frac{\|f(a+h) - f(a) - Ah\|}{\|h\|} = \left\| \sum_{i=1}^n \left(\frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i) \right) \frac{h_i}{\|h\|} \right\| \quad (154)$$

$$= \left\| \sum_{i=1}^n \left(\frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i) \right) \right\| \quad (155)$$

$$\leq \sum_{i=1}^n \left\| \frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i) \right\| \quad (156)$$

But from continuity of partials, we know that for $\epsilon > 0$, there exists a $\delta > 0$ s.t.

$$\|h\| < \delta \implies \left| \frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(a + \phi_k(h)) \right| < \frac{\epsilon}{n} \quad (157)$$

$$\implies \left| \frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i) \right| < \frac{\epsilon}{n} \quad (158)$$

since $c_i \in (a, a + \phi_k(h))$. Therefore,

$$\frac{\|f(a+h) - f(a) - Ah\|}{\|h\|} = \sum_{i=1}^n \left| \frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i) \right| < \epsilon \quad (159)$$

While continuous partial derivatives guarantee differentiability, the converse is not true. A function can be differentiable even if its partial derivatives are not continuous. However, if they are not continuous, we cannot use the previous theorem to determine differentiability and must revert to the definition.

Example 1.13 (Non-Continuous Partials and Not Differentiable)

Recall $f(x, y) = \begin{cases} \frac{xy}{x^2+y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$. The partial derivative $\frac{\partial f}{\partial x}$ for $(x, y) \neq (0, 0)$ is:

$$\frac{\partial f}{\partial x} = \frac{y(x^2 + y^2) - xy(2x)}{(x^2 + y^2)^2} = \frac{y^3 - x^2y}{(x^2 + y^2)^2}.$$

At the origin, we found $\frac{\partial f}{\partial x}(0, 0) = 0$. However, $\lim_{(x,y) \rightarrow (0,0)} \frac{y^3 - x^2y}{(x^2 + y^2)^2}$ does not exist (it depends on the path). Thus, the partial derivatives are not continuous at the origin. As shown before, the function is not differentiable because it is not continuous.

Example 1.14 (Non-Continuous Partials and Differentiable)

Consider $f(x, y) = \begin{cases} (x^2 + y^2) \sin\left(\frac{1}{\sqrt{x^2 + y^2}}\right) & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$. At the origin, the partial derivatives exist and are 0:

$$\frac{\partial f}{\partial x}(0, 0) = \lim_{h \rightarrow 0} \frac{h^2 \sin(1/|h|) - 0}{h} = \lim_{h \rightarrow 0} h \sin(1/|h|) = 0.$$

Away from the origin, $\frac{\partial f}{\partial x}$ involves a $\cos(1/\sqrt{x^2 + y^2}) \cdot \frac{1}{x^2 + y^2}$ term which oscillates wildly as $(x, y) \rightarrow (0, 0)$. Thus, the partials are not continuous.

However, f is differentiable at the origin. Using the definition with $Df(0,0) = 0$:

$$\lim_{h \rightarrow 0} \frac{|(h_1^2 + h_2^2) \sin(1/\|h\|) - 0 - 0|}{\|h\|} = \lim_{h \rightarrow 0} \|h\| \sin(1/\|h\|) = 0.$$

This satisfies the definition of the total derivative.

Theorem 1.9 (C^1 Space)

The set of all continuously differentiable functions $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}$, denoted $C^1(E)$, is a vector space.

Proof. The proof is trivial since Df is continuous. Therefore, sums and scalar multiples of are continuous as well.

Note that from now, whenever we talk about differentiating a function f , we will assume that it is C^1 . This is overkill, since the set of all k -times differentiable functions is a subset of C^k , but it is conventional to work with C^k functions.

1.5 Mean Value Theorem

Lemma 1.10 (Mean Value Theorem for Paths)

If $f : E \subset \mathbb{R} \rightarrow \mathbb{R}^m$ is continuous on $[a, b]$ and differentiable on (a, b) , then there exists $x \in (a, b)$ s.t.

$$\|f(b) - f(a)\| \leq (b - a)\|f'(x)\| \quad (160)$$

Proof. The general idea is that we want to convert this into a scalar-valued function in a clever way.

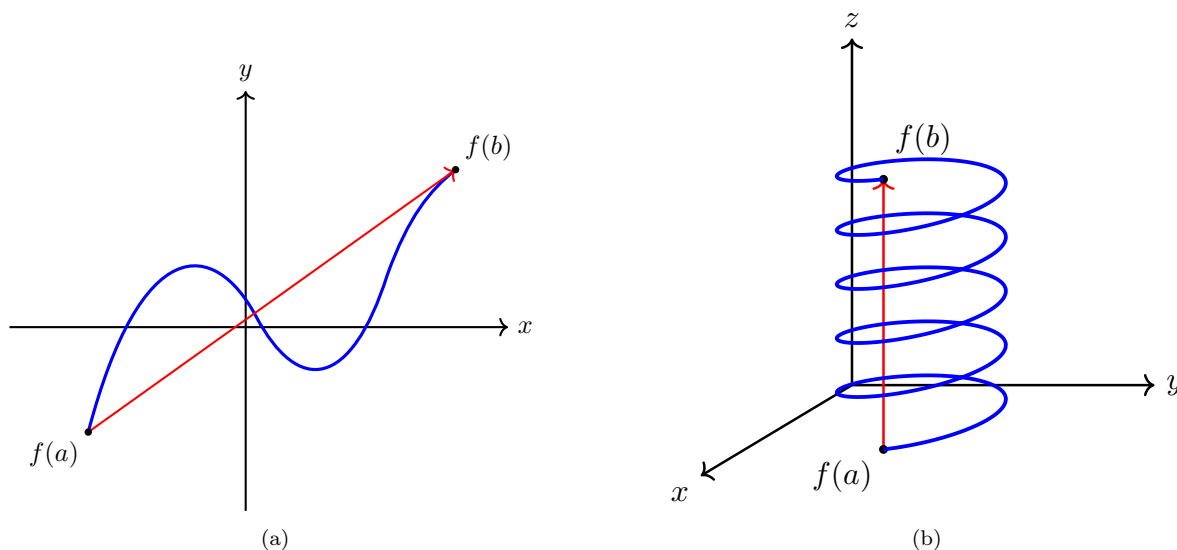


Figure 8: If we sketch out a path in $\mathbb{R}^2$, we might speculate that at some point, the tangent line of the curve must be parallel to the secant vector $f(b) - f(a)$. Perhaps defining $\phi(x) = \langle f(b) - f(a), \frac{Df_x}{\|Df_x\|} \rangle$ and trying to maximize it using EVT or something might help. However, there is no guarantee that Df_x is continuous, and unfortunately, this assumption turns out not to be true as seen in the solenoid.

After some thought (see figure caption), we might think of looking at the inner product of the actual

vectors on the curve with the secant vector, defining

$$\phi(x) = \langle f(b) - f(a), f(x) \rangle \quad (161)$$

ϕ is differentiable on (a, b) and continuous on $[a, b]$, so by MVT, there exists some $x \in (a, b)$ such that

$$\phi(b) - \phi(a) = D\phi_x(b - a) = \langle f(b) - f(a), Df_x \rangle (b - a) \quad (162)$$

But $\phi(b) - \phi(a) = \langle f(b) - f(a), f(b) - f(a) \rangle = \|f(b) - f(a)\|^2$, so by Cauchy Schwarz, we get

$$\|f(b) - f(a)\|^2 = \langle f(b) - f(a), Df_x \rangle (b - a) \leq \|f(b) - f(a)\| \|Df_x\| (b - a) \quad (163)$$

Therefore, $\|f(b) - f(a)\| \leq \|Df_x\| (b - a)$.

There is a nice reason why equality doesn't hold in higher dimensions. For my first attempt at this proof, I tried to replicate my work for proving that C^1 implies differentiable by dividing f into component-wise paths. I defined

$$\phi_k(f) = (f_1(a), \dots, f_k(a), f_{k+1}(b), \dots, f_m(b))^T \quad (164)$$

Therefore, I was hoping to bound $\|f(b) - f(a)\| \leq \sum_{k=1}^m \|\phi_k(f) - \phi_{k-1}(f)\| = \sum_{k=1}^m |f_k(b) - f_k(a)|$. But invoking the univariate MVT for the real-valued function f_k leads to finding multiple $x_k \in (a, b)$. This is a problem since we need to find one x .

Example 1.15 (MVT Inequality Does Not Hold on Circle)

Let

$$f : [0, 2\pi] \rightarrow \mathbb{R}^2, \quad f(x) = (\cos(x), \sin(x)) \quad (165)$$

Then, $f(2\pi) - f(0) = 0$, but

$$Df_x = \begin{pmatrix} -\sin(x) \\ \cos(x) \end{pmatrix} \quad (166)$$

is never 0 over $x \in [0, 2\pi]$.

Theorem 1.11 (Mean Value Theorem)

Let $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ be differentiable on convex open set E . If there exists a $M \in \mathbb{R}$ s.t. $\|Df_x\| \leq M$ for all $x \in E$, then

$$\|f(b) - f(a)\| \leq M \|b - a\| \quad (167)$$

for all $a, b \in E$.

Proof. Since E is convex, for any $a, b \in E$, we can consider

$$\gamma : [0, 1] \rightarrow E, \quad \gamma(t) := ta + (1 - t)b \quad (168)$$

Then, $f \circ \gamma$ is a path function that is continuous on $[0, 1]$ and differentiable on $(0, 1)$. Therefore, there exists a $t_0 \in (0, 1)$ (and hence, corresponding $x_0 = \gamma(t_0) \in E$) such that

$$\|(f \circ \gamma)(1) - (f \circ \gamma)(0)\| \leq \|D(f \circ \gamma)_{t_0}\| \|1 - 0\| \quad (169)$$

$$= \|Df_{x_0} D\gamma_{t_0}\| \quad (170)$$

$$\leq \|Df_{x_0}\| \|a - b\| \quad (171)$$

$$\leq \|Df_{x_0}\| \|a - b\| \quad (172)$$

$$\leq M \|b - a\| \quad (173)$$

where in the second step, we used the chain rule, in the third step, we computed the derivative $D\gamma = a - b$, and in the fourth step, we used the operator norm bound. The left hand side is really $\|f(b) - f(a)\|$, and we have our desired result.

The following corollary is pretty obvious.

Corollary 1.12 (Vanishing Derivative Implies Constant)

Let $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ be differentiable on convex open set E . If $\|Df_x\| = 0$ for all $x \in E$, then f is constant on E .

Proof. Set $M = 0$ in the proof above.

1.6 Inverse Function Theorem

Definition 1.8 (Contraction)

Given a metric space (X, d) and some $c < 1$, a **contraction** is a mapping $\phi : X \rightarrow X$ such that

$$d(\phi(x), \phi(y)) \leq cd(x, y) \quad (174)$$

Theorem 1.13 (Banach Fixed Point Theorem)

Given a complete metric space (X, d) and a contraction ϕ on X , there exists a unique point $x \in X$ such that $\phi(x) = x$.

Proof. We give a constructive proof on how to find such a point. Let us pick any $x_0 \in X$ and set $x_k = \phi^k(x_0)$ to be the output of the k -fold composition of ϕ . Then,

$$d(x_k, x_{k+1}) \leq c^k d(x_0, x_1) \quad (175)$$

So given any $m, n \in \mathbb{N}$, WLOG let $m < n$. Then,

$$d(x_m, x_n) \leq \sum_{k=m}^{n-1} d(x_k, x_{k+1}) \quad (176)$$

$$\leq \sum_{k=m}^{n-1} c^k d(x_0, x_1) \quad (177)$$

$$= d(x_0, x_1) \sum_{k=m}^{n-1} c^k \quad (178)$$

$$= d(x_0, x_1) \sum_{k=m}^{\infty} c^k \quad (179)$$

Since the series converges from the geometric test, the tail sum $\sum_{k=m}^{\infty} c^k$ converges to 0 as $m \rightarrow +\infty$. So, $(x_n)_n$ is a Cauchy sequence and therefore converges to some $x \in X$.

Since ϕ is a contraction, ϕ is continuous. In fact it is uniformly continuous, since given $\epsilon > 0$, we can set $\delta = \frac{\epsilon}{c}$ and see that for all $x, y \in X$,

$$d(x, y) < \delta = \frac{\epsilon}{c} \implies d(\phi(x), \phi(y)) < c \frac{\epsilon}{c} = \epsilon \quad (180)$$

Therefore,

$$\phi(x) = \lim_{n \rightarrow +\infty} \phi(x_n) = \lim_{n \rightarrow +\infty} x_{n+1} = x \quad (181)$$

To prove uniqueness, assume that $x, y \in X$ are both fixed points. Then,

$$d(x, y) = d(\phi(x), \phi(y)) \leq cd(x, y) \implies d(x, y) = 0 \implies x = y \quad (182)$$

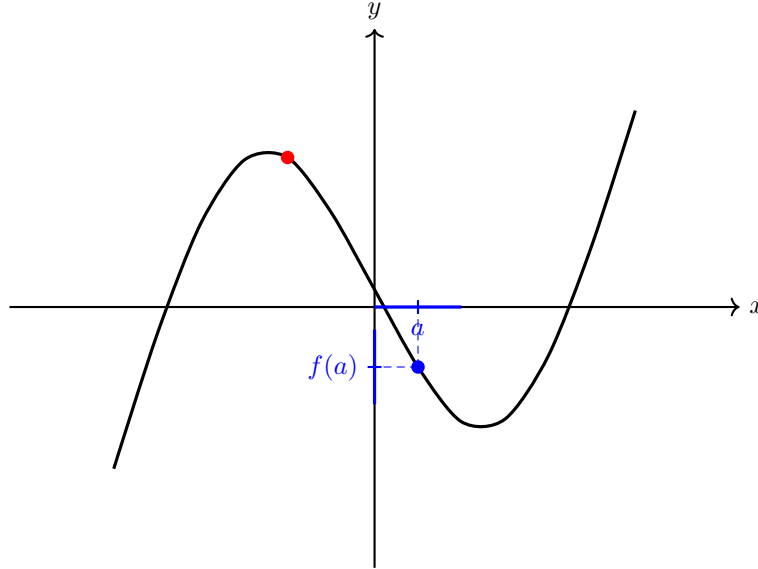


Figure 9: A cubic can fail to be locally invertible at a turning point, but behave regularly at nearby non-turning points.

Theorem 1.14 (Inverse Function Theorem)

Let $U \subset \mathbb{R}^n$ be an open neighborhood of a , and let $f : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a C^1 function. If Df_a is invertible, then there exist open neighborhoods V of a and W of $f(a)$, with $V \subset U$, such that

$$f|_V : V \rightarrow W \quad (183)$$

is a bijection. Moreover, the inverse map $(f|_V)^{-1} : W \rightarrow V$ is C^1 , and for every $y \in W$,

$$D(f^{-1})_y = (Df_{f^{-1}(y)})^{-1}. \quad (184)$$

Proof. The idea is to linearize $f(x)$. By definition of differentiability at $x \in E$, there exists a linear map $Df_a : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and infinitesimal function $\Phi : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that $f(x) = f(a) + Df_a(x - a) + \Phi(x - a)\|x - a\|$ within some neighborhood U of x . Therefore, we can approximate

$$f(x) \approx f(a) + Df_a(x - a) \quad (185)$$

Given our center $(a, f(a))$, this is a formula to approximate $f(x)$ from x . We must go backwards and approximate x from $y = f(x)$. We can write the inverse of this affine function as

$$x = a + Df_a^{-1}(f(x) - f(a)) \quad (186)$$

So for all $y_0 \in U$, we define the function $\phi_y : U \rightarrow U$.

$$\phi_{y_0}(x) = x + Df_a^{-1}(y_0 - f(x)) \quad (187)$$

which attempts to estimate the inverse of y_0 with respect to center $(x, f(x))$. The idea is that if we input the preimage of y_0 , then the point we are estimating is the preimage of y_0 , so

$$\phi_{y_0}(f^{-1}(y_0)) = f^{-1}(y_0) + Df_a^{-1}(y_0 - f(f^{-1}(y_0))) = f^{-1}(y_0) \quad (188)$$

The idea is to prove that ϕ_{y_0} is a contraction of some subset that is a complete metric space, and so (by the Banach fixed point theorem) the unique fixed point—call it $x_0 \in U$ —is the inverse. Let's do this step by step.

1. ϕ_0 is a contraction. $\phi_{y_0}(x) = x + Df_a^{-1}(y_0 - f(x))$, which implies that

$$D(\phi_{y_0})_x = I - Df_a^{-1}Df_x = Df_a^{-1}(Df_a - Df_x) \quad (189)$$

But since $x \rightarrow Df_x$ is continuous at a , there exists an open neighborhood U containing a s.t.

$$x \in U \implies \|Df_x - Df_a\| < \frac{1}{2\|Df_a^{-1}\|} \quad (190)$$

Therefore, when $x \in U$,

$$\|D(\phi_{y_0})_x\| = \|Df_a^{-1}\|\|Df_a - Df_x\| \quad (191)$$

$$\leq \|Df_a^{-1}\|\frac{1}{2\|Df_a^{-1}\|} = \frac{1}{2} \quad (192)$$

Therefore, by the MVT, for all $x_1, x_2 \in U$,

$$\|\phi_{y_0}(x_1) - \phi_{y_0}(x_2)\| \leq \frac{1}{2}\|x_1 - x_2\| \quad (193)$$

and so, ϕ_{y_0} is a contraction on U .

2. f is injective over U . Say that x_1, x_2 are both fixed points of U . Then,

$$\|x_1 - x_2\| = \|\phi_{y_0}(x_1) - \phi_{y_0}(x_2)\| \leq \frac{1}{2}\|x_1 - x_2\| \implies x_1 = x_2 \quad (194)$$

But note that

$$f(x_0) = y_0 \iff \phi_{y_0}(x_0) = x_0 + Df_a^{-1}(y_0 - f(x_0)) = x_0 \quad (195)$$

$$\iff x_0 \text{ is a fixed point of } \phi_{y_0} \quad (196)$$

So, since there is at most 1 fixed point, there can be at most one $x_0 \in U$ s.t. $f(x_0) = y_0$. Thus, f is injective. Note that we did *not* use Banach fixed point theorem yet.

3. $V = f(U)$ is open. This is much trickier than it looks. First, pick $y_1 \in V$ and so $y_1 = f(x_1)$ for some $x_1 \in U$. We wish to show that there exists a δ -neighborhood around y_1 , which establishes that V is open. The choose δ , let us choose $\epsilon > 0$ so small that $\overline{B_\epsilon(x_1)} \subset U$. Then, if $x \in B_\epsilon(x_1)$,

$$\|\phi_{y_0}(x) - x_1\| \leq \|\phi_{y_0}(x) - \phi_{y_0}(x_1)\| + \|\phi_{y_0}(x_1) - x_1\| \quad (197)$$

$$\leq \frac{1}{2}\|x - x_0\| + \|Df_a^{-1}(y_0 - f(x_1))\| \quad (198)$$

$$\leq \frac{\epsilon}{2} + \|Df_a^{-1}\|\|y_0 - f(x_1)\| \quad (199)$$

$$= \frac{\epsilon}{2} + \|Df_a^{-1}\|\|y_0 - y_1\| \quad (200)$$

Therefore, this hints that we need to bound $\|y_0 - y_1\|$, so we can choose y_0 (and hence ϕ_{y_0}) s.t.

$$\|y_0 - y_1\| < \frac{\epsilon}{2\|Df_a^{-1}\|} \quad (201)$$

This implies that $\|\phi_{y_0}(x) - x_1\| < \epsilon$. That is, as long as the center x is close enough to x_1 and y_0 is close enough to y_1 , the approximate inverse of y_0 is close to x_1 . Therefore, $\phi_{y_0}(x) \in B_\epsilon(x_1)$. So, if $x \in B_\epsilon(x_1)$ and $\|y_0 - y_1\| < \frac{\epsilon}{2\|Df_a^{-1}\|}$, then $\phi_{y_0}(x) \in B_\epsilon(x_1)$. Since ϕ_{y_0} is a contraction over U , it is also a contraction over $\overline{B_\epsilon(x_1)}$.

Now since $\overline{B_\epsilon(x_1)}$ is closed in $\mathbb{R}^n$, it is complete, so by Banach fixed point theorem, ϕ_{y_0} has a unique fixed point, which turns out to be $x_0 \in \overline{B_\epsilon(x_1)}$. So, $y_0 = f(x_0)$, and therefore

$$x_0 \in \overline{B_\epsilon(x_1)} \implies y_0 = f(x_0) \in f(B_\epsilon(x_1)) \subset f(U) = V \quad (202)$$

And so, by choosing $y_0 \in B_{\epsilon/(2\|Df_a^{-1}\|)}(y_1)$, we have $y_0 \in V \implies B_{\epsilon/(2\|Df_a^{-1}\|)}(y_1) \subset V$.

Great, now we wish to show that for $f : U \rightarrow V$, $D(f^{-1})_b = (Df_a)^{-1}$.

Example 1.16 (A Univariate Example)

Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = x^2. \quad (203)$$

Its derivative is

$$Df_x = f'(x) = 2x. \quad (204)$$

Therefore, at any point $x_0 \neq 0$, we have

$$Df_{x_0} = 2x_0 \neq 0. \quad (205)$$

Hence Df_{x_0} is invertible as a linear map from $\mathbb{R}$ to $\mathbb{R}$.

By the inverse function theorem, for every $x_0 \neq 0$, there exists an open neighborhood V of x_0 such that

$$f|_V : V \rightarrow f(V) \quad (206)$$

is invertible.

For example, at $x_0 = 1$, we can take a small interval around 1, such as

$$V = \left(\frac{1}{2}, \frac{3}{2}\right). \quad (207)$$

On this interval, $f(x) = x^2$ is injective, so it has a well-defined inverse

$$f^{-1}(y) = \sqrt{y}. \quad (208)$$

However, f is not invertible on all of $\mathbb{R}$, since

$$f(1) = 1 = f(-1). \quad (209)$$

Thus the inverse function theorem guarantees local invertibility near points where $f'(x_0) \neq 0$, but it does not guarantee global invertibility.

Example 1.17

Consider the vector-valued function $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$f(x_1, x_2) = \begin{pmatrix} e^{x_1} \cos(x_2) \\ e^{x_1} \sin(x_2) \end{pmatrix}$$

The total derivative/Jacobian matrix is

$$Jf(x_1, x_2) = \begin{pmatrix} e^{x_1} \cos(x_2) & -e^{x_1} \sin(x_2) \\ e^{x_1} \sin(x_2) & e^{x_1} \cos(x_2) \end{pmatrix} \implies \det Jf(x_1, x_2) = e^{2x_1} \cos^2(x_2) + e^{2x_1} \sin^2(x_2) = e^{2x_1}$$

Since the determinant e^{2x_1} is nonzero everywhere, Df_x is nonsingular. Thus, the theorem guarantees that for every point $x_0 \in \mathbb{R}^2$, there exists a neighborhood about x_0 over which f is invertible. However, this does not mean f is invertible over its entire domain: in this case f isn't even injective since it is periodic: e.g. the preimage of $(e, 0)$ contains $(1, 0)$ and $(1, 2\pi)$.

Example 1.18 (A Function from $\mathbb{R}^3$ to $\mathbb{R}^3$)

Consider the function $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by

$$f(x_1, x_2, x_3) = \begin{pmatrix} e^{x_1} \cos(x_2) \\ e^{x_1} \sin(x_2) \\ x_3 \end{pmatrix}. \quad (210)$$

The total derivative/Jacobian matrix is

$$Jf(x_1, x_2, x_3) = \begin{pmatrix} e^{x_1} \cos(x_2) & -e^{x_1} \sin(x_2) & 0 \\ e^{x_1} \sin(x_2) & e^{x_1} \cos(x_2) & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (211)$$

Therefore,

$$\det Jf(x_1, x_2, x_3) = e^{2x_1} \cos^2(x_2) + e^{2x_1} \sin^2(x_2). \quad (212)$$

Hence

$$\det Jf(x_1, x_2, x_3) = e^{2x_1}. \quad (213)$$

Since $e^{2x_1} \neq 0$ for all $x_1 \in \mathbb{R}$, the derivative Df_x is invertible at every point $x \in \mathbb{R}^3$.

Thus, by the inverse function theorem, for every point $x_0 \in \mathbb{R}^3$, there exists an open neighborhood V of x_0 such that

$$f|_V : V \rightarrow f(V) \quad (214)$$

is invertible.

However, f is not invertible on all of $\mathbb{R}^3$, since it is periodic in the x_2 -variable. For example,

$$f(1, 0, 5) = \begin{pmatrix} e \\ 0 \\ 5 \end{pmatrix} = f(1, 2\pi, 5). \quad (215)$$

Therefore, f is locally invertible everywhere, but not globally invertible.

1.7 Implicit Function Theorem

Remember that given an explicit representation of a set $y = f(x)$, we can easily find the implicit form as $F(x, y) = y - f(x) = 0$. What about the other way around? That is, given an implicit representation of some surface, what conditions must be met so that it can be represented as the graph of a function? The implicit function theorem is a tool that allows relations between points in $\mathbb{R}^n$ to be converted to functions of several real variables. That is, it states that for sufficiently "nice" points on a n -dimensional surface defined as $F(x, y) = 0$ (where $F : \mathbb{R}^{n+m} \rightarrow \mathbb{R}^m$), we can locally pretend that this surface is a graph of a function $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ whose graph $(x, g(x))$ is precisely the set of all (x, y) s.t. $f(x, y) = 0$. When $m = 1$, it basically states that if an implicit surface suffices the vertical line test in a neighborhood, then it can be written as a function.

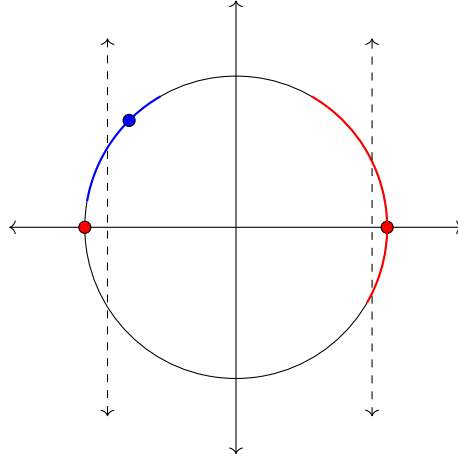


Figure 10: Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by $f(x, y) = x^2 + y^2 - 1$. The level set at $z = 0$ would be the set of points satisfying $x^2 + y^2 - 1 = 0$, i.e. the unit circle. The derivative of f with respect to y is 0 at the points $(-1, 0)$ and $(1, 0)$, meaning that in any neighborhood of these points, we cannot define a function of y with respect to x . This is true, indeed, since any such function would fail the vertical line test, which can be seen in the red neighborhood around $(1, 0)$. However, the blue neighborhood of the point $(-\sqrt{2}/2, \sqrt{2}/2)$ does indeed define a function of y with respect to x satisfying the vertical line test.

Theorem 1.15 (Implicit Function Theorem)

Let $F : \mathbb{R}^n \times \mathbb{R}^c \rightarrow \mathbb{R}^c$ be a C^1 function. Write points in $\mathbb{R}^n \times \mathbb{R}^c$ as (x, y) , where $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^c$. If

1. $F(a, b) = 0$, and
2. $D_y F_{(a, b)}$ is invertible,

then there exist open neighborhoods U of a and V of b , and a unique C^1 function

$$h : U \rightarrow V \quad (216)$$

such that $h(a) = b$ and, for every $(x, y) \in U \times V$,

$$F(x, y) = 0 \iff y = h(x). \quad (217)$$

Moreover, for every $x \in U$,

$$Dh_x = -(D_y F_{(x, h(x))})^{-1} D_x F_{(x, h(x))}. \quad (218)$$

Proof. Define a function

$$G : \mathbb{R}^n \times \mathbb{R}^c \rightarrow \mathbb{R}^n \times \mathbb{R}^c \quad (219)$$

by

$$G(x, y) = (x, F(x, y)). \quad (220)$$

Since $F(a, b) = 0$, we have

$$G(a, b) = (a, 0). \quad (221)$$

We compute the derivative of G . Since the first component of G is just x , and the second component is $F(x, y)$, we have

$$DG_{(x, y)} = \begin{pmatrix} I_n & 0 \\ D_x F_{(x, y)} & D_y F_{(x, y)} \end{pmatrix}. \quad (222)$$

In particular,

$$DG_{(a, b)} = \begin{pmatrix} I_n & 0 \\ D_x F_{(a, b)} & D_y F_{(a, b)} \end{pmatrix}. \quad (223)$$

This is a block lower triangular matrix. Therefore,

$$\det DG_{(a,b)} = \det(I_n) \det(D_y F_{(a,b)}) = \det(D_y F_{(a,b)}). \quad (224)$$

Since $D_y F_{(a,b)}$ is invertible, we have

$$\det DG_{(a,b)} \neq 0. \quad (225)$$

Hence $DG_{(a,b)}$ is invertible.

By the inverse function theorem, there exist open neighborhoods N of (a, b) and P of $(a, 0)$ such that

$$G|_N : N \longrightarrow P \quad (226)$$

is a bijection with a C^1 inverse

$$(G|_N)^{-1} : P \longrightarrow N. \quad (227)$$

Since N is an open neighborhood of (a, b) , we may choose open neighborhoods U_0 of a and V of b such that

$$U_0 \times V \subset N. \quad (228)$$

Since P is an open neighborhood of $(a, 0)$, we may choose an open neighborhood U_1 of a such that

$$U_1 \times \{0\} \subset P. \quad (229)$$

For $x \in U_1$, define

$$h(x) = \pi_y((G|_N)^{-1}(x, 0)), \quad (230)$$

where π_y denotes projection onto the $\mathbb{R}^c$ component. In other words, $h(x)$ is the unique vector such that

$$(G|_N)^{-1}(x, 0) = (x, h(x)). \quad (231)$$

Since $(G|_N)^{-1}$ is C^1 , the function h is also C^1 .

Also,

$$h(a) = b, \quad (232)$$

because

$$G(a, b) = (a, 0). \quad (233)$$

Shrinking U_1 if necessary, we may assume that

$$U_1 \subset U_0 \quad (234)$$

and

$$h(U_1) \subset V. \quad (235)$$

Set

$$U = U_1. \quad (236)$$

We now show that $F(x, y) = 0$ is equivalent to $y = h(x)$ on $U \times V$. First, if $x \in U$, then

$$G(x, h(x)) = (x, 0). \quad (237)$$

Since

$$G(x, h(x)) = (x, F(x, h(x))), \quad (238)$$

it follows that

$$F(x, h(x)) = 0. \quad (239)$$

Conversely, suppose that $(x, y) \in U \times V$ and

$$F(x, y) = 0. \quad (240)$$

Then

$$G(x, y) = (x, 0). \quad (241)$$

But also

$$G(x, h(x)) = (x, 0). \quad (242)$$

Since $(x, y) \in U \times V \subset N$ and $(x, h(x)) \in U \times V \subset N$, and since $G|_N$ is injective, we conclude that

$$(x, y) = (x, h(x)). \quad (243)$$

Therefore,

$$y = h(x). \quad (244)$$

Hence, for every $(x, y) \in U \times V$,

$$F(x, y) = 0 \iff y = h(x). \quad (245)$$

This also proves uniqueness. Indeed, if another function $\tilde{h} : U \rightarrow V$ satisfies

$$F(x, \tilde{h}(x)) = 0 \quad (246)$$

for all $x \in U$, then by the equivalence above,

$$\tilde{h}(x) = h(x) \quad (247)$$

for all $x \in U$.

It remains to compute the derivative of h . Since

$$F(x, h(x)) = 0 \quad (248)$$

for every $x \in U$, we differentiate both sides with respect to x . By the chain rule,

$$D_x F_{(x, h(x))} + D_y F_{(x, h(x))} Dh_x = 0. \quad (249)$$

Therefore,

$$D_y F_{(x, h(x))} Dh_x = -D_x F_{(x, h(x))}. \quad (250)$$

Since $D_y F_{(a, b)}$ is invertible, and invertibility is an open condition, we may shrink U and V if necessary so that $D_y F_{(x, h(x))}$ is invertible for every $x \in U$. Thus,

$$Dh_x = - (D_y F_{(x, h(x))})^{-1} D_x F_{(x, h(x))}. \quad (251)$$

This proves the theorem.

Example 1.19 (Circle Example)

Let $n = c = 1$, and consider the function $F : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$F(x_1, x_2) = x_1^2 + x_2^2 - 1. \quad (252)$$

The level set

$$F(x_1, x_2) = 0 \quad (253)$$

is the unit circle

$$x_1^2 + x_2^2 = 1. \quad (254)$$

We would like to determine at which points of the circle we can locally write x_2 as a function of x_1 .

The total derivative of F is

$$DF_{(x_1, x_2)} = (\partial_{x_1} F \quad \partial_{x_2} F) = (2x_1 \quad 2x_2). \quad (255)$$

In this case, if we want to solve for x_2 as a function of x_1 , then we look at the partial derivative with respect to x_2 :

$$D_{x_2} F_{(x_1, x_2)} = 2x_2. \quad (256)$$

This is invertible as a 1×1 matrix precisely when

$$x_2 \neq 0. \quad (257)$$

Therefore, by the implicit function theorem, at every point of the circle with $x_2 \neq 0$, we can locally write the circle in the form

$$x_2 = g(x_1). \quad (258)$$

For example, consider the point

$$\left(\frac{4}{5}, \frac{3}{5}\right) \quad (259)$$

on the unit circle. Since $\frac{3}{5} \neq 0$, we can locally solve for x_2 as a function of x_1 . Near this point, the appropriate branch is

$$x_2 = g(x_1) = \sqrt{1 - x_1^2}. \quad (260)$$

On the other hand, at the points $(1, 0)$ and $(-1, 0)$, we have

$$D_{x_2} F_{(x_1, x_2)} = 2x_2 = 0. \quad (261)$$

So the implicit function theorem does not allow us to solve for x_2 as a function of x_1 near these points. Indeed, near $x_1 = 1$ or $x_1 = -1$, the circle fails to be the graph of a function of x_1 , since nearby x_1 -values correspond to two possible x_2 -values.

The derivative of the implicit function g is given by the theorem:

$$Dg = -(D_{x_2} F)^{-1} D_{x_1} F. \quad (262)$$

Therefore,

$$Dg = -(2x_2)^{-1}(2x_1) = -\frac{x_1}{x_2}. \quad (263)$$

Since $x_2 = g(x_1)$, this becomes

$$g'(x_1) = -\frac{x_1}{g(x_1)}. \quad (264)$$

Similarly, if we want to solve for x_1 as a function of x_2 , then we look at the partial derivative with respect to x_1 :

$$D_{x_1} F_{(x_1, x_2)} = 2x_1. \quad (265)$$

This is invertible precisely when

$$x_1 \neq 0. \quad (266)$$

Hence, at every point of the circle with $x_1 \neq 0$, we can locally write the circle in the form

$$x_1 = h(x_2). \quad (267)$$

The derivative of h is

$$Dh = -(D_{x_1} F)^{-1} D_{x_2} F. \quad (268)$$

Therefore,

$$Dh = -(2x_1)^{-1}(2x_2) = -\frac{x_2}{x_1}. \quad (269)$$

Since $x_1 = h(x_2)$, this becomes

$$h'(x_2) = -\frac{x_2}{h(x_2)}. \quad (270)$$

Example 1.20 (Sphere Example)

Let $n = 2$ and $c = 1$, and consider the function $F : \mathbb{R}^2 \times \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$F(x_1, x_2, y) = x_1^2 + x_2^2 + y^2 - 1. \quad (271)$$

The level set

$$F(x_1, x_2, y) = 0 \quad (272)$$

is the unit sphere

$$x_1^2 + x_2^2 + y^2 = 1. \quad (273)$$

We would like to determine at which points of the sphere we can locally write y as a function of (x_1, x_2) . The total derivative of F is

$$DF_{(x_1, x_2, y)} = (\partial_{x_1} F \quad \partial_{x_2} F \quad \partial_y F) = (2x_1 \quad 2x_2 \quad 2y). \quad (274)$$

If we want to solve for y as a function of (x_1, x_2) , then we look at the partial derivative with respect to y :

$$D_y F_{(x_1, x_2, y)} = 2y. \quad (275)$$

This is invertible as a 1×1 matrix precisely when

$$y \neq 0. \quad (276)$$

Therefore, by the implicit function theorem, at every point of the sphere with $y \neq 0$, we can locally write the sphere in the form

$$y = g(x_1, x_2). \quad (277)$$

For example, consider the point

$$\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}} \right) \quad (278)$$

on the unit sphere. Since $\frac{1}{\sqrt{2}} \neq 0$, we can locally solve for y as a function of (x_1, x_2) . Near this point, the appropriate branch is

$$y = g(x_1, x_2) = \sqrt{1 - x_1^2 - x_2^2}. \quad (279)$$

On the other hand, at points on the equator where $y = 0$, the implicit function theorem does not allow us to solve for y as a function of (x_1, x_2) . Geometrically, near such points, the sphere has both an upper and a lower branch over the same nearby (x_1, x_2) -values.

The derivative of the implicit function g is given by

$$Dg = -(D_y F)^{-1} D_x F. \quad (280)$$

In this case,

$$D_x F = (2x_1 \quad 2x_2) \quad (281)$$

and

$$D_y F = 2y. \quad (282)$$

Therefore,

$$Dg = -(2y)^{-1} (2x_1 \quad 2x_2) = \left(-\frac{x_1}{y} \quad -\frac{x_2}{y} \right). \quad (283)$$

Since $y = g(x_1, x_2)$, this becomes

$$Dg_{(x_1, x_2)} = \left(-\frac{x_1}{g(x_1, x_2)} \quad -\frac{x_2}{g(x_1, x_2)} \right). \quad (284)$$

Example 1.21 (A System of Two Equations)

Let $n = 2$ and $c = 2$, and consider the function $F : \mathbb{R}^2 \times \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ defined by

$$F(x_1, x_2, y_1, y_2) = \begin{pmatrix} y_1^2 + y_2 - x_1 \\ y_1 + e^{y_2} - x_2 \end{pmatrix}. \quad (285)$$

We would like to locally solve the system

$$F(x_1, x_2, y_1, y_2) = 0 \quad (286)$$

for (y_1, y_2) as a function of (x_1, x_2) .

First observe that

$$F(1, 2, 1, 0) = \begin{pmatrix} 1^2 + 0 - 1 \\ 1 + e^0 - 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (287)$$

Thus the point

$$(x_1, x_2, y_1, y_2) = (1, 2, 1, 0) \quad (288)$$

lies on the zero set of F .

The derivative of F with respect to the y -variables is

$$D_y F_{(x_1, x_2, y_1, y_2)} = \begin{pmatrix} \partial_{y_1} F_1 & \partial_{y_2} F_1 \\ \partial_{y_1} F_2 & \partial_{y_2} F_2 \end{pmatrix} = \begin{pmatrix} 2y_1 & 1 \\ 1 & e^{y_2} \end{pmatrix}. \quad (289)$$

At the point $(1, 2, 1, 0)$, this becomes

$$D_y F_{(1, 2, 1, 0)} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}. \quad (290)$$

Its determinant is

$$\det D_y F_{(1, 2, 1, 0)} = 2 \cdot 1 - 1 \cdot 1 = 1. \quad (291)$$

Therefore $D_y F_{(1, 2, 1, 0)}$ is invertible.

By the implicit function theorem, there exist open neighborhoods U of $(1, 2)$ and V of $(1, 0)$, and a unique C^1 function

$$h : U \longrightarrow V \quad (292)$$

such that

$$h(x_1, x_2) = \begin{pmatrix} h_1(x_1, x_2) \\ h_2(x_1, x_2) \end{pmatrix} \quad (293)$$

and

$$F(x_1, x_2, h_1(x_1, x_2), h_2(x_1, x_2)) = 0 \quad (294)$$

for all $(x_1, x_2) \in U$.

In other words, near $(x_1, x_2) = (1, 2)$, the system

$$\begin{cases} y_1^2 + y_2 = x_1, \\ y_1 + e^{y_2} = x_2 \end{cases} \quad (295)$$

implicitly defines (y_1, y_2) as a function of (x_1, x_2) .

To compute the derivative of h , we use

$$Dh_x = - (D_y F_{(x, h(x))})^{-1} D_x F_{(x, h(x))}. \quad (296)$$

In this example,

$$D_x F_{(x_1, x_2, y_1, y_2)} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (297)$$

Therefore,

$$Dh_x = (D_y F_{(x, h(x))})^{-1}. \quad (298)$$

At the point $(x_1, x_2) = (1, 2)$, we have $h(1, 2) = (1, 0)$, so

$$Dh_{(1,2)} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}^{-1}. \quad (299)$$

Hence

$$Dh_{(1,2)} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}. \quad (300)$$

1.8 Extrema

Definition 1.9 (Local Extrema)

Given a function $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}$,

1. $a \in D$ is a **local minimum** if there exists a neighborhood U of a such that

$$f(x) \geq f(a) \text{ for every } x \in U \quad (301)$$

2. $a \in D$ is a *local maximum* if there exists a neighborhood U of x such that

$$f(x) \leq f(a) \text{ for every } x \in U \quad (302)$$

Theorem 1.16 (1st Derivative Test)

If a is a local extremum of a differentiable function f , then $Df_a = 0$.

Note that even though the converse of this theorem is not true, we can use the contrapositive to determine that every point that has a nonzero derivative cannot be a extremum.

Example 1.22 (Function with an Infinite Number of Critical Points)

A function may also have an infinite amount of critical points (e.g. if they lie in a circle).

Unfortunately, determining whether a point a is a local extremum requires us to define an open neighborhood around a . This means that we can only determine local extrema within open sets in $\mathbb{R}^n$. Therefore, we must modify our procedure when looking for extrema on functions defined over closed bounded sets. We now describe a method of computing the global extrema. Let $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be a multivariable function defined on a closed and bounded set $E = E^\circ \cup \partial E$, where the E° is the interior and ∂E is the boundary of E . To find the global extrema on E , we find all local extrema of

1. f defined over the interior of E° , which is open, by locating all points where $Df = 0$, and
2. f defined over ∂E .

In the second point, we could reparamaterize ∂E cleverly by another function $\phi : \mathbb{R}^k \rightarrow \mathbb{R}^n$ (e.g. a path function for 1-dimensional manifolds ∂E) where the image coincides with ∂E , but this can be tedious. A more flexible method is to introduce a system of equality constraints of the form

$$g : \mathbb{R}^n \rightarrow \mathbb{R}^c, \quad g(x) = \begin{pmatrix} g_1(x) \\ \vdots \\ g_c(x) \end{pmatrix} = 0 \quad (303)$$

which really just represents a level set of g at 0, i.e. a set described by an implicit representation.¹

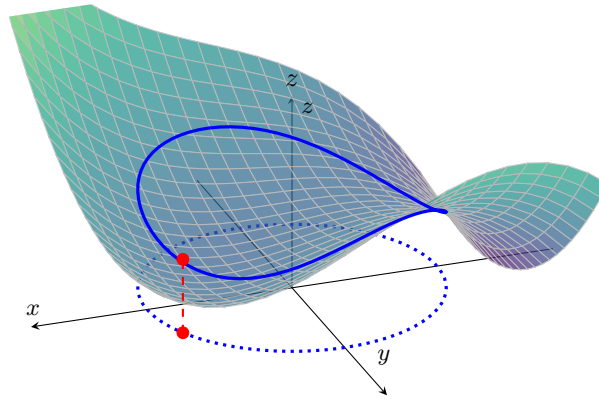


Figure 11: An example of a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ constrained to the unit circle, where $g(x, y) = x^2 + y^2 - 1 = 0$. The blue curve in the xy -plane is the constraint set, and the upper blue curve is its image on the graph of f .

To solve this constraint problem, we use the method of Lagrange multipliers. The basic idea is to convert a constrained problem into a form such that the derivative test of an unconstrained problem can still be applied.

Theorem 1.17 (Lagrange Multipliers Theorem)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $g : \mathbb{R}^n \rightarrow \mathbb{R}^c$ be C^1 functions. If

1. a is a local maximum of f restricted to $M = \{x \in \mathbb{R}^n : g(x) = 0\}$, and
2. $\text{rank}(Dg_a) = c$,

Then there exists a unique row vector $\lambda \in \mathbb{R}^{1 \times c}$ —called the **Lagrange multipliers**—such that

$$Df_a = \lambda Dg_a \quad (304)$$

Proof. Let

$$M = \{x \in \mathbb{R}^n : g(x) = 0\}.$$

Since $\text{rank}(Dg_a) = c$, the implicit function theorem tells us that near a , the constraint set M is a smooth surface of dimension $n - c$. Its tangent space at a is

$$T_a M = \ker(Dg_a).$$

Since a is a local maximum of f restricted to M , the derivative of f must vanish in every tangent direction to M . Therefore, for every $v \in T_a M$,

$$Df_a v = 0.$$

But $T_a M = \ker(Dg_a)$, so

$$v \in \ker(Dg_a) \implies Df_a v = 0.$$

Hence

$$\ker(Dg_a) \subseteq \ker(Df_a).$$

¹In physics, these types of "well-behaved" constraints are known as *holonomic constraints*.

Now we use a linear algebra fact: if a row vector annihilates $\ker(Dg_a)$, then it lies in the row space of Dg_a . Therefore Df_a lies in the row space of Dg_a . Thus there exists a row vector $\lambda \in \mathbb{R}^{1 \times c}$ such that

$$Df_a = \lambda Dg_a.$$

Finally, since $\text{rank}(Dg_a) = c$, the rows of Dg_a are linearly independent. Therefore the representation of Df_a as a linear combination of those rows is unique. Hence λ is unique.

By transposing Equation 304, we can get a matrix realization of

$$\begin{pmatrix} \frac{\partial f}{\partial x_1}(a) \\ \vdots \\ \frac{\partial f}{\partial x_n}(a) \end{pmatrix} = \begin{pmatrix} \frac{\partial g_1}{\partial x_1}(a) & \cdots & \frac{\partial g_c}{\partial x_1}(a) \\ \vdots & \ddots & \vdots \\ \frac{\partial g_1}{\partial x_n}(a) & \cdots & \frac{\partial g_c}{\partial x_n}(a) \end{pmatrix} \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_c \end{pmatrix} \quad (305)$$

$$= \lambda_1 \begin{pmatrix} \frac{\partial g_1}{\partial x_1}(a) \\ \vdots \\ \frac{\partial g_1}{\partial x_n}(a) \end{pmatrix} + \lambda_2 \begin{pmatrix} \frac{\partial g_2}{\partial x_1}(a) \\ \vdots \\ \frac{\partial g_2}{\partial x_n}(a) \end{pmatrix} + \cdots + \lambda_c \begin{pmatrix} \frac{\partial g_c}{\partial x_1}(a) \\ \vdots \\ \frac{\partial g_c}{\partial x_n}(a) \end{pmatrix} \quad (306)$$

This equation tells us that at any critical points a of f evaluated under the equality constraints, the total derivative of f at a can be expressed as a unique linear combination of the total derivatives of the constraints $D(g_i)_a$, with the Lagrange multipliers acting as coefficients. Therefore, finding the critical points a of f constrained with g is equivalent to solving the system of $c + n$ equations for the n unknowns in x and c unknowns in λ .

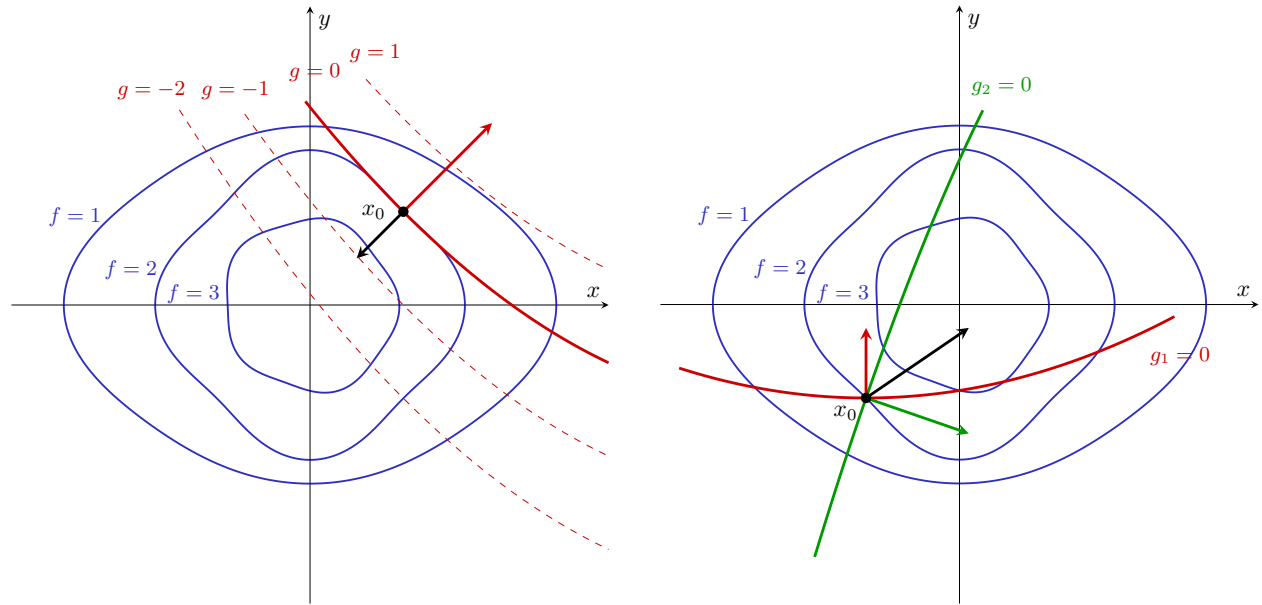
$$g(x) = 0 \iff \begin{cases} g_1(x) &= 0 \\ \dots &= 0 \\ g_c(x) &= 0 \end{cases} \quad (307)$$

$$Df_a = \lambda Dg_a \iff \begin{cases} \frac{\partial f}{\partial x_1}(a) &= \lambda_1^* \frac{\partial g_1}{\partial x_1}(a) + \lambda_2^* \frac{\partial g_2}{\partial x_1}(a) + \cdots + \lambda_c^* \frac{\partial g_c}{\partial x_1}(a) \\ \dots &= \dots \\ \frac{\partial f}{\partial x_n}(a) &= \lambda_1^* \frac{\partial g_1}{\partial x_n}(a) + \lambda_2^* \frac{\partial g_2}{\partial x_n}(a) + \cdots + \lambda_c^* \frac{\partial g_c}{\partial x_n}(a) \end{cases} \quad (308)$$

This relationship between the gradient of the function and gradients of the constraints rather naturally leads to a reformulation of the original problem, known as the *Lagrangian function*. That is, in order to find the maximum/minimum of f subjected to the equality constraint $g(x) = 0$, we form the Lagrangian function

$$\mathcal{L}(x, \lambda) = f(x) - \lambda^T g(x) \quad (309)$$

and find the stationary points of $\mathcal{L}$ considered as a function of $x \in E$ and the Lagrange multiplier $\lambda \in \mathbb{R}$. The main advantage to this method is that it allows the optimization to be solved without explicit parameterization in terms of the constraints.



(a) When there is a single constraint $g : \mathbb{R}^n \rightarrow \mathbb{R}$, $\nabla f(x_0)$ can be expressed as a linear combination of $\nabla g(x_0)$.

(b) When there are multiple constraints g_1, g_2 , $\nabla f(x_0)$ can be expressed as a linear combination of $\nabla g_1(x_0)$ and $\nabla g_2(x_0)$.

Figure 12: From the properties of the gradient, $\nabla f(x_0)$ is orthogonal to the level set of points satisfying $f(x) = f(x_0)$ at point x_0 . We can see that the point where the contour line of $g(x) = 0$ tangentially touches the contours of f is the maximum. Since it intersects it tangentially, the gradient vector at that point $\nabla g(x_0)$ is parallel to $\nabla f(x_0)$.

From the properties of the gradient introduced before, $\nabla f(x_0)$ is orthogonal to the level set of points satisfying $f(x) = c$ at the point x_0 . But this level set $f(x) = c$ actually intersects the level set determined by $g(x) = c$ at the point x_0 and is indistinguishable from each other at x_0 . This means that $\nabla g(x_0)$ is normal the level set of $g(x) = c$ at $x_0 \iff$ it is normal to the level set of $f(x) = c$ at x_0 . But $\nabla f(x_0)$ is also normal at that point, so $\nabla f(x_0)$ must be parallel to $\nabla g(x_0)$.

2 Higher Order Derivatives

So far, we have talked about derivatives as linear maps. However, this language is a bit too restrictive because it doesn't allow us to represent the derivatives of higher order using matrices. It turns out that these higher order derivatives are tensors.

2.1 Second Order Derivatives

Definition 2.1 (Second Derivative)

Let $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be a function that is differentiable at a . Then, $Df_a : \mathbb{R}^n \rightarrow \mathbb{R}$. Then, the **second total derivative** is a bilinear map

Definition 2.2 (Hessian)

Given $f : E \subset V \rightarrow \mathbb{R}$ twice-differentiable at x , the **Hessian** of f at x is the matrix realization of the total derivative, denoted Hf_x . The entries of the Hessian matrix are precisely the partials.

$$Hf_x := \begin{pmatrix} \partial_{x_1 x_1}(a) & \cdots & \partial_{x_1 x_n}(a) \\ \vdots & \ddots & \vdots \\ \partial_{x_n x_1}(a) & \cdots & \partial_{x_n x_n}(a) \end{pmatrix} \text{ and } Hf := \begin{pmatrix} \partial_{x_1 x_1} & \cdots & \partial_{x_1 x_n} \\ \vdots & \ddots & \vdots \\ \partial_{x_n x_1} & \cdots & \partial_{x_n x_n} \end{pmatrix} \quad (310)$$

Proof.

So, we can see that the Hessian matrix must be symmetric for a twice-differentiable function.

Theorem 2.1 (Clairut's Theorem)

Given $f \in C^2$ at point a , its second iterated partials are equal.

$$\partial_{x_i x_j} f(a) = \partial_{x_j x_i} f(a) \text{ for } i, j = 1, 2, \dots, n \quad (311)$$

Proof. For clarity, denote x_i, x_j as x, y and ignore the rest of the variables. Then, the partial derivatives $\partial_{xy} f$ and $\partial_{yx} f$ at a point (x_0, y_0) can be expressed as double limits:

$$\partial_{xy} f(x_0, y_0) = \lim_{y \rightarrow y_0} \frac{\partial_x f(x_0, y) - \partial_x f(x_0, y_0)}{y - y_0} \quad (312)$$

where $\partial_x f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}$. We can use the two limit definitions of partial derivatives

$$\partial_x f(x_0, y) = \lim_{x \rightarrow x_0} \frac{f(x, y) - f(x_0, y)}{x - x_0} \text{ and } \partial_x f(x_0, y_0) = \lim_{x \rightarrow x_0} \frac{f(x, y_0) - f(x_0, y_0)}{x - x_0} \quad (313)$$

and substitute them to get the two partials

$$\partial_{xy} f(x_0, y_0) = \lim_{y \rightarrow y_0} \frac{\lim_{x \rightarrow x_0} \frac{f(x, y) - f(x_0, y)}{x - x_0} - \lim_{y \rightarrow y_0} \frac{f(x_0, y) - f(x_0, y_0)}{x - x_0}}{y - y_0} \quad (314)$$

$$= \lim_{y \rightarrow y_0} \lim_{x \rightarrow x_0} \left(\frac{f(x, y) - f(x_0, y) - f(x, y_0) + f(x_0, y_0)}{(x - x_0)(y - y_0)} \right) \quad (315)$$

$$\partial_{yx} f(x_0, y_0) = \lim_{x \rightarrow x_0} \frac{\lim_{y \rightarrow y_0} \frac{f(x, y) - f(x, y_0)}{y - y_0} - \lim_{y \rightarrow y_0} \frac{f(x_0, y) - f(x_0, y_0)}{y - y_0}}{x - x_0} \quad (316)$$

$$= \lim_{x \rightarrow x_0} \lim_{y \rightarrow y_0} \left(\frac{f(x, y) - f(x, y_0) - f(x_0, y) + f(x_0, y_0)}{(y - y_0)(x - x_0)} \right) \quad (317)$$

Now invoking our assumption that f is C^2 , the two limits, which approach (x_0, y_0) along different paths, both exist and are equal to

$$\lim_{(x,y) \rightarrow (x_0,y_0)} \frac{f(x,y) - f(x_0,y) - f(x,y_0) + f(x_0,y_0)}{(x-x_0)(y-y_0)} \quad (318)$$

and therefore $\partial_{xy}f = \partial_{yx}f$.

Once we have computed the Hessian, let's take the eigendecomposition of it. Since Hf_a is a real symmetric matrix, by the spectral theorem, it will have n real eigenvalues $\lambda_1, \dots, \lambda_n$ (in descending values) and corresponding orthonormal eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$. Now given the gradient $\nabla f(a)$ at a , we can approximate $\nabla f(a + \mathbf{h})$ at $a + \mathbf{h}$ using its total derivative as

$$\nabla f(a + \mathbf{h}) \approx \nabla f(a) + D\nabla f_a \mathbf{h} = \nabla f(a) + Hf(a)\mathbf{h} \quad (319)$$

The eigenvalues of $Hf(a)$ will tell us how "fast" the gradient changes at a . That is, given a small displacement vector $\mathbf{h}$, we can take an orthonormal decomposition of it in the form

$$\mathbf{h} = \sum_i h_i \mathbf{v}_i \quad (320)$$

and now the approximate gradient can be written as

$$\nabla f(a + \mathbf{h}) \approx \nabla f(a) + \sum_i h_i \lambda_i \mathbf{v}_i \quad (321)$$

Therefore, bigger λ_i 's will contribute to a greater change in $f(a)$, and smaller ones will contribute less. We can use this information to speed up convergence by scaling along different axes of $\mathbf{h}$ when sampling.

2.2 Convexity

Recall what a convex function is. For smooth functions $f : \mathbb{R}^n \rightarrow \mathbb{R}$, we have an equivalent formulation of convexity.

Theorem 2.2 (Convexity of C^1 Functions)

Let D be a convex set and $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be C^1 . Then, f is convex over E if and only if

$$f(a) + Df_a(b - a) \leq f(b) \quad (322)$$

for all $a, b \in E$.

This theorem for C^1 functions is quite intuitive, since for a convex function, the tangent plane on its graph must never be "above" the graph. In other words, the first order approximation must be a global underestimate of f .

Theorem 2.3 (Convexity of C^2 Functions)

Let $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be a C^2 -function over E . Then, f is convex over E if and only if Hf is positive semidefinite over all interior points of E .

In order to determine whether a critical point a is a relative maximum, minimum, or neither, we use the second derivative test.

Theorem 2.4 (2nd Derivative Test)

Let a be a critical point of C^2 function $f : \mathbb{R}^n \rightarrow \mathbb{R}$. That is, $Df_a = \mathbf{0}$. Then,

1. a is a local minimum if Hf_a is positive definite.
2. a is a local maximum if Hf_a is negative definite.
3. a is a *saddle point* if Hf_a is not positive definite nor negative definite.

Visually, this makes sense since given a critical point a , the derivative matrix would be 0, meaning that the 2nd degree Taylor expansion of f near a would be in form

$$f(x) \approx f(a) + \frac{1}{2}(x-a)^T Hf_a(x-a) \quad (323)$$

If Hf_a is positive definite, then by definition $\frac{1}{2}(x-a)^T Hf_a(x-a) > 0$ for all x near a , and so f would increase in every direction within the neighborhood of a . The logic follows similarly for negative definite matrix Hf_a . If Hf_a is not positive nor negative definite, then $\frac{1}{2}(x-a)^T Hf_a(x-a)$ could be positive or negative, depending on which direction vector $\mathbf{h} = x-a$ we choose for computing the directional derivative. Therefore, f will increase for certain $\mathbf{h}$ and decrease for other $\mathbf{h}$.

2.3 k-Times Continuously Differentiable Functions

Recall from linear algebra that a linear map $f : V \rightarrow W$ is an element of the tensor product space $V^\dagger \otimes W$, where $V^\dagger$ represents the dual space of V . That is,

$$\text{Hom}(V, W) \simeq V^\dagger \otimes W \quad (324)$$

Definition 2.3 (Frechet Derivative)

A function $f : V \rightarrow W$ is **differentiable** at $x \in V$ if there exists a unique tensor $Df_x \in V^\dagger \otimes W$ —called the **total derivative** or **Frechet derivative**—such that

$$\lim_{h \rightarrow 0} \frac{\|f(x+h) - f(x) - Df_x h\|}{\|h\|} = 0 \quad (325)$$

It's a simple—almost trivial—change, but extremely powerful. Note that if f is differentiable over V , then the transformation $Df : V \rightarrow (V^\dagger \otimes W)$ that maps x_1 to its Frechet derivative Df_{x_1} can also be analyzed. Supposing that it is differentiable, the derivative of it at x is

$$D^2 f_x := D(Df)_x \in V^\dagger \otimes (V^\dagger \otimes W) \simeq (V^\dagger)^{\otimes 2} \otimes W \quad (326)$$

by the definition above. Intuitively this also make sense: if we want to take the second derivative of f , we must first specify the point in the domain of Df (our velocity) plus the point in the domain of f itself (our position) to compute it (the acceleration). Continuing this pattern, we get the following recursive definition.

Definition 2.4 (kth Frechet Derivative)

For $k > 1$, a function $f : V \rightarrow W$ is **k-times differentiable** at $x \in V$ if

1. f is $k-1$ times differentiable at $x \in V$, and
2. there exists a unique rank $(k, 1)$ tensor $D^k f_x \in (V^\dagger)^{\otimes k} \otimes W$ —called the **kth total derivative** or **kth Frechet derivative**—such that

$$\lim_{h \rightarrow 0} \frac{\|D^{k-1} f_x(x+h) - D^{k-1} f_x(x) - D^k f_x h\|}{\|h\|} = 0 \quad (327)$$

where $D^{k-1}f_x$ is the $(k-1)$ th derivative of f at x .

Generally, multivariate calculus courses do not go into tensor algebra, so these higher order derivatives are omitted; you only work with the Hessian for scalar-valued functions. But we won't be scared off and will present this in full generality, and this gives us two major benefits:

1. We can compute higher order derivative as we have seen.
2. We can compute functions that also input or output tensors, such as matrix-valued functions. This knowledge becomes very useful for optimization, e.g. when you optimize a neural network with respect to matrices (in fully-connected layers) and 3-tensors (e.g. convolutions) and is a must-know for building autograd libraries.

Definition 2.5 (k th Partial Derivative)

Directional derivatives lose meaning in these higher order regimes, and we just work with iterated partials.

Definition 2.6 (C^k Functions)

A function $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}$ is said to be a C^k function if all k -times iterated partial derivatives

$$\partial_{x_{i_1} x_{i_2} \dots x_{i_k}} f \quad (328)$$

exist and are continuous. The vector space of all C^k functions is denoted $C^k(D; \mathbb{R})$, or $C^k(D)$.

Whenever we want to get the k th iterated partial derivative of f , we will assume that $f \in C^k$. Again, this is overkill, but it is conventional since we don't really work with the set of functions with existing partial derivatives. Note that mathematicians throw around the word "smooth" a lot. Usually, it means one of three things: it is of class C^1 , C^∞ , or C^k where k is however high it needs to be to satisfy our assumptions. For example, if I say let us differentiate smooth f two times, then I am assuming that $f \in C^2(\mathbb{R}^n)$.

Visualizing C^k -functions is easy for low orders. A C^0 function produces a graph that isn't "ripped" or "punctured," since this is exactly what a discontinuity would look like. A C^1 function requires the surface to be smooth in such a way that there is a well defined affine tangent subspace at every point. This means that there cannot be any sharp "points" or "edges" on the graph since a tangent subspace cannot be well defined.

Corollary 2.5 (Symmetry in k th Iterated Partials)

Given $f \in C^k$, its k th iterated partials are equal. That is, given any permutation σ ,

$$\partial_{x_{i_1} x_{i_2} \dots x_{i_k}} f = \partial_{x_{\sigma(i_1)} x_{\sigma(i_2)} \dots x_{\sigma(i_k)}} f \text{ for } i_1, \dots, i_k = 1, \dots, n \quad (329)$$

Proof. By induction.

2.4 Taylor Series

To talk about convergence, the big-O notation is very useful.

Definition 2.7 (Classes of Infinitesimal Functions)

A function $\alpha : \mathbb{R}^n \rightarrow \mathbb{R}$ is **infinitesimal** if $\alpha \rightarrow 0$ as $x \rightarrow a$. There are multiple "levels" of infinitesimal functions, i.e. how fast they converge to 0. We can classify them by comparing their limits to polynomials.

1. α is of class $O(1)$ if

$$\lim_{x \rightarrow a} \frac{\alpha(x)}{1} = 0$$

This means that $\alpha(x)$ tends to 0 infinitely faster than 1 (which just means that it tends to 0).

2. α is of class $O(h)$ if

$$\lim_{x \rightarrow a} \frac{\alpha(x)}{\|x - a\|} = 0$$

This means that $\alpha(x)$ tends to 0 infinitely faster than the linear $\|\mathbf{h}\|$, where $\mathbf{h} = x - a$.

3. α is of class $O(h^2)$ if

$$\lim_{x \rightarrow a} \frac{\alpha(x)}{\|x - a\|^2} = 0$$

This means that $\alpha(x)$ tends to 0 infinitely faster than the quadratic $\|\mathbf{h}\|^2$, where $\mathbf{h} = x - a$.

4. α is of class $O(h^k)$ if

$$\lim_{x \rightarrow a} \frac{\alpha(x)}{\|x - a\|^k} = 0$$

This means that $\alpha(x)$ tends to 0 infinitely faster than the k th-order $\|\mathbf{h}\|^k$, where $\mathbf{h} = x - a$. Clearly, $O(h^k) \supset O(h^{k+1})$.

Now given a $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}$, we present some polynomial approximations of f at $a \in D$:

1. If $f \in C^0$, the zeroth (constant) approximation is just

$$P_0(x) = f(a)$$

This is not interesting at all, since it is just constant. Furthermore, the error term $\epsilon_0(x) = f(x) - P_0(x)$ is an infinitesimal function as $x \rightarrow a$ and is of class $O(1)$, since

$$\lim_{x \rightarrow \mathbf{x}_0} \frac{f(x) - P_0(x)}{1} = 0$$

2. If $f \in C^1$, the first (linear) approximation requires us to use our total derivative:

$$P_1(x) = f(a) + Df_a(x - a)$$

and we know that the error $\epsilon_1(x) = f(x) - P_1(x)$ is infinitesimal as $x \rightarrow a$ and is of class $O(h)$, since

$$\lim_{x \rightarrow a} \frac{f(x) - P_1(x)}{\|x - a\|} = 0$$

3. If $f \in C^2$, the second (quadratic) approximation requires us to use a quadratic term (i.e. a bilinear form of $\mathbf{h} = x - a$) centered at a . Call it $H_a : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$, and our estimation is

$$P_2(x) = f(a) + Df_a(x - a) + \frac{1}{2}H(x - a, x - a)$$

which we would like the error term $\epsilon_2(x) = f(x) - P_2(x)$ to be $O(h^2)$, or in limit terms, P_2 must satisfy

$$\lim_{x \rightarrow a} \frac{f(x) - P_2(x)}{\|x - a\|^2} = 0$$

We show that this form H is precisely the Hessian matrix.

Theorem 2.6 (2nd Order Taylor Polynomial)

The second order approximation of a C^2 -differentiable function f about a point a is

$$f(x) = f(a) + Df_a(x - a) + \frac{1}{2}(x - a)^T Hf_a(x - a) + O(h^2)$$

where Df_a is the total derivative at a and Hf_a is the Hessian matrix at a .

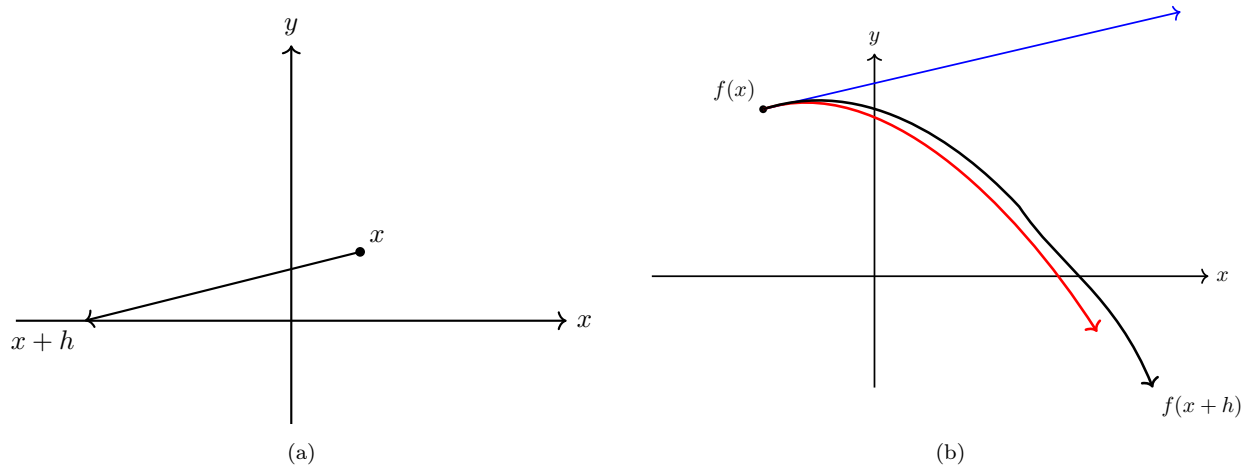


Figure 13: As we change $x \rightarrow x+h$, the actual function deviates from $f(x)$ to $f(x+h)$. The first order approximation (blue) and second order approximation (red) are shown.

2.5 Matrix Calculus

Now we will take a look at functions that have either an input or output as matrices. Essentially, matrices are also vectors, so there is nothing new here to learn, but having a concrete set of notation is useful. First, note that when we talk about a total derivative Df_a , we can interpret this as a linear map that takes in some small perturbation $\mathbf{h}$ and gives us the result $Df_a(\mathbf{h})$. In our column-vector setting, this just corresponded to left matrix multiplication:

$$Df_a(\mathbf{h}) = D\mathbf{f}_a \mathbf{h} \quad (330)$$

This is not the case in the matrix setting. Let us compare the following:

1. The derivative of $f : \mathbb{R} \rightarrow \mathbb{R}$ at a is a linear function $Df_a : \mathbb{R} \rightarrow \mathbb{R}$ satisfying $f(a+h) \approx f(a) + Df_a(h) + O(h^2)$. But linearity reduces Df_a to simply a scalar, and so our condition reduces to

$$f(a+h) \approx f(a) + Df_a h + O(h^2)$$

2. The derivative of a path function $\mathbf{f} : \mathbb{R} \rightarrow \mathbb{R}^m$ is a linear function $D\mathbf{f}_a : \mathbb{R} \rightarrow \mathbb{R}^m$ satisfying $\mathbf{f}(a+h) = \mathbf{f}(a) + D\mathbf{f}_a(h) + O(h^2)$. Linearity implies that $D\mathbf{f}_a$ is a rank-1 linear map, which reduces to it being a column vector, and our condition reduces to

$$\underbrace{\mathbf{f}(a+h)}_{m \times 1} \approx \underbrace{\mathbf{f}(a)}_{m \times 1} + \underbrace{D\mathbf{f}_a}_{m \times 1} \underbrace{h}_{1 \times 1}$$

3. The derivative of a matrix function $\mathbf{F} : \mathbb{R} \rightarrow \mathbb{R}^{m \times n}$ is a linear map $D\mathbf{F}_a : \mathbb{R} \rightarrow \mathbb{R}^{m \times n}$ satisfying $\mathbf{F}(a+h) \approx \mathbf{F}(a) + D\mathbf{F}_a(h)$. This time, linearity does not reduce it to simple left-hand matrix multiplication.

We could just say that this is a left-hand scalar multiplication, but this doesn't generalize well, so we are stuck with just saying that $D\mathbf{F}_a$ is a linear map.

$$\underbrace{\mathbf{F}(a+h)}_{m \times n} \approx \underbrace{\mathbf{F}(a)}_{m \times n} + \underbrace{D\mathbf{F}_a(h)}_{m \times n}$$

Now let us take a look at when we have matrix inputs.

1. The derivative of $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ is a linear function $Df_{\mathbf{A}} : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ satisfying $f(\mathbf{A} + \mathbf{H}) \approx f(\mathbf{A}) + Df_{\mathbf{A}}(\mathbf{H})$. We could let $Df_{\mathbf{A}}$ be some linear map like $\mathbf{M} \mapsto \mathbf{v}^T \mathbf{M} \mathbf{u}$, where $\mathbf{v}, \mathbf{u}$ is fixed. But in generality, we just have the condition

$$\underbrace{f(\mathbf{A} + \mathbf{H})}_{1 \times 1} \approx \underbrace{f(\mathbf{A})}_{1 \times 1} + \underbrace{Df_{\mathbf{A}}(\mathbf{H})}_{1 \times 1}$$

2. The derivative of $\mathbf{f} : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^d$ is some linear map $D\mathbf{f}_{\mathbf{A}} : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^d$ satisfying $\mathbf{f}(\mathbf{A} + \mathbf{H}) \approx \mathbf{f}(\mathbf{A}) + D\mathbf{f}_{\mathbf{A}}(\mathbf{H})$. Again, we could construct some form that would give us a linear map in terms of some matrix multiplication, but in generality, we have the condition

$$\underbrace{\mathbf{f}(\mathbf{A} + \mathbf{H})}_{d \times 1} \approx \underbrace{\mathbf{f}(\mathbf{A})}_{d \times 1} + \underbrace{D\mathbf{f}_{\mathbf{A}}(\mathbf{H})}_{d \times 1}$$

Now we present some theorems on basic differentiation. Proving these just requires us to expand the function and compute the derivatives component-wise.

Theorem 2.7 (Derivative of Affine Map)

Given $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined $f(x) = \mathbf{A}x + \mathbf{b}$ (where $\mathbf{A}, \mathbf{b}$ is not dependent on x), its derivative is

$$Df = \mathbf{A}$$

Theorem 2.8

Given the scalar α defined by

$$\alpha = \mathbf{y}^T \mathbf{A} x$$

where $\mathbf{y} \in \mathbb{R}^{m \times 1}, \mathbf{A} \in \mathbb{R}^{m \times n}$, and $x \in \mathbb{R}^{n \times 1}$, then

$$\frac{\partial \alpha}{\partial x} = \mathbf{y}^T \mathbf{A} : \mathbb{R}^n \rightarrow \mathbb{R}$$

and

$$\frac{\partial \alpha}{\partial \mathbf{y}} = x^T \mathbf{A}^T : \mathbb{R}^m \rightarrow \mathbb{R}$$

Rewritten in the total derivative notation, we can interpret α as a function of both x and $\mathbf{y}$ and write

$$D\alpha_{(x, \mathbf{y})} = \begin{pmatrix} \mathbf{y}^T \mathbf{A} \\ x^T \mathbf{A}^T \end{pmatrix} : \mathbb{R}^{n+m} \rightarrow \mathbb{R}$$

Theorem 2.9

Given the scalar α defined by the quadratic form

$$\alpha = x^T \mathbf{A} x \tag{331}$$

where $x \in \mathbb{R}^{n \times 1}$ and $\mathbf{A} \in \mathbb{R}^{n \times n}$, then

$$\frac{\partial \alpha}{\partial x} = x^T (\mathbf{A} + \mathbf{A}^T) \quad (332)$$

or in the total derivative notation,

$$D\alpha_{\mathbf{a}} = \mathbf{a}^T (\mathbf{A} + \mathbf{A}^T) : \mathbb{R}^n \rightarrow \mathbb{R} \quad (333)$$

3 Vector Fields

3.1 Gradients

Definition 3.1 (Gradient)

The **gradient** of a C^1 scalar-valued function f is the vector field $\nabla f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ defined

$$\nabla f(a) := \begin{pmatrix} \partial_{x_1}(a) \\ \vdots \\ \partial_{x_n}(a) \end{pmatrix} \quad (334)$$

Note that the gradient is a vector field (a bundle of vectors), while the total derivative is a covector field (a bundle of covectors). Since $\mathbb{R}^n$ is an inner product space, we can invoke Riesz Representation theorem and see that they are related in the way that

$$Df_a v = \nabla f(a) \cdot v \quad (335)$$

where $\cdot$ represents the dot product. At this point, it's a bit hard to see the difference between these two, but in more abstract spaces the total derivative generalizes much better than the gradient, which exists for inner product spaces. From this, we can write a coordinate independent definition of the gradient.

Definition 3.2 (Gradient)

The gradient of a scalar valued function $f \in C^1$ is the unique vector field whose dot product with any vector v at each point is the directional derivative of f along v . That is,

$$\nabla f_a \cdot v = Df_a v \text{ for all } a \in D \quad (336)$$

Theorem 3.1 (Gradient as Direction of Fastest Increase)

Let f be a real-valued function such that $\nabla f(x) \neq 0$. Then, at the point x , $\nabla f(x)$ points in the direction along which f is increasing the fastest. Equivalently, $-\nabla f(x)$ points in the direction along which f is decreasing the fastest.

Proof. Note that this is a coordinate-independent proof. Given a directional vector v , we can normalize it since we are only interested in direction. Evaluating it with the total derivative at x gives us $D_a f v$. But by definition,

$$\nabla f_a \cdot v = Df_a v \quad (337)$$

which means that

$$\sup_{\|v\|=1} \{Df_a v\} = \sup_{\|v\|=1} \{\nabla f_a \cdot v\} \quad (338)$$

$$= \sup_{\|v\|=1} \{\|\nabla f_a\| \|v\| \cos(\theta)\} \quad (339)$$

$$= \sup \{\|\nabla f_a\| \cos(\theta)\} \quad (340)$$

$$= \|\nabla f_a\| \text{ when } \theta = 0 \quad (341)$$

Therefore, v must point in the direction of ∇f_a .

Therefore, we can interpret the gradient evaluated at a point as the tangent vector that points in the direction of fastest increase. We can also interpret the gradient ∇f itself as the vector field that determines some

sort of "flow" in the domain $\mathbb{R}^n$. Therefore, if we drop a point in this field, the point will flow through $\mathbb{R}^n$ through a current determined by ∇f and will eventually end up at a local maximum.

Definition 3.3 (Del Operator)

For convenience, we use the del operator to denote the gradient. The **del operator** $\nabla : f \mapsto \nabla f$ takes in a differentiable function and outputs the gradient of it.

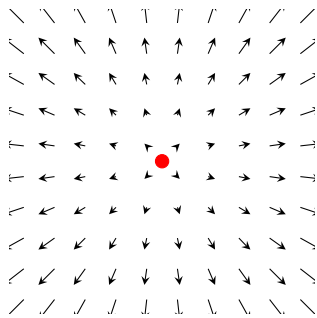
3.2 Divergence

Colloquially, the divergence is an operator div that operates on a vector field and produces a scalar field which provides the quantity of the vector field's source at each point. Technically, the divergence represents the volume density of the outward flux of a vector field from an infinitesimal volume around a given point.

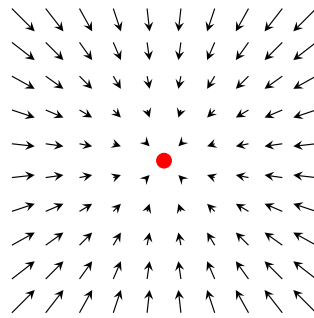
Definition 3.4 (Divergence)

The **divergence** of a vector field $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a scalar field defined

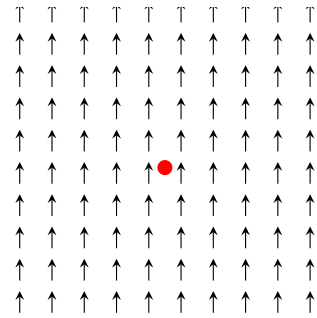
$$\text{div } F := \nabla \cdot F = \begin{pmatrix} \frac{\partial}{\partial x_1} \\ \vdots \\ \frac{\partial}{\partial x_n} \end{pmatrix} \cdot \begin{pmatrix} F_1 \\ \vdots \\ F_n \end{pmatrix} = \sum_i \frac{\partial F_i}{\partial x_i} \quad (342)$$



(a) $\nabla \cdot F > 0$ (Source)



(b) $\nabla \cdot F < 0$ (Sink)



(c) $\nabla \cdot F = 0$ (Constant)

Figure 14: Imagine that the vector field F represents fluid flow in $\mathbb{R}^n$. Divergence is then the measure of the net amount of fluid flowing in and out of an infinitesimally small region, labeled at each point. If the net fluid flow is positive (i.e. more fluid is flowing in than out) at point x_0 , then $\text{div } F(x_0) > 0$. If the net fluid flow is negative (i.e. more fluid is flowing out than in) at point x_0 , then $\text{div } F(x_0) < 0$. Therefore, each point either acts as a "source" of fluid emanating from it or as a "sink" that sucks in more fluid than it puts out.

When $n = 1$, F reduces to a regular function and $\text{div } F$ reduces to the ordinary derivative.

Lemma 3.2 (Linearity of Divergence)

By linearity of partials, div is also a linear operator. That is, given two vector fields F, G and two scalars α, β ,

$$\text{div}(\alpha F + \beta G) = \alpha \text{div } F + \beta \text{div } G \quad (343)$$

Proof.

Lemma 3.3 (Product Rule)

Divergence satisfies the product rule: Given a vector field $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and a scalar function $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}$.

$$\nabla \cdot (\varphi F) = \nabla \varphi \cdot F + \varphi (\nabla \cdot F) \quad (344)$$

Proof.

Lemma 3.4 (Divergence in Cylindrical Coordinates)

For vector field $F : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ expressed in cylindrical coordinates as

$$F = \begin{pmatrix} F_r \\ F_\theta \\ F_z \end{pmatrix} \quad (345)$$

the divergence is

$$\operatorname{div} F = \nabla \cdot F = \frac{1}{r} \frac{\partial}{\partial r} (r F_r) + \frac{1}{r} \frac{\partial F_\theta}{\partial \theta} + \frac{\partial F_z}{\partial z} \quad (346)$$

Note that the condition of locality is important, since in general a global cylindrical coordinate system would be inconsistent.

Proof.

Lemma 3.5 (Divergence in Spherical Coordinates)

For vector field $F : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ expressed in spherical coordinates (r, θ, ϕ) , the divergence is

$$\operatorname{div} F = \nabla \cdot F = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 F_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta F_\theta) + \frac{1}{r \sin \theta} \frac{\partial F_\phi}{\partial \phi} \quad (347)$$

Proof.

3.3 Curl

Colloquially, the curl is a vector operator that describes the infinitesimal circulation of a vector field in 3-dimensional Euclidean space, where the curl at each point is represented by a vector whose length and direction denote the magnitude and axis as the maximum circulation. That is, if one drops a twig or a ball with its center of mass at a certain point, the curl measures how much it will spin. In physics, the rotation of a rigid body in 3-dimensions can be described by a vector ω along the axis of rotation. ω is called the *angular velocity vector*, with $\|\omega\|$ denoting the angular speed of the body. The curl of this vector field measured at the center of mass of the body is measured as 2ω . That is, the curl outputs *twice* the angular velocity vector of any rigid body. Note that unlike the gradient and divergence operators, curl does not generalize as simply to other dimensions.

Definition 3.5 (Curl)

The **curl** of a 3-dimensional C^k vector field $F : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is an operator

$$\operatorname{curl} F \equiv \nabla \times F := \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \times \begin{pmatrix} F_1 \\ F_2 \\ F_3 \end{pmatrix} = \begin{pmatrix} \frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \\ \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \\ \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \end{pmatrix} \quad (348)$$

Definition 3.6 (Irrotational Vector Fields)

A vector field F is **irrotational** if

$$\operatorname{curl} F = 0 \quad (349)$$

Visually, this indicates that there are no "whirlpools" everywhere, meaning that any rigid body placed anywhere, while it may travel along a path, will not rotate around its own axis.

It has been shown that fluid draining from a tub is usually irrotational except for right at the center, which is surprising since the fluid itself is "rotating" around the drain.

Theorem 3.6 (Vanishing Divergence of Curl)

For any C^2 vector field F ,

$$\operatorname{div} \operatorname{curl} F = \nabla \cdot (\nabla \times F) = 0 \quad (350)$$

That is, the divergence of any curl is 0.

Proof. Proved by equality of mixed partials.

Definition 3.7 (Laplacian)

The **Laplacian** of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is the divergence of the gradient.

$$\nabla^2 f \equiv \nabla \cdot (\nabla f) \equiv \sum_{i=1}^n \frac{\partial^2 f}{\partial x_i^2} \quad (351)$$

3.4 Conservative, Solenoidal Vector Fields**Definition 3.8 (Conservative Vector Fields)**

A vector field $F : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a **conservative vector field** if and only if there exists a scalar field $f : U \subset \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$F = \nabla f \quad (352)$$

on U .

Conservative vector fields appear naturally in mechanics: they are vector fields representing forces of physical systems in which energy is conserved.

Theorem 3.7

Given a C^2 -function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$,

$$\nabla \times (\nabla f) = 0 \quad (353)$$

That is, the curl of any gradient vector field is the zero vector.

Proof. $\nabla \times \nabla f$ can be expanded to

$$\begin{pmatrix} \frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial z \partial y} \\ \frac{\partial^2 f}{\partial z \partial x} - \frac{\partial^2 f}{\partial x \partial z} \\ \frac{\partial^2 f}{\partial x \partial y} - \frac{\partial^2 f}{\partial y \partial x} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad (354)$$

by equality of mixed partials.

Definition 3.9 (Solenoidal Vector Fields)

A vector field $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is **solenoidal**, or **incompressible**, if

$$\operatorname{div} F = \nabla \cdot F = 0 \quad (355)$$

at all point in the field. That is, the field has no sources or sinks.

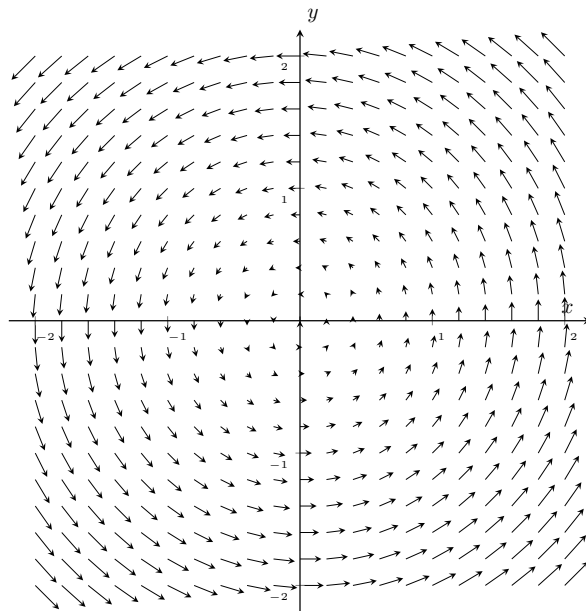
Example 3.1 (2D Solenoidal Vector Field)

Figure 15: The vector field $F : (x, y) \mapsto (y, -x)$ is solenoidal.

4 Riemann Integration

We have done integration over closed intervals $[a, b]$. The natural extension is to define integration over *boxes* $B = \prod_i [a_i, b_i]$. Essentially the construction is exactly the same.

Definition 4.1 (Partition/Mesh)

Let $B = \prod_i [a_i, b_i] \subset \mathbb{R}^n$ be a *box*. Then, a **partition**—or **mesh**—of B is a finite set of points $P = \{x_{ij}\}_{1 \leq i \leq n, 0 \leq j \leq m_i}$ ^a s.t.

$$a_i = x_{i,0} \leq x_{i,1} \leq x_{i,2} \leq \dots \leq x_{i,m_i-1} \leq x_{i,m_i} = b_i \quad \forall i = 1, \dots, n \quad (356)$$

We denote each **box** within the partition as

$$\Delta_j = \Delta_{j_1, j_2, \dots, j_n} = \prod_i [x_{i, j_i-1}, x_{i, j_i}] \quad \forall j_i = 1, \dots, m_i \quad (357)$$

of volume $|\Delta_j| := \prod_i (x_{i, j_i} - x_{i, j_i-1})$.

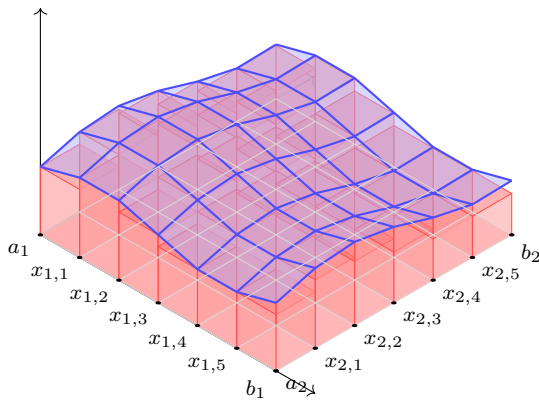
^aTherefore, for each dimension i , we want to take the interval $[a_i, b_i]$ and subdivide it into $m_i + 1$ subintervals.

This nearly² partitions B into a grid of $\prod_i m_i$ smaller boxes, and can be seen as a discretization of the integral which we will construct.

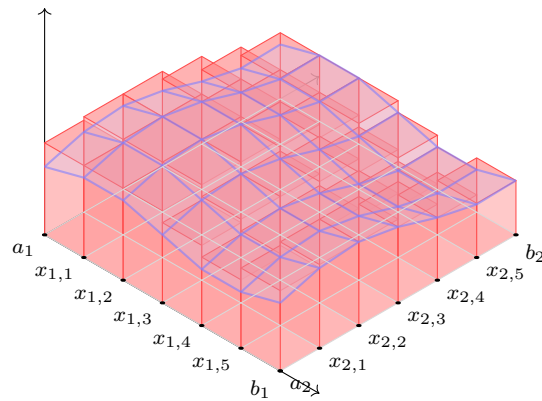
Definition 4.2 (Darboux Sums with Respect to Partition)

Let $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be a bounded function, let $B \subset E$ be a box, and P be a partition of B . Then, the **upper and lower Darboux integrals** of f with respect to P are defined

$$U(P, f) := \sum_J \sup_{x \in \Delta_J} f(x) |\Delta_J|, \quad L(P, f) := \sum_J \inf_{x \in \Delta_J} f(x) |\Delta_J| \quad (358)$$



(a) Lower Riemann sum.



(b) Upper Riemann sum.

Figure 16

²since boundaries overlap

Definition 4.3 (Darboux Integral)

Let $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be a bounded function, and let $B \subset E$ be a box. Then, the **upper and lower Darboux integrals** of f are defined

$$\overline{\int}_B f(x) dx := \inf_P U(P, f), \quad \underline{\int}_B f(x) dx := \sup_P L(P, f) \quad (359)$$

If the upper and lower Riemann integrals are equal then f is said to be **Darboux integrable** over B and the **Darboux integral** of f is

$$\int_B f(x) dx \quad (360)$$

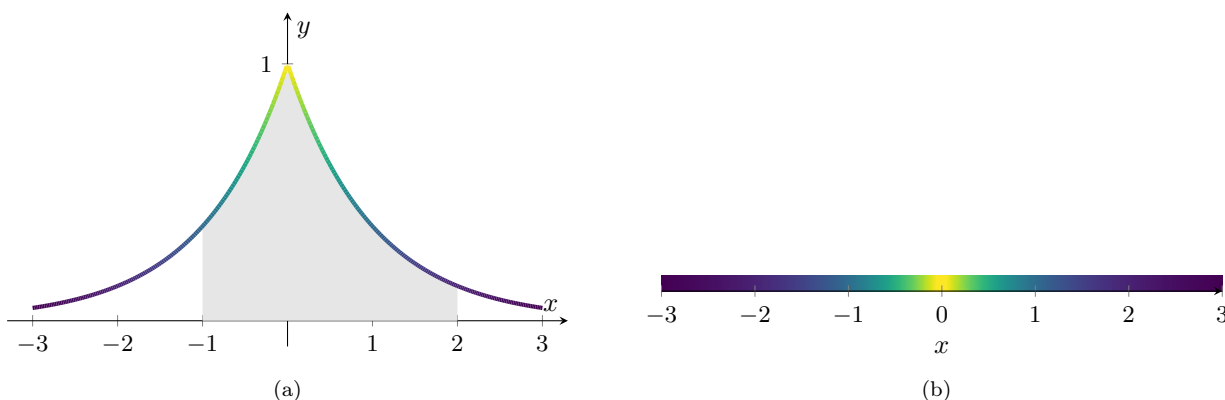


Figure 17: Two interpretations of the integral $\int_{-1}^2 e^{-|x|} dx$. On the left, we view it as the area under a curve. On the right, we view it as the mass of a 1-dimensional rod lying on $[-1, 2]$ of varying density defined by function $f(x)$.

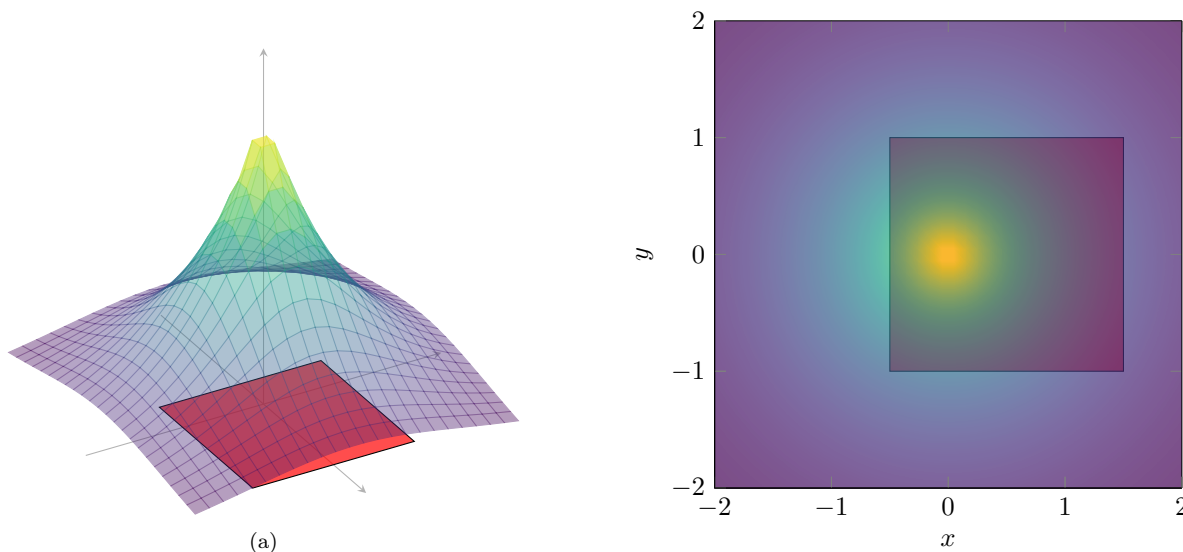


Figure 18: Two interpretations of double integral $\iint_B e^{-\sqrt{x^2+y^2}} dx dy$. On the left, we can view it as the area under a surface. On the right, we view it as the mass of a 2-dimensional sheet lying on $[-0.5, 1.5] \times [-1, 1]$ of varying density defined by function $f(x, y)$.

4.1 Conditions for Integrability

Theorem 4.1 (Continuous Implies Integrable)
$f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}$ continuous means f is integrable over B .
<i>Proof.</i>

However, there are some functions with discontinuities that are in fact integrable.

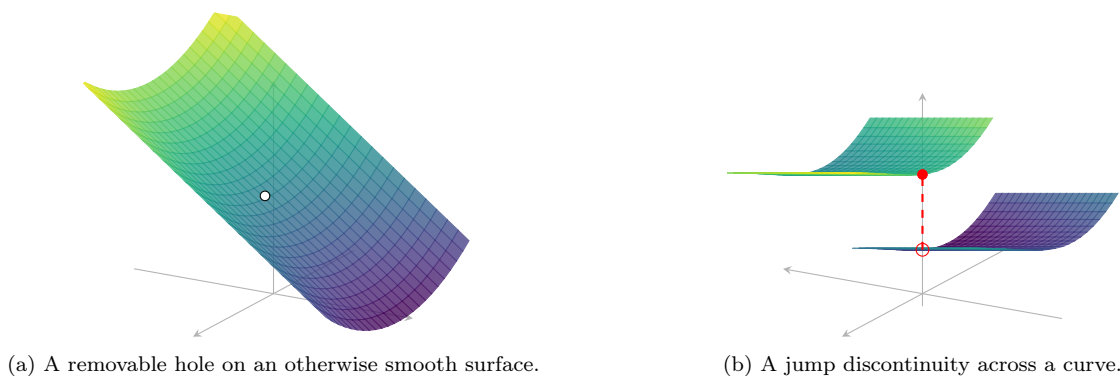


Figure 19: Two typical discontinuities that can still occur in integrable functions: a removable hole and a jump discontinuity.

1. Given that there is a subset N in B with volume 0 over which f is not defined, we can integrate over $B \setminus N$. In the one and two dimensional cases,

$$\int_{B \setminus N} f(x) dx \text{ and } \iint_{B \setminus N} f(x) dA$$

are well-defined. Visually, this corresponds to a surface with a puncture over a set of measure zero.

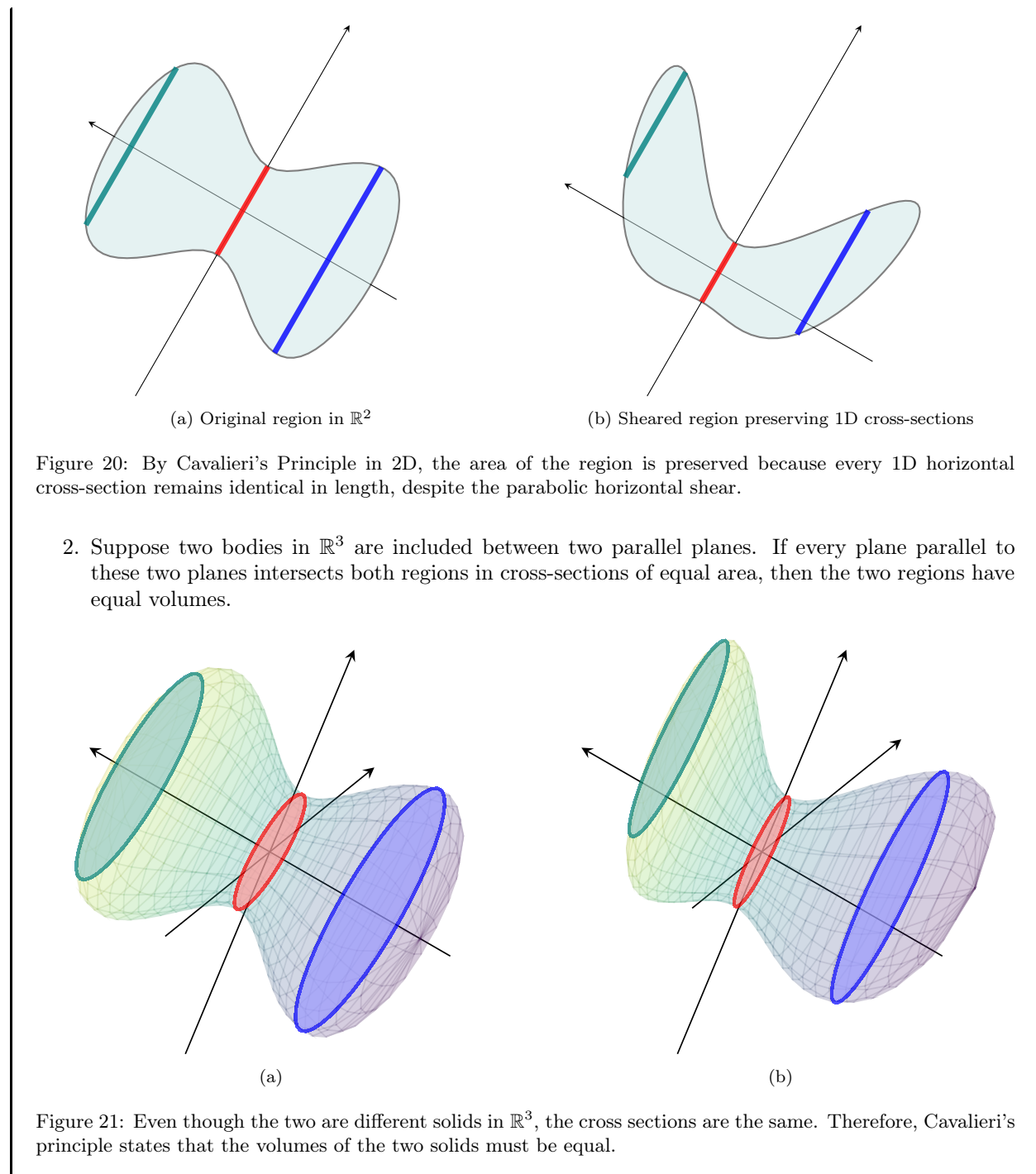
2. The function is defined for all values in the region, but there is a jump in the value of the function. Here the domain remains intact, but the graph splits into two sheets separated by a finite jump.

Informally, if we can visualize the Riemann sum converging to a well-defined area as the rectangles get thinner and thinner, then a discontinuous function is integrable. Indeed, all continuous functions (over a bounded set) are integrable since their Riemann sums are well defined.

4.2 Iterated Integrals

The construction of the integral is one step, but we should know how to practically compute such an integral. We can do this by recursively “reducing” the dimension of the region of the integral until we can work with one dimension at which point we can use the univariate Riemann integral. This is informally known as *Cavalieri’s principle*, which states the following.

Theorem 4.2 (Cavalieri’s Principle)
1. Suppose two shapes in $\mathbb{R}^2$ are included between two parallel lines in that plane. If every line parallel to these two lines intersects both regions in line segments of equal length, then the two regions have equal areas.



In the case when S is a box in $\mathbb{R}^n$, Fubini's theorem states that whether we cut S up along the x_1 -axis, x_2 -axis, ..., or the x_n -axis, the symmetry in volume is always preserved. This theorem is really just a specific case of this general symmetry in volume.

Theorem 4.3 (Euler-Fubini Theorem)

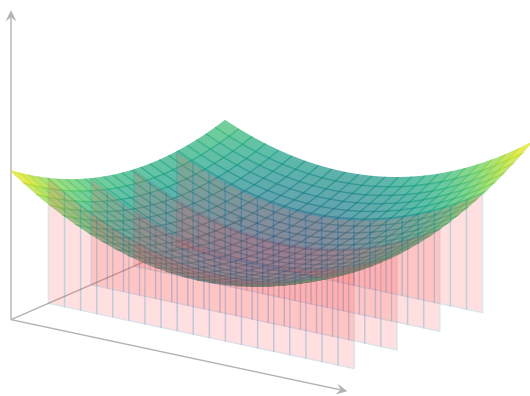
Given a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, let $B := \prod_{i=1}^n [\alpha_i, \beta_i]$ and let p be any permutation of the elements $\{x_1, x_2, \dots, x_n\}$. Then

$$\int_B f dV = \int_{\alpha_n}^{\beta_n} \dots \int_{\alpha_1}^{\beta_1} f dx_1 \dots dx_n = \int_{p(\alpha_1)}^{p(\beta_1)} \dots \int_{p(\alpha_n)}^{p(\beta_n)} f dp(x_1) \dots dp(x_n) \quad (361)$$

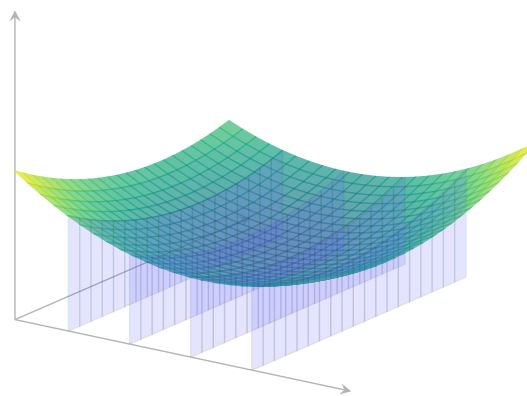
Proof.

In the two dimensional case, we have

$$\iint_B f dA = \int_c^d \int_a^b f(x, y) dx dy = \int_a^b \int_c^d f(x, y) dy dx \quad (362)$$



(a) Slicing at fixed values of y .



(b) Slicing at fixed values of x .

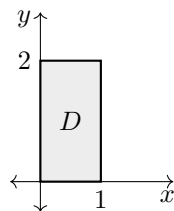
and in the three dimensional case, we can use any of the 6 permutations. Computation of these integrals is simple. You do the innermost integral first with respect to the corresponding variable, while treating the rest of the variables constant. Evaluating each integral outputs a formula for a higher dimensional cross section of the solid S . It is clear that computing iterated integrals is really just doing Cavalieri's principle repeatedly.

Example 4.1 (Double Integral over a Rectangle: Easy)

Let

$$D = [0, 1] \times [0, 2]$$

and consider the function $f(x, y) = x + 2y$. Then



Since D is a rectangle, we may write

$$\begin{aligned}
 \iint_D (x + 2y) \, dA &= \int_0^2 \int_0^1 (x + 2y) \, dx \, dy \\
 &= \int_0^2 \left[\frac{x^2}{2} + 2yx \right]_{x=0}^{x=1} dy \\
 &= \int_0^2 \left(\frac{1}{2} + 2y \right) dy \\
 &= \left[\frac{y}{2} + y^2 \right]_{y=0}^{y=2} \\
 &= 1 + 4 = 5.
 \end{aligned}$$

Therefore,

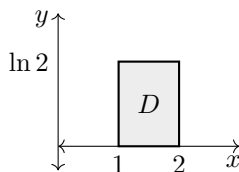
$$\iint_D (x + 2y) \, dA = 5.$$

Example 4.2 (Double Integral over a Rectangle: Hard)

Let

$$D = [1, 2] \times [0, \ln 2]$$

and let $f(x, y) = xe^y$. Then



Again, the region is rectangular, so

$$\begin{aligned}
 \iint_D xe^y \, dA &= \int_0^{\ln 2} \int_1^2 xe^y \, dx \, dy \\
 &= \int_0^{\ln 2} e^y \left[\frac{x^2}{2} \right]_{x=1}^{x=2} dy \\
 &= \int_0^{\ln 2} \frac{3}{2} e^y \, dy \\
 &= \frac{3}{2} [e^y]_{y=0}^{y=\ln 2} \\
 &= \frac{3}{2} (2 - 1) = \frac{3}{2}.
 \end{aligned}$$

Hence,

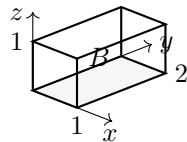
$$\iint_D xe^y \, dA = \frac{3}{2}.$$

Example 4.3 (Triple Integral over a Box: Easy)

Let

$$B = [0, 1] \times [0, 2] \times [0, 1]$$

and let $f(x, y, z) = x + y + z$. Then



Using the order $dz\,dy\,dx$, we get

$$\begin{aligned}
 \iiint_B (x + y + z) \, dV &= \int_0^1 \int_0^2 \int_0^1 (x + y + z) \, dz \, dy \, dx \\
 &= \int_0^1 \int_0^2 \left[(x + y)z + \frac{z^2}{2} \right]_{z=0}^{z=1} dy \, dx \\
 &= \int_0^1 \int_0^2 \left(x + y + \frac{1}{2} \right) dy \, dx \\
 &= \int_0^1 \left[xy + \frac{y^2}{2} + \frac{y}{2} \right]_{y=0}^{y=2} dx \\
 &= \int_0^1 (2x + 3) \, dx \\
 &= \left[x^2 + 3x \right]_{x=0}^{x=1} \\
 &= 1 + 3 = 4.
 \end{aligned}$$

Therefore,

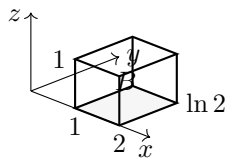
$$\iiint_B (x + y + z) \, dV = 4.$$

Example 4.4 (Triple Integral over a Box: Hard)

Let

$$B = [1, 2] \times [0, \ln 2] \times [0, 1]$$

and let $f(x, y, z) = xe^y(1 + z^2)$. Then



With the order $dz\,dy\,dx$, we obtain

$$\begin{aligned}
 \iiint_B x e^y (1 + z^2) \, dV &= \int_1^2 \int_0^{\ln 2} \int_0^1 x e^y (1 + z^2) \, dz \, dy \, dx \\
 &= \int_1^2 \int_0^{\ln 2} x e^y \left[z + \frac{z^3}{3} \right]_{z=0}^{z=1} dy \, dx \\
 &= \int_1^2 \int_0^{\ln 2} \frac{4}{3} x e^y \, dy \, dx \\
 &= \int_1^2 \frac{4}{3} x [e^y]_{y=0}^{y=\ln 2} dx \\
 &= \int_1^2 \frac{4}{3} x \, dx \\
 &= \frac{4}{3} \left[\frac{x^2}{2} \right]_{x=1}^{x=2} \\
 &= \frac{4}{3} \cdot \frac{3}{2} = 2.
 \end{aligned}$$

Hence,

$$\iiint_B x e^y (1 + z^2) \, dV = 2.$$

4.3 Integration over Regions between Curves

Definition 4.4 (Simple Regions w.r.t. a Variable)

A bounded region D in $\mathbb{R}^n$ is said to be x_i -simple if it is bounded by the graphs of two continuous functions $u_1, u_2 : \mathbb{R}^{n-1} \rightarrow \mathbb{R}$ of the variables

$$x_1, x_2, \dots, x_{i-1}, x_{i+1}, \dots, x_n$$

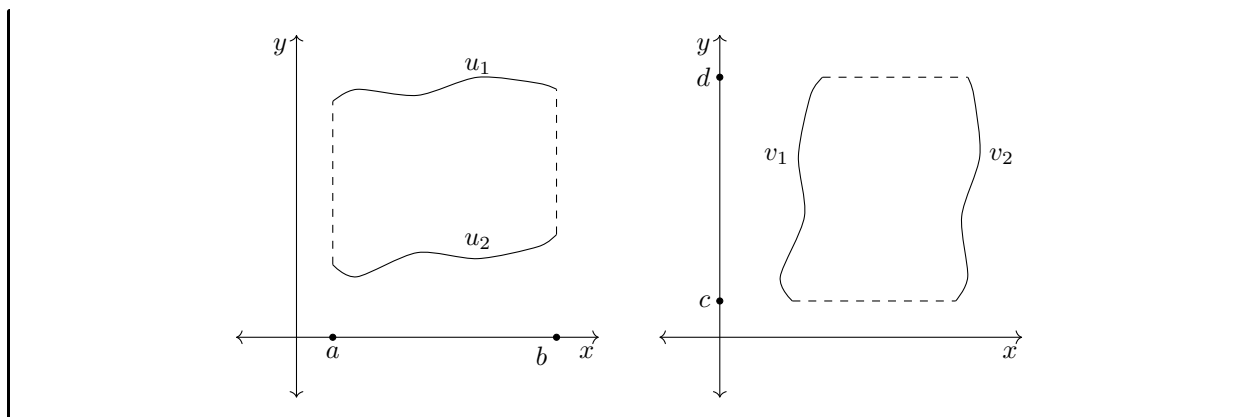
That is, D can be expressed in the form

$$\{x \in \mathbb{R}^n \mid u_1(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \leq x_i \leq u_2(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)\}$$

If a region is simple in all of its variables, it is simply called *simple*. Note that n -dimensional boxes are simple regions.

Example 4.5

In $\mathbb{R}^2$, the region on the left graph is an y -simple region and the region on the right is a x -simple region.



We now describe the method of calculating double integrals over elementary regions.

Theorem 4.4 (Iterated Integrals over Simple Regions)

The double integral over a y -simple region D bounded by functions u_1 and u_2 in $\mathbb{R}^2$ and the x -values a and b (as shown in the left graph of example 2.1) is

$$\iint_D f(x, y) = \int_a^b \int_{u_2(x)}^{u_1(x)} f(x, y) dy dx \quad (363)$$

The double integral over an x -simple region D bounded by functions v_1 and v_2 in $\mathbb{R}^2$ and the y -values c and d (shown in the right of graph of example 2.1) is

$$\iint_D f(x, y) = \int_c^d \int_{v_2(y)}^{v_1(y)} f(x, y) dx dy \quad (364)$$

Example 4.6 (Unit Disk)

Integrating $f(x, y)$ over the unit disk would have the form

$$\int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} f(x, y) dy dx \text{ or } \int_{-1}^1 \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} f(x, y) dx dy \quad (365)$$

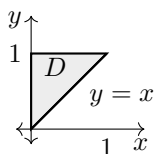
Note that the unit disk is both x and y simple.

Example 4.7 (Double Integral over a Non-Rectangular Region: Easy)

Consider the triangular region

$$D = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 1, x \leq y \leq 1\}$$

and let $f(x, y) = x + y$. Then



Since D is y -simple,

$$\begin{aligned}
 \iint_D (x+y) dA &= \int_0^1 \int_x^1 (x+y) dy dx \\
 &= \int_0^1 \left[xy + \frac{y^2}{2} \right]_{y=x}^{y=1} dx \\
 &= \int_0^1 \left(x + \frac{1}{2} - x^2 - \frac{x^2}{2} \right) dx \\
 &= \int_0^1 \left(x + \frac{1}{2} - \frac{3x^2}{2} \right) dx \\
 &= \left[\frac{x^2}{2} + \frac{x}{2} - \frac{x^3}{2} \right]_{x=0}^{x=1} \\
 &= \frac{1}{2}.
 \end{aligned}$$

Therefore,

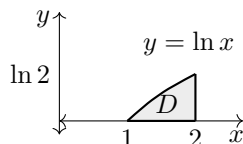
$$\iint_D (x+y) dA = \frac{1}{2}.$$

Example 4.8 (Double Integral over a Non-Rectangular Region: Hard)

Let

$$D = \{(x, y) \in \mathbb{R}^2 \mid 1 \leq x \leq 2, 0 \leq y \leq \ln x\}$$

and let $f(x, y) = ye^y$. Then



This region is not a rectangle because its upper boundary depends on x . We compute

$$\begin{aligned}
 \iint_D ye^y dA &= \int_1^2 \int_0^{\ln x} ye^y dy dx \\
 &= \int_1^2 [(y-1)e^y]_{y=0}^{y=\ln x} dx \\
 &= \int_1^2 (x \ln x - x + 1) dx \\
 &= \left[\frac{x^2}{2} \ln x - \frac{x^2}{4} - \frac{x^2}{2} + x \right]_{x=1}^{x=2} \\
 &= (2 \ln 2 - 1) - \frac{1}{4} \\
 &= 2 \ln 2 - \frac{5}{4}.
 \end{aligned}$$

Thus,

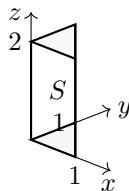
$$\iint_D ye^y dA = 2 \ln 2 - \frac{5}{4}.$$

Example 4.9 (Triple Integral over a Non-Rectangular Region: Easy)

Consider the solid

$$S = \{(x, y, z) \in \mathbb{R}^3 \mid 0 \leq x \leq 1, 0 \leq y \leq 1 - x, 0 \leq z \leq 2\}$$

and let $f(x, y, z) = x + y + z$. Then



The base in the xy -plane is triangular, so the bounds are not independent. We compute

$$\begin{aligned} \iiint_S (x + y + z) \, dV &= \int_0^1 \int_0^{1-x} \int_0^2 (x + y + z) \, dz \, dy \, dx \\ &= \int_0^1 \int_0^{1-x} \left[(x + y)z + \frac{z^2}{2} \right]_{z=0}^{z=2} dy \, dx \\ &= \int_0^1 \int_0^{1-x} (2x + 2y + 2) \, dy \, dx \\ &= \int_0^1 [2xy + y^2 + 2y]_{y=0}^{y=1-x} dx \\ &= \int_0^1 (3 - 2x - x^2) \, dx \\ &= \left[3x - x^2 - \frac{x^3}{3} \right]_{x=0}^{x=1} \\ &= 3 - 1 - \frac{1}{3} = \frac{5}{3}. \end{aligned}$$

Hence,

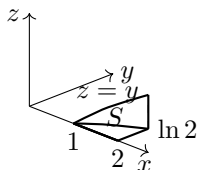
$$\iiint_S (x + y + z) \, dV = \frac{5}{3}.$$

Example 4.10 (Triple Integral over a Non-Rectangular Region: Hard)

Let

$$S = \{(x, y, z) \in \mathbb{R}^3 \mid 1 \leq x \leq 2, 0 \leq y \leq \ln x, 0 \leq z \leq y\}$$

and let $f(x, y, z) = e^y(1 + z)$. Then



Here both the y - and z -bounds depend on earlier variables. Using the order $dz\,dy\,dx$,

$$\begin{aligned}
 \iiint_S e^y(1+z)\,dV &= \int_1^2 \int_0^{\ln x} \int_0^y e^y(1+z)\,dz\,dy\,dx \\
 &= \int_1^2 \int_0^{\ln x} e^y \left[z + \frac{z^2}{2} \right]_{z=0}^{z=y} dy\,dx \\
 &= \int_1^2 \int_0^{\ln x} e^y \left(y + \frac{y^2}{2} \right) dy\,dx \\
 &= \int_1^2 \left[\frac{y^2}{2} e^y \right]_{y=0}^{y=\ln x} dx \\
 &= \int_1^2 \frac{x(\ln x)^2}{2} dx \\
 &= \left[\frac{x^2}{4} (\ln x)^2 - \frac{x^2}{4} \ln x + \frac{x^2}{8} \right]_{x=1}^{x=2} \\
 &= \left((\ln 2)^2 - \ln 2 + \frac{1}{2} \right) - \frac{1}{8} \\
 &= (\ln 2)^2 - \ln 2 + \frac{3}{8}.
 \end{aligned}$$

Therefore,

$$\iiint_S e^y(1+z)\,dV = (\ln 2)^2 - \ln 2 + \frac{3}{8}.$$

4.4 Change of Basis

Sometimes, integrating a region over a different basis would make the integral computation much more simpler. In this case, we may be able to transform more complicated regions into elementary regions. We first introduce a change of basis in 2 dimensions and then generalize it into higher dimensions.

Let $\mathbb{R}^2$ have the standard orthonormal basis e_1, e_2 , commonly known as the x, y basis. Now, let us construct new basis vectors of $\mathbb{R}^2$, denoted f_1, f_2 such that f_1, f_2 are functions of e_1, e_2 . Since they are both bases that span $\mathbb{R}^2$, we can equally represent e_1, e_2 as functions of f_1, f_2 .

$$\begin{aligned}
 e_1 &= g(f_1, f_2) \\
 e_2 &= h(f_1, f_2)
 \end{aligned}$$

Note that this change of basis does not necessarily have to be linear, as in the context of passive transformation in linear algebra. Then, every point (x, y) in the (e_1, e_2) -basis can be rewritten as

$$\begin{aligned}
 (x, y) &= xe_1 + ye_2 \\
 &= xg(f_1, f_2) + yh(f_1, f_2) \\
 &= uf_1 + vf_2
 \end{aligned}$$

Note that it is customary to denote x, y as the coefficients in the e_1, e_2 basis and u, v as the coefficients in the new f_1, f_2 basis. This way, we can not only write e_1 and e_2 as functions of f_1 and f_2 , but we can also write the coefficients x, y as functions of the coefficients u, v ! That is,

$$\begin{aligned}
 x &= x(u, v) \\
 y &= y(u, v)
 \end{aligned}$$

which is really just a function

$$B : \mathbb{R}^2 \longrightarrow \mathbb{R}^2, \quad B(u, v) = \begin{pmatrix} x(u, v) \\ y(u, v) \end{pmatrix}$$

Notice that B changes the u, v coordinates to the x, y coordinates, and B^{-1} changes the x, y coordinates to the u, v coordinates.

$$B^{-1} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2, \quad B^{-1}(x, y) = \begin{pmatrix} u(x, y) \\ v(x, y) \end{pmatrix}$$

Note that these coefficients actually change *contravariantly*, that is, they change inversely with respect to how the basis vectors are changed. In vector calculus, it is conventional to represent a change of basis with functions that relate the coefficients x, y with u, v , rather than the bases f_1, f_2 with e_1, e_2 .

Theorem 4.5 (Integration over Change of Bases in $\mathbb{R}^2$)

Let $\mathbb{R}^2$ have the standard orthonormal basis e_1, e_2 . Now, let us construct new basis vectors of $\mathbb{R}^2$, denoted f_1, f_2 such that the coefficients of the vectors in $\mathbb{R}^2$ are related by the change of basis function

$$B = \begin{pmatrix} x \\ y \end{pmatrix} \implies B(u, v) = \begin{pmatrix} x(u, v) \\ y(u, v) \end{pmatrix}$$

Given region $D \subset \mathbb{R}^2$ and $S = B(D)$ is the region transformed by B , the integral of function $f(x, y)$ over region D can be expressed as

$$\iint_D f(x, y) dA = \iint_S f(x(u, v), y(u, v)) |JB(u, v)| d\bar{A}$$

where $|JB(u, v)|$ is the determinant of the Jacobian matrix of B . Expanding the Jacobian determinant gives

$$|JB(u, v)| = \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u}$$

Theorem 4.6 (Integration over Change of Bases in $\mathbb{R}^3$)

Given that we have the change of basis function

$$B : \mathbb{R}^3 \longrightarrow \mathbb{R}^3, \quad B(u, v, w) = \begin{pmatrix} x(u, v, w) \\ y(u, v, w) \\ z(u, v, w) \end{pmatrix}$$

a region $D \in \mathbb{R}^3$ and $S = B(D)$, the region transformed by B , the integral of $f(x, y, z)$ over region D can be expressed as

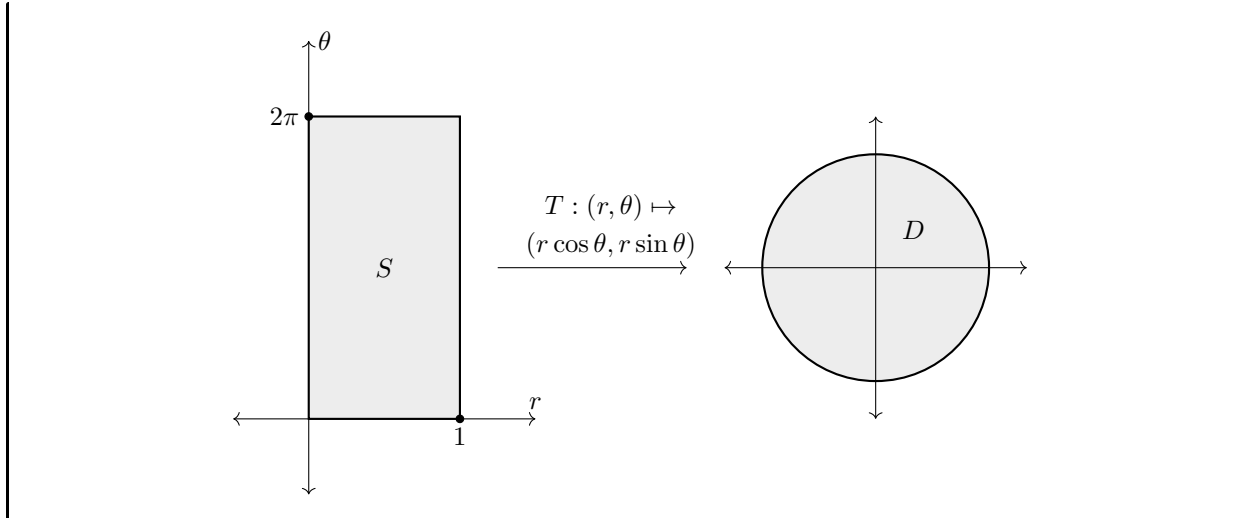
$$\iiint_D f(x, y, z) dV = \iiint_S f(x(u, v, w), y(u, v, w), z(u, v, w)) |JB(u, v, w)| d\bar{V}$$

where $|JB(u, v, w)|$ is the Jacobian determinant of B .

Example 4.11

Given a real-valued function f defined over the region $D \subset \mathbb{R}^2$, we can perform a change of basis of the x, y coordinates into polar ones within a new region S . The change of basis

$$\begin{aligned} x &= r \cos \theta \\ y &= r \sin \theta \end{aligned}$$


Theorem 4.7 (Integration over Change of Bases in $\mathbb{R}^n$)

Let $\mathbb{R}^n$ have the standard orthonormal basis $e_1, e_2, \dots, e_n$, and let us construct a new basis $f_1, f_2, \dots, f_n$ such that the coefficients of the vectors in $\mathbb{R}^n$ are related with the functions

$$B : \mathbb{R}^n \longrightarrow \mathbb{R}^n, \quad B(u_1, u_2, \dots, u_n) = \begin{pmatrix} x_1(u_1, \dots, u_n) \\ x_2(u_1, \dots, u_n) \\ \vdots \\ x_n(u_1, \dots, u_n) \end{pmatrix}$$

Given that the region $D \subset \mathbb{R}^n$ is transformed into a new region $S = B(D) \subset \mathbb{R}^n$ under this basis transformation, the integral of function $f(x_1, \dots, x_n)$ over region D can be expressed as

$$\int_D f(x) dH = \int_S f(x_1(u), x_2(u), \dots, x_n(u)) |JB(u_1, \dots, u_n)| d\bar{H}$$

where the integral on both the left and right hand side represents integration over an n -dimensional region, x represents the n -tuple $(x_1, \dots, x_n)$, u represents the n -tuple $(u_1, \dots, u_n)$, and $|JB(u_1, \dots, u_n)|$ represents the Jacobian determinant of function B .

We now describe some common change of basis formulas for polar, cylindrical, and spherical coordinates.

Theorem 4.8 (Integration in Polar Coordinates)

$$\iint_D f(x, y) dx dy = \iint_S f(r \cos \theta, r \sin \theta) r dr d\theta \quad (366)$$

Definition 4.5 (Cylindrical, Spherical Coordinates)

In $\mathbb{R}^3$, *cylindrical coordinates* have the following relation to rectangular coordinates.

$$x = r \cos \theta \quad (367)$$

$$y = r \sin \theta \quad (368)$$

$$z = z \quad (369)$$

In $\mathbb{R}^3$, *spherical coordinates* have the following relation to rectangular coordinates.

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

Corollary 4.9 (Integration in Cylindrical Coordinates)

$$\iiint_D f(x, y, z) \, dx \, dy \, dz = \iiint_S f(r \cos \theta, r \sin \theta, z) r \, dr \, d\theta \, dz$$

Corollary 4.10 (Integration in Spherical Coordinates)

$$\iiint_D f(x, y, z) \, dx \, dy \, dz = \iiint_S f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \rho^2 \sin \theta \, d\rho \, d\theta \, d\phi$$

Example 4.12 (Gaussian Integral)

The following is the (un-normalized) probability distribution function of the Gaussian distribution.

$$\int_{-\infty}^{\infty} e^{-x^2} \, dx = \sqrt{\pi}$$

4.5 Improper Integrals

There are generally two types of improper integrals.

1. The region D integrated over is unbounded.
2. The function f that is integrated is unbounded within the region D .

These types of improper integrals are usually evaluated using a limiting process. When the interval I is unbounded, say $(1, \infty)$, the integral can be evaluated as

$$\int_1^{\infty} \frac{1}{x^2} \, dx = \lim_{b \rightarrow \infty} \int_1^b \frac{1}{x^2} \, dx = \lim_{b \rightarrow \infty} \left(1 - \frac{1}{b} \right) = 1$$

In case 2, we can add a limit at the point where the function f diverges as such.

$$\int_0^1 \frac{1}{\sqrt{x}} \, dx = \lim_{a \rightarrow 0} \int_a^1 \frac{1}{\sqrt{x}} \, dx = \lim_{a \rightarrow 0} (2 - 2\sqrt{a}) = 2$$

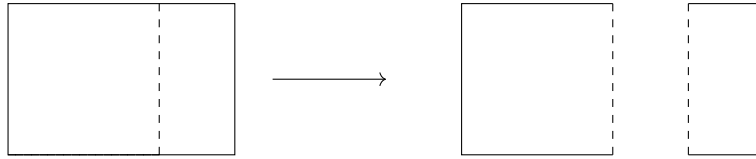
We now describe how to integrate over a certain path p embedded in a higher dimensional space $\mathbb{R}^n$, possibly with a scalar or vector field f . We must first go over oriented paths.

Extending the previous case, we use a multivariate limiting process in $\mathbb{R}^2$. We will first work with case 2, when f is unbounded within the region D . Let us define an elementary region D in $\mathbb{R}^2$; without loss of generality, we will make it y -simple, meaning that D can be expressed as

$$D \equiv \{(x, y) \in \mathbb{R}^2 \mid a \leq x \leq b, \phi_1(x) \leq y \leq \phi_2(x)\}$$

We can actually assume that the region in which f is unbounded lies in the boundary ∂D . This is because if it lied in the interior of D , we could split D into pieces across a path that intersects this region with divergent

values, evaluate the integrals over the pieces separately, and then sum the integrals. For example, in the rectangular region below, let the dashed line represent the values where the function f diverges. Then, we can split the region into two rectangular regions shown in the right.



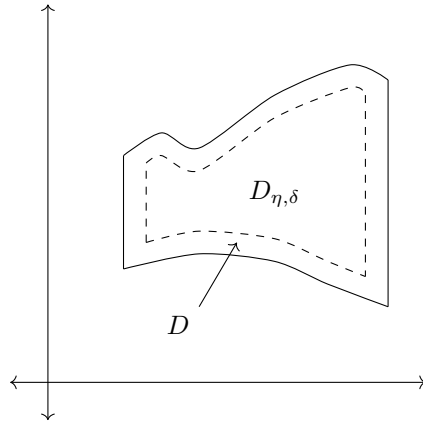
Therefore, assuming that f is unbounded in ∂D , we can construct a new region

$$D_{\eta,\delta} \equiv \{(x, y) \in \mathbb{R}^2 \mid a + \eta \leq x \leq b - \eta, \phi_1(x) + \delta \leq y \leq \phi_2(x) - \delta\}$$

for some arbitrarily small numbers $\eta, \delta > 0$, meaning that the integral (reduced to iterated integrals using Fubini's theorem)

$$F(\eta, \delta) \equiv \iint_{D_{\eta,\delta}} f(x, y) dA = \int_{a+\eta}^{b-\eta} \int_{\phi_1(x)+\delta}^{\phi_2(x)-\delta} f(x, y) dy dx$$

is well defined.



Clearly, the function $F(\eta, \delta)$ is a function of two variables η and δ . So, if the limit

$$\lim_{(\eta,\delta) \rightarrow (0,0)} F(\eta, \delta)$$

is well defined, then so is the improper integral. For it to exist, the iterated limits must both equal to a well-defined real number $\mathcal{L}$ (and to each other). That is,

$$\lim_{\eta \rightarrow 0} \lim_{\delta \rightarrow 0} F(\eta, \delta) = \lim_{\delta \rightarrow 0} \lim_{\eta \rightarrow 0} F(\eta, \delta) = \mathcal{L} \implies \lim_{(\eta,\delta) \rightarrow (0,0)} F(\eta, \delta) = \mathcal{L}$$

It is also worthwhile to note that functions unbounded at isolated points can be evaluated using the methods above using a change of basis. Consider the example below.

Example 4.13

In the unit disk $D \subset \mathbb{R}^2$, let the function f be defined as

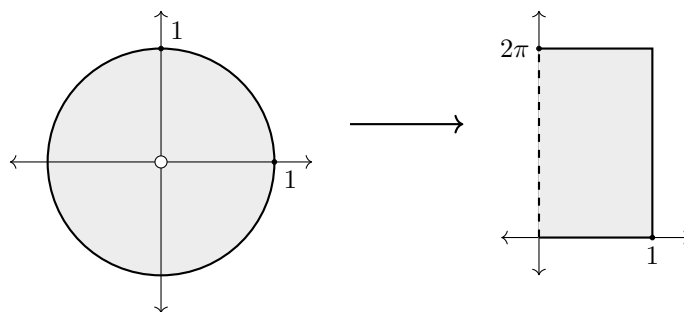
$$f(x, y) \equiv \frac{1}{\sqrt{x^2 + y^2}}$$

Clearly, f is continuous at every point except $0 = (0, 0)$, meaning that

$$\iint_{D \setminus \{0\}} f(x, y) dA$$

is well-defined. In order to solve the integral over the entire disk, we convert to polar coordinates and evaluate the limit

$$\iint_{D \setminus \{0\}} f(x, y) dA = \lim_{\delta \rightarrow 0} \int_{\delta}^1 \int_0^{2\pi} r f(r \cos \theta, r \sin \theta) d\theta dr$$



If we are given an unbounded region $D \subset \mathbb{R}^2$, we can first create a bounded region and expand that region using a limit to cover all of D .

5 Integration of Forms

Definition 5.1 (Parameterization)

Given some subset $E \subset \mathbb{R}^n$, a **k-parameterization** of E is a continuous surjective map

$$\varphi : D \subset \mathbb{R}^k \rightarrow E \quad (370)$$

where D is a compact subset of $\mathbb{R}^k$.

We usually want continuity since if we don't, this allows for very pathological maps.

Example 5.1 (Space-Filling Curve in $\mathbb{R}^2$)

Continuity alone is not enough to guarantee that a parameterized set behaves like a curve in the usual geometric sense. In fact, there exists a continuous surjective map

$$\gamma : [0, 1] \rightarrow [0, 1]^2. \quad (371)$$

Such a map is called a **space-filling curve**.

We sketch one construction. Let $C \subset [0, 1]$ be the Cantor set. Every point $x \in C$ has a ternary expansion using only the digits 0 and 2:

$$x = 0.a_1a_2a_3 \cdots, \quad a_i \in \{0, 2\}. \quad (372)$$

Define binary digits

$$b_i = \frac{a_i}{2} \in \{0, 1\}. \quad (373)$$

Now split the binary digits into odd and even positions, and define

$$\Gamma(x) = (0.b_1b_3b_5 \cdots, 0.b_2b_4b_6 \cdots). \quad (374)$$

Equivalently,

$$\Gamma(x) = \left(\sum_{j=1}^{\infty} \frac{b_{2j-1}}{2^j}, \sum_{j=1}^{\infty} \frac{b_{2j}}{2^j} \right). \quad (375)$$

This defines a continuous map

$$\Gamma : C \rightarrow [0, 1]^2. \quad (376)$$

We claim that Γ is surjective. Indeed, take any point

$$(y, z) \in [0, 1]^2. \quad (377)$$

Write y and z in binary:

$$y = 0.c_1c_2c_3 \cdots, \quad z = 0.d_1d_2d_3 \cdots, \quad (378)$$

where $c_i, d_i \in \{0, 1\}$. Now interleave these digits to define

$$b_1 = c_1, \quad b_2 = d_1, \quad b_3 = c_2, \quad b_4 = d_2, \quad \dots \quad (379)$$

Then define

$$x = 0.(2b_1)(2b_2)(2b_3)(2b_4) \cdots. \quad (380)$$

Since the ternary digits of x are only 0 and 2, we have $x \in C$. By construction,

$$\Gamma(x) = (y, z). \quad (381)$$

Hence $\Gamma(C) = [0, 1]^2$.

Finally, we extend Γ from C to all of $[0, 1]$. The complement $[0, 1] \setminus C$ is a union of disjoint open intervals. On each such gap (a, b) , define γ by linearly interpolating between the endpoint values $\Gamma(a)$ and $\Gamma(b)$:

$$\gamma(t) = \frac{b-t}{b-a}\Gamma(a) + \frac{t-a}{b-a}\Gamma(b), \quad t \in (a, b). \quad (382)$$

On the Cantor set itself, define

$$\gamma(t) = \Gamma(t), \quad t \in C. \quad (383)$$

This gives a continuous map

$$\gamma : [0, 1] \rightarrow [0, 1]^2. \quad (384)$$

Since γ agrees with Γ on C , and since

$$\Gamma(C) = [0, 1]^2, \quad (385)$$

we also have

$$\gamma([0, 1]) = [0, 1]^2. \quad (386)$$

Therefore γ is a continuous surjection from $[0, 1]$ onto the unit square. This shows that a continuous 1-parameterization can have a 2-dimensional image. Thus continuity alone is not enough for a useful geometric theory of curves and surfaces.

Definition 5.2 (Reparameterization)

Let $E \subset \mathbb{R}^n$ with parameterization $\varphi : D \subset \mathbb{R}^k \rightarrow E$. Given **reparameterization** of E is a continuous bijective map $h : D' \subset \mathbb{R}^k \rightarrow D$, where D' is a compact subset of $\mathbb{R}^k$. Therefore,

$$\varphi \circ h : D' \rightarrow E \quad (387)$$

is another parameterization of E .

Note that a parameterization is not necessarily a homeomorphism. It may fail to be injective, and its image may have self-intersections or other singular behavior. If φ is a continuous bijection from compact D onto $E \subset \mathbb{R}^n$, then φ is a homeomorphism onto E . In general, however, a parameterized set should not automatically be regarded as a manifold.

We must make sure that the set E is smooth in the sense that (informally) there aren't any wrinkles, points, folds, or self-intersections in such a way that the tangent plane to the surface is not well-defined.

Definition 5.3 (Regular Parameterization)

Let $E \subset \mathbb{R}^n$. A **regular k -parameterization** of E is a continuously differentiable surjective map

$$\varphi : D \subset \mathbb{R}^k \longrightarrow E, \quad (388)$$

where D is compact and where $D\varphi_p : \mathbb{R}^k \rightarrow \mathbb{R}^n$ has rank k for every interior point $u \in D$.

There are many special cases of surfaces, most notably 1-dimensional paths in $\mathbb{R}^2$ and $\mathbb{R}^3$, along with 2-dimensional surfaces in $\mathbb{R}^3$.

Definition 5.4 (Curve)

A **parameterized curve** is a C^1 surjective map

$$\varphi : [a, b] \subset \mathbb{R} \rightarrow E\mathbb{R}^n \quad (389)$$

A **curve** is the image $\varphi([a, b]) \subset \mathbb{R}^n$.

1. φ is **closed** if $\varphi(a) = \varphi(b)$.
2. φ is **simple** if the restriction of φ to $[a, b]$ is injective.^a

^aVisually, the curve doesn't intersect itself except possible at the endpoints.

Definition 5.5 (Surface)

A **parameterized surface** is a C^1 surjective map

$$\varphi : [a_1, b_1] \times [a_2, b_2] \subset \mathbb{R}^2 \rightarrow \mathbb{R}^n \quad (390)$$

A **surface** is the image E .

5.1 Tangent Spaces and Orientation

In lower dimensions, we can find some motivation for the tangent space of a parameterized set.

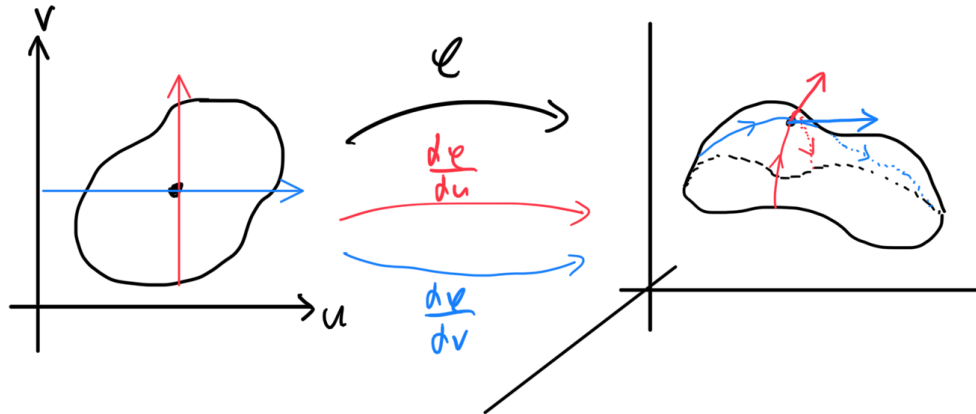


Figure 23: When $\varphi : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}^3$, we can see visually that given $(u_0, v_0) \in D$, there should be two linearly independent tangent vectors T_u, T_v sticking out from $\varphi(u_0, v_0)$ on the surface E . So every vector in the “tangent plane” should be a linear combination of T_u and T_v . In fact, these two vectors are precisely $T_u = \frac{\partial \varphi}{\partial u}(u_0, v_0)$ and $T_v = \frac{\partial \varphi}{\partial v}(u_0, v_0)$, so the tangent plane is really just $\varphi(u_0, v_0)$ plus the vectors in the column space of $D\varphi(u_0, v_0)$!

Definition 5.6 (Tangent Space)

Let $\varphi : D \subset \mathbb{R}^k \rightarrow \mathbb{R}^n$ be a regular parameterization, and let $u \in D$ be an interior point. The **tangent space** to the parameterized set at $\varphi(u)$ is the k -dimensional subspace

$$T_{\varphi(u)}E := \text{Im } D\varphi_u \subset \mathbb{R}^n. \quad (391)$$

The corresponding **affine tangent plane** is

$$\varphi(u) + T_{\varphi(u)}E = \{\varphi(u) + v : v \in T_{\varphi(u)}E\}. \quad (392)$$

If $k = n - 1$, then the normal space is one-dimensional. If $N(u)$ is a nonzero vector satisfying $N(u) \perp T_{\varphi(u)}E$, then the affine tangent hyperplane can also be written as

$$\{x \in \mathbb{R}^n : \langle x - \varphi(u), N(u) \rangle = 0\}. \quad (393)$$

Often, we work with 2-parameterizations of some $E \subset \mathbb{R}^3$. Therefore, the definition for a regularity at a point (u_1, u_2) is equivalent to

$$\frac{\partial \varphi}{\partial u_1}(u_1, u_2) \times \frac{\partial \varphi}{\partial u_2}(u_1, u_2) \neq 0 \quad (394)$$

where $\times$ is the Euclidean cross product.

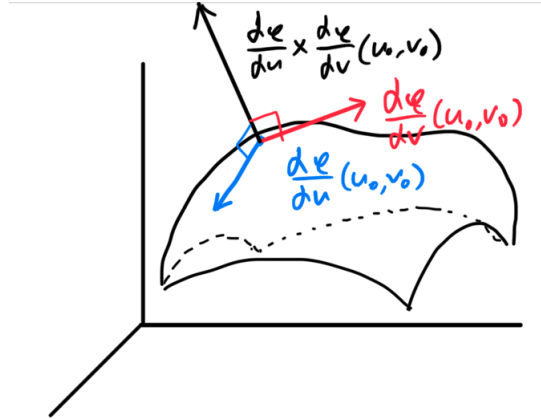


Figure 24: Note that the cross product $\frac{\partial \varphi}{\partial u}(u_0, v_0) \times \frac{\partial \varphi}{\partial v}(u_0, v_0) \neq 0$ if and only if $D\varphi(u_0, v_0)$ is full rank, i.e. the tangent plane exists.

A path has a natural orientation defined by the function. We would like to determine a similar notion of orientation by defining one “side” of a surface to be positive and the other to be negative. Surprisingly, a parameterization does not have an intrinsic global orientation. Rather, we determine the orientation ourselves by choosing a unit vector that generally points towards the outside of the surface S . Again, this choice is arbitrary, but it is customary to choose a vector that generally points “out.”

Definition 5.7 (Orientation of a Regular Parameterized Set)

Let $\varphi : D \subset \mathbb{R}^k \rightarrow \mathbb{R}^n$ be a regular parameterization. The ordered tangent basis

$$D\varphi_p(e_1), \dots, D\varphi_p(e_k) \quad (395)$$

determines an orientation of the tangent space

$$T_{\varphi(p)}E = \text{Im}(D\varphi_p). \quad (396)$$

An **orientation** of E is a continuous choice of such positive tangent bases at each point of E .

Example 5.2 (Orientation of Surface)

When $k = 2$ and $n = 3$, an orientation can equivalently be described by a continuous choice of unit normal vector

$$n(\varphi(u, v)) = \pm \frac{\varphi_u(u, v) \times \varphi_v(u, v)}{\|\varphi_u(u, v) \times \varphi_v(u, v)\|}. \quad (397)$$

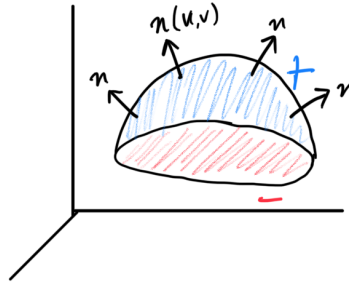


Figure 25

Example 5.3 (Determining Orientation of an Ellipsoid)

Consider the ellipsoid

$$E = \left\{ (x, y, z) \in \mathbb{R}^3 : \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \right\}, \quad (398)$$

where $a, b, c > 0$. A standard parameterization is

$$\varphi(u, v) = (a \sin u \cos v, b \sin u \sin v, c \cos u), \quad (399)$$

where

$$0 \leq u \leq \pi, \quad 0 \leq v \leq 2\pi. \quad (400)$$

The tangent vectors are

$$\varphi_u(u, v) = (a \cos u \cos v, b \cos u \sin v, -c \sin u) \quad (401)$$

and

$$\varphi_v(u, v) = (-a \sin u \sin v, b \sin u \cos v, 0). \quad (402)$$

Their cross product is

$$\varphi_u(u, v) \times \varphi_v(u, v) = (bc \sin^2 u \cos v, ac \sin^2 u \sin v, ab \sin u \cos u). \quad (403)$$

To determine whether this points outward or inward, compare it with the position vector

$$\varphi(u, v) = (a \sin u \cos v, b \sin u \sin v, c \cos u). \quad (404)$$

Compute the dot product:

$$(\varphi_u \times \varphi_v) \cdot \varphi = abc \sin^3 u \cos^2 v + abc \sin^3 u \sin^2 v + abc \sin u \cos^2 u \quad (405)$$

$$= abc \sin u (\sin^2 u \cos^2 v + \sin^2 u \sin^2 v + \cos^2 u) \quad (406)$$

$$= abc \sin u. \quad (407)$$

For $0 < u < \pi$, we have $\sin u > 0$, so

$$(\varphi_u \times \varphi_v) \cdot \varphi > 0. \quad (408)$$

Hence $\varphi_u \times \varphi_v$ generally points away from the origin. Therefore, this parameterization induces the outward orientation on the ellipsoid.

The corresponding unit normal vector field is

$$n(u, v) = \frac{\varphi_u(u, v) \times \varphi_v(u, v)}{\|\varphi_u(u, v) \times \varphi_v(u, v)\|}. \quad (409)$$

If we wanted the inward orientation, we would use

$$-n(u, v) = -\frac{\varphi_u(u, v) \times \varphi_v(u, v)}{\|\varphi_u(u, v) \times \varphi_v(u, v)\|}. \quad (410)$$

Example 5.4 (Möbius Strip is Not Orientable)

The Möbius strip can be parameterized by

$$\varphi(u, v) = \left(\left(1 + \frac{v}{2} \cos \frac{u}{2} \right) \cos u, \left(1 + \frac{v}{2} \cos \frac{u}{2} \right) \sin u, \frac{v}{2} \sin \frac{u}{2} \right), \quad (411)$$

where

$$0 \leq u \leq 2\pi, \quad -1 \leq v \leq 1. \quad (412)$$

Here u moves around the central circle, while v moves across the width of the strip. The factors $\cos(u/2)$ and $\sin(u/2)$ encode the half-twist.

Notice that when $u = 0$,

$$\varphi(0, v) = \left(1 + \frac{v}{2}, 0, 0 \right), \quad (413)$$

while when $u = 2\pi$,

$$\varphi(2\pi, v) = \left(1 - \frac{v}{2}, 0, 0 \right) = \varphi(0, -v). \quad (414)$$

Thus, after going once around the strip, the transverse coordinate v is reversed.

This reversal is the geometric reason the Möbius strip is not orientable. If we start with a normal vector n at some point and move it continuously once around the strip, then after returning to the same point, the normal vector points in the opposite direction, namely $-n$. Therefore, there is no continuous way to choose a unit normal vector on the entire Möbius strip.

Hence the Möbius strip is not orientable.

5.2 Line Integrals**Definition 5.8 (Scalar Line Integral)**

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$, which can be interpreted as a scalar field. Now define a C^1 path function

$$c : [a, b] \subset \mathbb{R} \rightarrow \mathbb{R}^n \quad (415)$$

such that the composition of functions

$$f \circ c : [a, b] \subset \mathbb{R} \rightarrow \mathbb{R}^n \quad (416)$$

is continuous. Then, the *path integral*, or *scalar line integral*, of f along the path c is defined

$$\int_c f \, ds = \int_a^b f(c(t)) \|c'(t)\| \, dt \quad (417)$$

$$= \int_a^b f(x_1(t), x_2(t), \dots, x_n(t)) \|c'(t)\| \, dt \quad (418)$$

If $c(t)$ is only piece-wise C^1 , we can define the path integral by breaking $[a, b]$ into pieces over which $f(c(t)) \|c'(t)\|$ is continuous and then summing the integrals over the pieces. That is,

$$\int_a^b f(c(t)) \|c'(t)\| \, dt = \sum_{i=0}^{n-1} \int_{\alpha_i}^{\alpha_{i+1}} f(c(t)) \|c'(t)\| \, dt \quad (419)$$

Note that since f is a scalar-valued function, we can interpret a path integral as the sum of infinitesimal segments of the path c having a weight determined by f at each section. If f is a constant function outputting

1 at every point, then the path integral just outputs the length of the path c in $\mathbb{R}^n$.

$$L = \int_a^b f(c(t)) \|c'(t)\| dt = \int_a^b \|c'(t)\| dt \quad (420)$$

Definition 5.9 (Vector Line Integral)

Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a vector field on $\mathbb{R}^n$ that is continuous on the C^1 oriented path $c : [a, b] \subset \mathbb{R} \rightarrow \mathbb{R}^n$. The *line integral* of F along c is defined by the formula

$$\int_c F \cdot ds = \int_a^b F(c(t)) \cdot c'(t) dt \quad (421)$$

where $\cdot$ represents the dot product of F with c' over the interval $[a, b]$. It is also commonly written in differential notation,

$$\int_c F \cdot ds = \int_c F \cdot (dx_1, \dots, dx_n) = \int_c F_1 dx_1 + F_2 dx_2 + \dots F_n dx_n \quad (422)$$

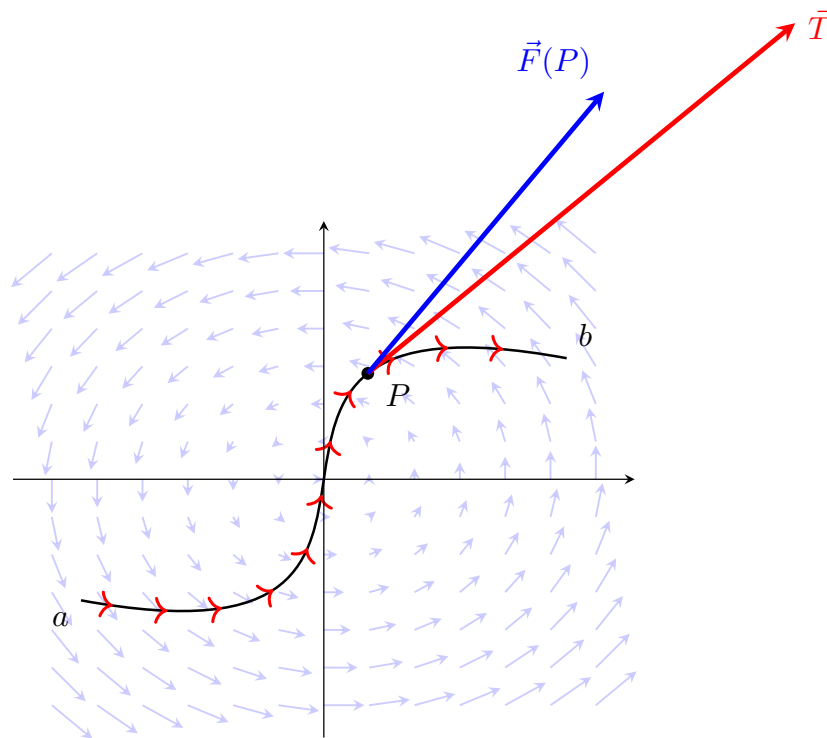


Figure 26: Visualizing a path integral. At a specific point P along the oriented curve, the red tangent vector $\vec{T}$ points along the path, while the solid blue vector $\vec{F}(P)$ represents the local vector field at that exact location. The path integral $\int_C \vec{F} \cdot d\vec{r}$ accumulates the dot product of these two vectors at every point along the curve.

Similarly with path integrals, we can also define line integrals as the sum of integrals over piece-wise continuous sections of c . That is, given an oriented curve C made up of several oriented component curves C_i , $i = 1, 2, \dots, k$, we can paramaterize C by paramaterizing the pieces C_i 's separately. Thus, we can treat $C = C_1 + \dots + C_k$ and get

$$\int_C F \cdot ds = \sum_{i=1}^k \int_{C_i} F \cdot ds \quad (423)$$

Note that a vector line integral is a generalization of scalar line integrals, so any results holding for vector line integrals also holds for their scalar counterpart.

Example 5.5 (Work)

In mechanics, **work** W is defined as

$$W = F \cdot d \quad (424)$$

where F is force and d is displacement. With this knowledge, the reader can easily see that the work done by vector field F on a particle traveling along a path c from time a to time b can be calculated by the line integral

$$W = \int_a^b F(c(t)) \cdot c'(t) dt \quad (425)$$

$$= \int_c F_1 dx + F_2 dy + F_3 dz \quad (426)$$

Theorem 5.1 (Invariance of Path Paramaterizations on Vector Line Integrals)

Let F be a vector field and f be a scalar field, both continuous on the C^1 path function $p : [a, b] \rightarrow \mathbb{R}^n$ and let $q : [\alpha, \beta] \rightarrow \mathbb{R}^n$ be a reparamaterization of p . Then,

$$\begin{aligned} q \text{ is orientation preserving} &\implies \int_p F \cdot ds = \int_q F \cdot ds \\ q \text{ is orientation reversing} &\implies \int_p F \cdot ds = - \int_q F \cdot ds \end{aligned}$$

We now introduce a fundamental theorem about line integrals over gradient fields. Recall the fundamental theorem of calculus. We can extend this to line integrals for functions mapping $\mathbb{R}^n$ to $\mathbb{R}$.

Theorem 5.2 (Invariance of Line Integrals in Conservative Vector Fields)

Given that $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a C^1 conservative vector field with $\nabla f = F$ for C^2 function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and path function $p : [a, b] \rightarrow \mathbb{R}^n$ is a piecewise C^1 path, then

$$\int_p F \cdot ds = \int_p \nabla f \cdot ds = f(p(b)) - f(p(a)) \quad (427)$$

That is, the line integral of any path in a conservative vector field is dependent on the value of f at the endpoints $p(a)$ and $p(b)$.

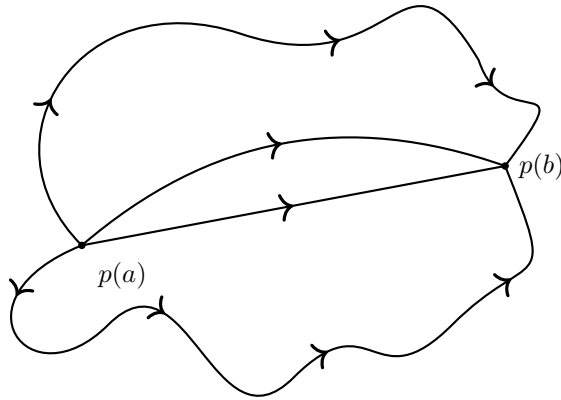


Figure 27

In physics, calculating the work done by a force represented by a vector field requires us to know the path that it travels through.

$$W = \int_p F \cdot ds \quad (428)$$

However, in many cases F is assumed to be conservative, so it is only necessary that we find the displacement of the particle from its endpoints, resulting in the simplification of the formula.

$$W = \int_p \nabla f \cdot ds = f(p(b)) - f(p(a)) \quad (429)$$

Corollary 5.3 (Equivalent Conditions for Vector Field to be Conservative)

The following conditions are equivalent:

1. $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a conservative vector field.
2. The line integral of $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ in curve C is path independent; that is, if C_1 and C_2 are two paramaterizations of C ,

$$\int_{C_1} F \cdot ds = \int_{C_2} F \cdot ds$$

3. Given that C is a closed loop, the line integral of $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ across C is 0.

$$\oint_C F \cdot ds = 0$$

4. The curl of $F : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ vanishes

$$\text{curl } F = \nabla \times F = \begin{pmatrix} \frac{\partial F}{\partial x} \\ \frac{\partial F}{\partial y} \\ \frac{\partial F}{\partial z} \end{pmatrix} \times \begin{pmatrix} F_1 \\ F_2 \\ F_3 \end{pmatrix} = 0$$

5. The following partial derivatives of $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ are equal

$$\frac{\partial F_1}{\partial y} = \frac{\partial F_2}{\partial x}$$

We can develop a bit of intuition to determine whether a vector field is conservative or not. If vector field F is conservative, then there exists a smooth scalar field f such that $\nabla f = F$. For each latitude and longitude on a certain map, we can give it an altitude as a function of those coordinates (picture a map with a bunch of hills and valleys). The gradient and thus the vector field is all the vectors that point in the direction of highest ascent. The vector field is all the vectors that point in the direction of highest ascent. Extending the metaphor the path integral is like starting on at a point and climbing the hills and valleys, creating work as you go up a hill (proportional to the steepness and thus the dot product of your motion vector with the gradient vector field in the path integral) and decreasing the work you put in by going down a hill. Since the path is closed, it is like you are going up and down the same amount overall, so the path integral is zero. Following this analogy, the vector field determined by this function (marked as arrows in the x, y plane) is conservative.

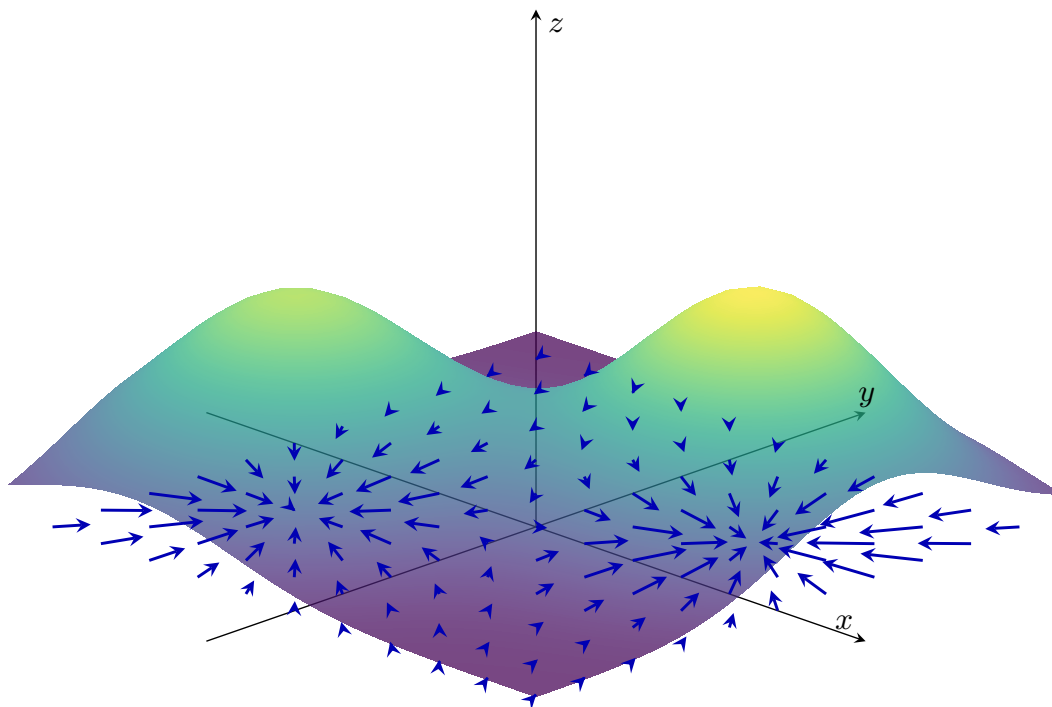


Figure 28: The vector field on the xy -plane is a conservative vector field since it is the gradient of the scalar field $f(x, y) = 2.2e^{-((x-1.2)^2+(y-0.8)^2)/1.6} + 1.9e^{-((x+1.3)^2+(y+0.9)^2)/1.8} + 0.12$. The arrows on the xy -plane point in the direction of steepest ascent on the surface.

If we can construct a closed loop around F where the line integral is nonzero, then it means that we have ended up at a "higher" or "lower" (altitude) at the same point. This means that rather than being a certain landscape, there exist different "levels" of values at one point, like a spiraling staircase. For example, look at the solenoidal vector field below, where we can construct a closed loop (a circle going around the origin counterclockwise). There is no "surface" that can be defined such that it contains the solenoid.

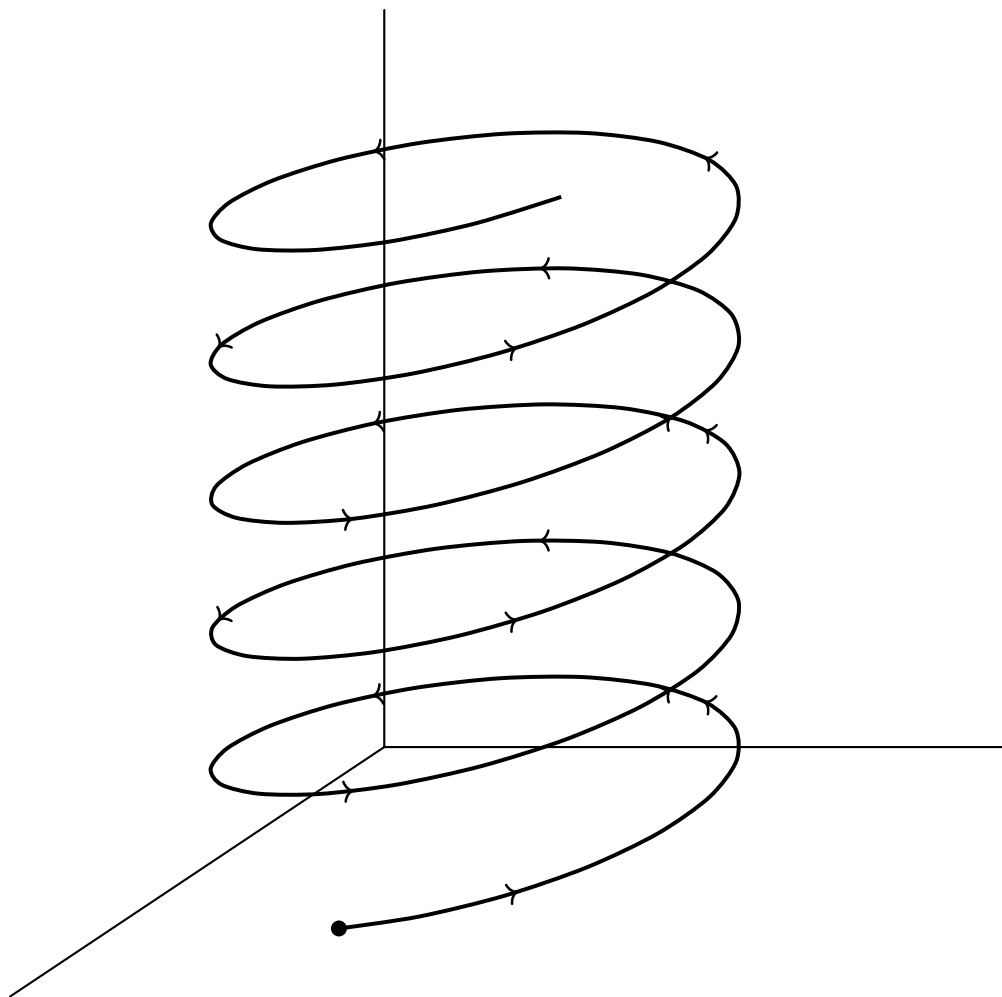


Figure 29

Clearly, as a particle travels through the vector field along the path, it does positive work while it has zero displacement, and clearly, there exists no function that can output both these values as determined by vector field F .

Theorem 5.4 (Helmholtz Decomposition)

Let $F : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a C^2 vector field. Then, F can be decomposed into a curl-free component and a divergence-free component. That is, there exists vector fields A and Φ

$$F = -\nabla \cdot \Phi + \nabla \times A \quad (430)$$

Definition 5.10 (Curvature at a Point)

Let $c : [a, b] \rightarrow C \subset \mathbb{R}^3$ be a unit-speed parameterization of C , meaning that $\|c'(t)\| = 1$ for all $t \in [a, b]$, and let $p = c(t_0)$ be a point in C . The *curvature* $\kappa(p)$ at p is a mapping defined

$$\kappa : C \rightarrow \mathbb{R}, \quad \kappa(p) := \|c''(t_0)\| \quad (431)$$

Notice that since we require a unit speed parameterization of C , we do not need to worry about how

a given curve is paramaterized.

Since the curvature is defined pointwise for each point in curve C , we can integrate over all the curvatures in C to define the total curvature.

Definition 5.11 (Total Curvature)

The *total curvature* of a curve $c : [a, b] \rightarrow C \subset \mathbb{R}^3$ is the scalar line integral

$$\int_C \kappa ds \quad (432)$$

Theorem 5.5 (Fary-Milnor Theorem)

Given a unit speed paramaterization $c : [a, b] \rightarrow C \subset \mathbb{R}^3$, if C is closed (that is, $c(a) = c(b)$), then

$$\oint_C \kappa ds \geq 2\pi \quad (433)$$

and equals 2π only when C is a circle. Furthermore, if C is a closed space curve with

$$\oint_C \kappa ds \leq 4\pi \quad (434)$$

then C is "unknotted." That is, C can be continuously deformed without ever intersecting itself into a planar circle. Therefore, for knotted curves C , we have

$$\oint_C \kappa ds > 4\pi \quad (435)$$

5.3 Surface Integrals

Surface integrals are the 2-dimensional analogue, or the double integral version, of line integrals. It is the integration of surfaces. A physical interpretation of a scalar surface integral is the weighted surface area of a certain surface.

Definition 5.12 (Scalar Surface Integrals)

Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be a C^1 scalar field defined on a paramaterized surface $S \subset \mathbb{R}^3$ with paramaterization $\varphi : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}^3$. That is, $\varphi(D) = S$. We define the integral f over S to be

$$\iint_S f dS = \iint_S f(x, y, z) dS \quad (436)$$

$$= \iint_D f(\varphi(u, v)) \left\| \frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v} \right\| du dv \quad (437)$$

Note that this will require us to transform f , a function of x, y, z , into the function $f \circ \varphi$ of u, v . Additionally, if the paramaterization of the surface S is not defined, then it one must be constructed. It is also clear that if S is a union of surfaces S_i , then its surface integral is the sum of the surface integrals of the S_i 's.

Letting the scalar field f be the constant field equal to 1, the scalar surface integral measures the surface

area of S .

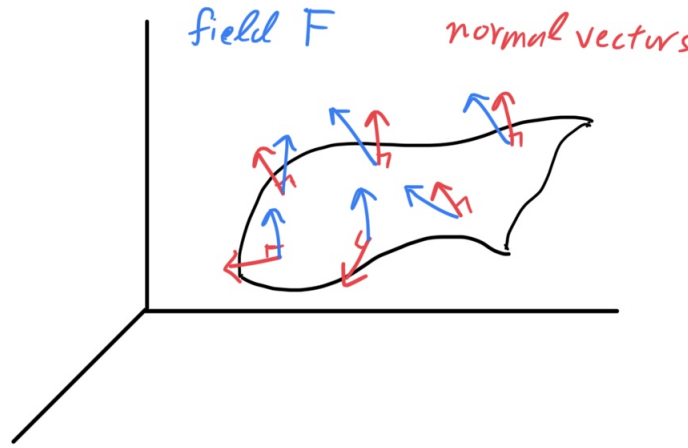
$$A(S) = \iint_S dS = \iint_D \left\| \frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v} \right\| du dv$$

It is easy to see that the orientation of the parameterization φ does not affect scalar surface integrals, since the sign of the orientation gets nullified by the absolute value sign over $\left\| \frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v} \right\|$.

Its physical interpretation is to measure the rate at which a fluid (determined by a vector field F) is crossing a given surface S . It also has many applications in electromagnetism.

Definition 5.13 (Vector Surface Integrals)

Let F be a vector field defined on surface S , the image of a parameterized surface φ . The *surface integral* of F over S is defined below, which is equivalent to summing up the dot product of the vector field and the normal vector to the surface.



It can be calculated with the following formulas by converting it into a scalar surface integral where the scalar field is the value of the dot product of the vector field with the normal vectors of the surface.

$$\iint_S F \cdot dS = \iint_S (F \cdot n) dS \quad (438)$$

$$= \iint_D \left(F(\varphi(u, v)) \cdot \frac{\frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v}}{\left\| \frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v} \right\|} \right) \left\| \frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v} \right\| du dv \quad (439)$$

$$= \iint_D F(\varphi(u, v)) \cdot \left(\frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v} \right) du dv \quad (440)$$

Since we are now talking about vector fields, the orientation of the parameterization is now significant. Visually, if the orientation of the surface S generally aligns with the vector field F , then the integral will be positive (since two vectors α, β generally pointing in the same direction implies that $\alpha \cdot \beta > 0$). The orientation of the parameterization, which is dependent on $\frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v}$, determines the direction of the normal vector n (since it is defined to be $(\frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v}) / \left\| \frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v} \right\|$). Therefore, changing the orientation of φ will reverse the direction of n , which will then reverse the sign of the integral since n now points in the opposite direction of the vector field F than it previously did (by reversing the sign of the dot products). This is formalized in the theorem below.

Theorem 5.6 (Invariance Vector Surface Integrals Under Parameterization)

Let S be an oriented surface and let φ_1 and φ_2 be two regular parameterizations with F a continuous vector field defined on S . Then, assuming φ_1 is orientation preserving,

$$\varphi_2 \text{ is orientation preserving} \implies \iint_{\varphi_1} F \cdot dS = \iint_{\varphi_2} F \cdot dS \quad (441)$$

$$\varphi_2 \text{ is orientation reversing} \implies - \iint_{\varphi_1} F \cdot dS = \iint_{\varphi_2} F \cdot dS \quad (442)$$

Theorem 5.7 (Surface Integral Formula for Graphs)

Given that we have the graph of a function $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ rather than a general surface, we can parameterize it simply as

$$\varphi(u, v) := (u, v, g(u, v)) \quad (443)$$

This means that

$$\frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v} = \begin{pmatrix} -\frac{\partial g}{\partial u} \\ -\frac{\partial g}{\partial v} \\ 1 \end{pmatrix} \implies \left\| \frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v} \right\| = \sqrt{1 + \left(\frac{\partial g}{\partial u}\right)^2 + \left(\frac{\partial g}{\partial v}\right)^2} \quad (444)$$

So we can simplify the equation for the surface area S of the graph of g over the region D in the xy -plane, as

$$A(S) = \iint_S dS = \iint_D \left\| \frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v} \right\| dA \quad (445)$$

$$= \iint_D \sqrt{1 + \left(\frac{\partial g}{\partial u}\right)^2 + \left(\frac{\partial g}{\partial v}\right)^2} du dv \quad (446)$$

With the same g , we can find the weighed surface area of S over the scalar function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ with the formula

$$\iint_S f dS = \iint_D f(u, v, g(u, v)) \sqrt{1 + \left(\frac{\partial g}{\partial u}\right)^2 + \left(\frac{\partial g}{\partial v}\right)^2} du dv \quad (447)$$

Finally, with the same graph g , the surface integral over the vector field F is

$$\iint_S F \cdot dS = \iint_D F(\varphi(u, v)) \cdot \left(\frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v} \right) du dv \quad (448)$$

$$= \iint_D \left(F_1(u, v) \left(-\frac{\partial g}{\partial u} \right) + F_2(u, v) \left(-\frac{\partial g}{\partial v} \right) + F_3(u, v) \right) du dv \quad (449)$$

5.4 Green Theorem

Recall the differential notation for writing line integrals. For 2 and 3 dimensions, it is written as

$$\int_C F \cdot ds = \int_C F \cdot (dx, dy) = \int_C F_1 dx + F_2 dy \quad (450)$$

$$\int_C F \cdot ds = \int_C F \cdot (dx, dy, dz) = \int_C F_1 dx + F_2 dy + F_3 dz \quad (451)$$

Green's Theorem gives the relationship between a line integral around a simple closed curve C and a double integral over the plane region D bounded by C .

Theorem 5.8 (Green's Theorem in $\mathbb{R}^2$)

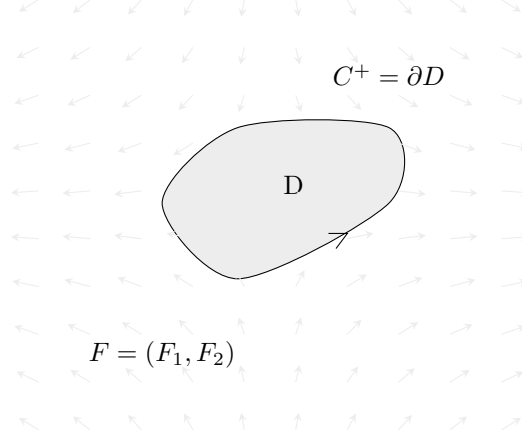
Let there be a 2-dimensional C^1 vector field F on $\mathbb{R}^2$ defined on a simple oriented closed piecewise-smooth curve C and its bounded region $D \subset \mathbb{R}^2$ (that is, $C = \partial D$). Let the orientation of the path of C be such that it is traveling *counterclockwise*, i.e. a point traveling through C would see the region D to its *left*, denoted as C^+ and the clockwise orientation as C^- . Then,

$$\oint_{C^+} F_1 dx + F_2 dy = \iint_D \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dx dy \quad (452)$$

By reversing the orientation, it is clear that we have

$$\oint_{C^-} F_1 dx + F_2 dy = - \iint_D \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dx dy \quad (453)$$

Note that this theorem is expressed in terms of the components of the vector field F .



Green's theorem has many applications in physics. For example, in order to solve two-dimensional flow integrals measuring the sum of fluid outflowing from a volume, Green's theorem allows us to calculate the total outflow summed about an enclosing area .

Corollary 5.9

Let D be a region for which Green's theorem applies with positively oriented boundary ∂D . Then, the area of D can be computed with the formula

$$A(D) = \frac{1}{2} \oint_{\partial D} x dy - y dx \quad (454)$$

Green's theorem can be used to determine the area of centroid of plane figures solely by integrating over the perimeter.

5.5 Stokes' Theorem

Green's theorem relates line integrals to double integrals. Stokes' theorem generalizes Green's theorem by relating line integrals to surface integrals of 2-dimensional surfaces embedded in $\mathbb{R}^3$.

Theorem 5.10 (Stokes' Theorem)

Let S be an oriented regular surface defined by parameterization $\varphi : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}^3$, and let the image of the boundary ∂D under φ be the boundary ∂S of S .

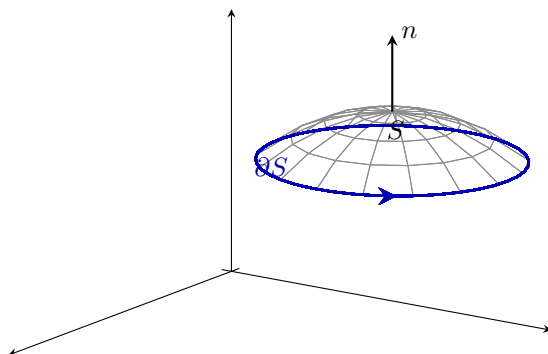


Figure 30: The orientation unit vector n of S induces the positive orientation of ∂S , denoted ∂S^+ . Visually, if you are walking along the curve with your head is pointing in the same direction as the unit normal vectors while the surface is on the left then you are walking in the positive direction on ∂S .

Given that F is a C^1 vector field defined on S , then

$$\iint_S \text{curl } F \cdot dS = \iint_S (\nabla \times F) \cdot dS = \oint_{\partial S^+} F \cdot ds \quad (455)$$

If S has no boundary, that is, if the image of $p' = \partial S$ is not a simple closed curve, then the integral is 0.

The above theorem implies that the vector surface integral of a surface without a boundary (i.e. a closed graph, such as a sphere) is always 0 along the curl of any C^1 field. Geometrically, this means that given a closed solid S with field $\nabla \times F$, the rate of flow of the vector field into S is equal to the flow out of S .

5.6 Gauss' Theorem

The divergence theorem relates the flux of a vector field through a closed surface to the divergence of the field in the volume enclosed.

Theorem 5.11 (Gauss' Divergence Theorem)

Let V be a subset of $\mathbb{R}^3$. Denote by ∂V the oriented closed surface that bounds V (with outward pointing normal orientation vectors), and let F be a C^1 vector field defined on a neighborhood of V . Then,

$$\iiint_V \text{div } F \, dV = \iiint_V (\nabla \cdot F) \, dV = \oiint_{\partial V} F \cdot dS = \oiint_{\partial V} (F \cdot n) \, dS \quad (456)$$

where the two left-most integrals are volume integrals, and the two right-most integrals are surface integrals.

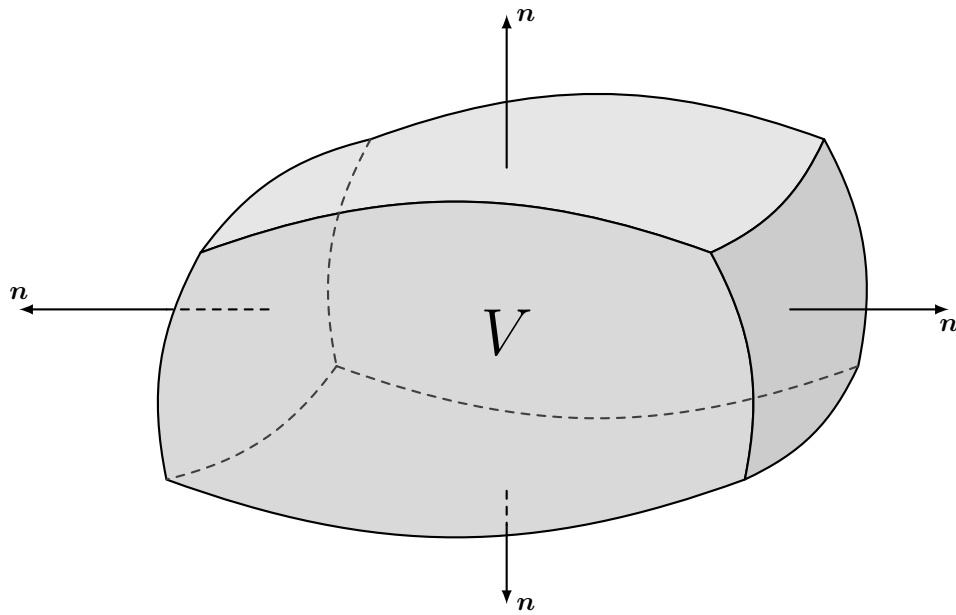


Figure 31

Intuitively, this makes sense; the volume integrals represent the total of the sources in volume V , and the right hand side represents the total flow across the boundary ∂V .

6 Sequences of Functions